

A Forbidden-Minor Characterization of Strictly Metrizable Graphs

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Abstract

A path system on a graph specifies one path between each pair of vertices and is consistent when every subpath of a specified path is itself specified. A graph is strictly metrizable if every consistent path system on it can be realized as the system of unique shortest paths for a positive edge weighting. We prove that a finite graph is strictly metrizable if and only if it has no minor isomorphic to any of six graphs: $K_{2,4}$, the triangular prism, $K_{3,3}$, the two subdivisions of K_4 obtained by subdividing respectively the edges of a triangle and the edges incident with one vertex, and $\Theta_{3,3,3}$. This resolves a conjecture of Chudnovsky, Cizma, and Linial. A support reduction turns a minor-minimal counterexample into a 2-connected graph carrying a neighborly path system. Structural reductions and a theorem of Lovász leave two finite cores and six parameterized subdivision families. We invoke the strict theta classification and construct positive strict realizations for the remaining five families. The W_5 case is certified by twelve explicit positive integer weightings, one for each dihedral orbit of consistent neighborly systems.

Keywords. consistent path system; unique shortest path; strict metrizability; graph minor; forbidden minor; graph subdivision.

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1 Introduction

Let $G = (V, E)$ be a finite graph. A path system on G prescribes one simple path P_{uv} for every pair of vertices u, v . It is consistent when every subpath of a prescribed path is itself the prescribed path between its endpoints. Every positive edge weighting with unique shortest paths produces a consistent path system, but the converse can fail. Following Cizma and Linial [4], we call G **strictly metrizable** if every consistent path system on G is the system of unique shortest paths for some positive edge weighting.

This paper determines exactly which graphs have that property. Let

$$\mathcal{F} = \{K_{2,4}, C_3 \square K_2, K_{3,3}, K_4^\Delta, K_4^*, \Theta_{3,3,3}\}, \quad (1.1)$$

where K_4^Δ is obtained from K_4 by subdividing once each edge of a triangle, K_4^* is obtained by subdividing once each of the three edges incident with one vertex, and $\Theta_{3,3,3}$ consists of three internally disjoint paths of length three with common endpoints. The six graphs are shown in Figure 1.

Theorem 1.1 (forbidden-minor characterization)

A finite graph G is strictly metrizable if and only if no member of $\mathcal{F}$ is a minor of G .

The six graphs in (1.1) were exhibited as non-strictly-metrizable minor obstructions by Chudnovsky, Cizma, and Linial, who conjectured that they form the complete list [2, Conjecture 5.1, Figure 14, and Appendix A]. Thus the main theorem resolves their conjecture. The necessity follows immediately from their theorem that strict metrizability is minor-closed. The work here is the converse: every graph with none of the six minors admits a positive strict realization for every consistent path system.

Previous work

Cizma and Linial introduced graph metrizability as a discrete analogue of geodesic geometry [4]. In the non-strict version, every prescribed path need only be shortest; ties are allowed. In the strict version, it must be the unique shortest path. They proved, among other results, that both properties are closed under topological minors, that cycles and outerplanar graphs are strictly metrizable, and that graph metrizability has a finite forbidden-topological-minor characterization.

Bodwin studied the related realization problem for partial systems of unique shortest paths and obtained a structural characterization by forbidden intersection patterns [1]. Cizma and Linial later constructed irreducible nonmetrizable path systems, showing that nonmetrizability need not decompose into smaller path-system obstructions [5]. Chudnovsky, Cizma, and Linial developed a structural description of metrizable graphs: apart from compliant edges, sufficiently large 2-connected metrizable graphs come from a small collection of subdivision cores [3].

The decisive development for the strict theory is the ordinary-minor closure theorem of Chudnovsky, Cizma, and Linial [2]. Together with the Graph Minor Theorem [9], it implies the existence of a finite forbidden minor set, but it does not identify that set. Their six-obstruction conjecture asks for precisely this identification.

Theta graphs illustrate why the remaining direction is not merely a finite check. For positive integers a, b, c , the graph $\Theta_{a,b,c}$ has two poles joined by three internally disjoint arms of lengths a, b, c . Wang has classified both ordinary and strict metrizability in this family: for $a \leq b \leq c$, the graph is metrizable if and only if $a \leq 2$ or $(a, b, c) = (3, 3, 3)$, and it is strictly metrizable if and only if $a \leq 2$ [10]. The strict realization of $\Theta_{2,b,c}$ used below is therefore cited as an external input and is not reproved here.

In a companion manuscript, Wang classifies ordinary and neighborly metrizability within the K_4 topological type [11]. That work contains non-strict precursors of the at-most-two-long and split-bridge realization theorems, together with Ferrers-potential and terminal-potential constructions. Sections 6 and 8 give the self-contained strict rational refinements required here. The H^{rr} , H^{sr} , and U realization theorems and the global ordinary-minor characterization are not consequences of that companion paper.

A complementary quantitative direction asks how frequent metrizable systems are on one fixed graph. Wang constructed a sequence of 2-connected graphs on which the proportion of metrizable, and hence of strictly metrizable, path systems tends to zero [12]. That enumerative result is logically independent of the universal graph classification proved here.

Contributions and proof outline

The proof separates the graph-structural reduction from the construction of edge weights. The following table records the provenance of the ingredients that are closest to the three companion manuscripts; this separation is also the division of claims used in the proofs below.

ingredient	status in this article	role
strict minor closure and the six negative certificates	external, [2]	necessity and minimal-counterexample reduction
strict theta classification	external, [10]	the family $\Theta_{2,b,c}$
ordinary/neighborly K_4 -subdivision constructions	non-strict precursors in [11]	Section 6 proves the strict refinements needed here
H^{rr} , H^{sr} , and U strict realizations	proved here	the three surviving wheel-derived families
K_5 and W_5	proved here	the two finite cores
asymptotic counting on Γ_m	independent, [12]	contextual only; not used in the proof

First suppose, for a contradiction, that an $\mathcal{F}$ -minor-free graph is not strictly metrizable, and choose such a graph minimal under ordinary minors. Strict metrizability decomposes over blocks. Moreover, for a fixed path system, every edge outside its support may be assigned a weight larger than the total weight of the support. These two observations show that the minimal counterexample is 2-connected and carries a non-realizable full-support path system. Full support and consistency force this system to be neighborly: every edge is the selected path between its endpoints. The compliant-edge reduction then shows that the graph has no compliant edge.

Next we refine a structural argument of Chudnovsky, Cizma, and Linial. If the minimal counterexample contained two vertex-disjoint cycles, the halo construction and explicit contractions would produce $K_{2,4}$ or the triangular prism as a minor. Parallel maximal flat paths are handled before degree-two vertices are suppressed; they leave only a subdivision of $K_{2,3}$. Otherwise suppression produces a simple graph of minimum degree at least three and with no two vertex-disjoint cycles. A theorem of Lov'asz [8] then reduces the possible cores to K_5 , W_5 , $K_{2,3}$, K_4 , W_4 , and $K_5 - e$.

The exclusion of the six proposed minors strongly restricts which core edges may be subdivided. The surviving subdivisions fall into six classes:

$$\Theta_{2,b,c}, \quad K_4 \text{ with at most two long core edges}, \quad P_4(p, 2, r), \\ H_{m,n}^{rr}, \quad H_{m,n}^{sr}, \quad U_{m,n}. \quad (1.2)$$

All parameters and labelled core edges are fixed in the preliminaries. The unit-parameter cases are included, so (1.2) has no missing boundary family. Figure 2 displays their common core geometry and the two deletion relations among the wheel-derived families.

For the theta class, the metric part invokes the external strict classification just stated. For each of the other five classes in (1.2), it constructs positive rational weights for every neighborly consistent path system. The common tools are strict order representations of Ferrers relations, endpoint-chord reduction, and strict lifting across triangle expansions and long ears. These constructions are symbolic and uniform in the path lengths.

The finite cores require separate arguments. Neighborliness makes the K_5 case immediate. On W_5 , consistency leaves exactly 57 cyclic words, forming twelve orbits under the dihedral group. For each orbit we give a positive integer edge weighting and an exact positive margin against every alternative simple path. The complete path catalogue and all integer data appear in the paper; the supplementary verifier only recomputes the displayed sums. Thus no finite experiment is used to infer an infinite-family statement, and no floating-point calculation is a premise of the proof.

Organization

Section 2 fixes the terminology for path systems, minors, flat paths, the six obstructions, and the six surviving families. Section 3 proves the minimal-counterexample and core reductions. Section 4 classifies the $\mathcal{F}$ -minor-free subdivisions and treats K_5 and W_5 . Section 5 records the common strict-realization tools and states the imported theta theorem. Sections 6–9 establish the remaining infinite-family realization theorems. Section 10 combines the structural and metric results to prove the main theorem.

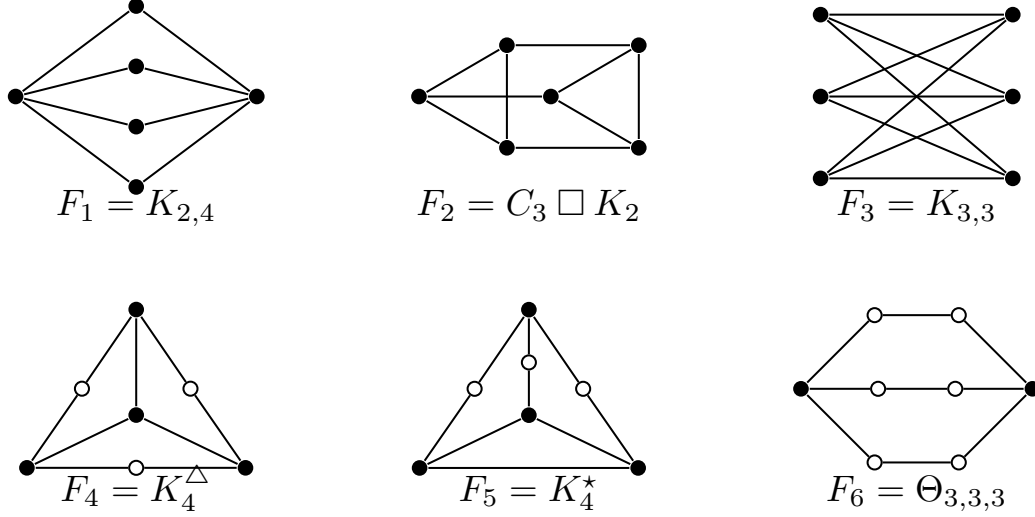


Figure 1: The six excluded minors. Black vertices belong to the named base graphs; white vertices mark the prescribed subdivisions in F_4, F_5, F_6 .

2 Preliminaries

All graphs are finite, undirected, and simple unless a suppressed multigraph is explicitly mentioned. Paths are simple, and their length is their number of edges. If P is a path containing vertices x, y , then $P[x, y]$ denotes its x - y subpath. For an edge weighting $w : E(G) \rightarrow \mathbb{R}$ and a subgraph H of G , write

$$w(H) = \sum_{e \in E(H)} w(e).$$

Path systems and strict realization

A **path system** on a connected graph G is a collection

$$\mathcal{P} = \{P_{uv} : \{u, v\} \in \binom{V(G)}{2}\},$$

where $P_{uv} = P_{vu}$ is a specified u - v path, read in either orientation. We take P_{uu} to be the trivial path when that notation is convenient. The system is **consistent** if

$$P_{uv}[x, y] = P_{xy} \tag{2.1}$$

whenever $x, y \in V(P_{uv})$. This is the subpath formulation of consistency used throughout [4].

A positive edge weighting $w : E(G) \rightarrow \mathbb{R}_{>0}$ **realizes** $\mathcal{P}$ if every P_{uv} is a shortest u - v path. It **strictly realizes** $\mathcal{P}$ if

$$w(P_{uv}) < w(Q) \tag{2.2}$$

for every pair u, v and every u - v path $Q \neq P_{uv}$. A path system is **metrizable**, respectively **strictly metrizable**, if it has a positive realization, respectively a positive strict realization. A graph is metrizable, respectively strictly metrizable, if every consistent path system on

it has the corresponding property. For a disconnected graph these properties are defined componentwise.

A path system is **neighborly** if

$$P_{uv} = uv \quad \text{for every } uv \in E(G).$$

Its **support** is the spanning subgraph with edge set

$$\text{supp}(\mathcal{P}) = \bigcup_{\{u,v\} \in \binom{V(G)}{2}} E(P_{uv}). \quad (2.3)$$

Equivalently, we also write $\text{supp}(\mathcal{P})$ for this edge set when no confusion can arise.

Lemma 2.1 (selected paths in a neighborly system are induced)

If $\mathcal{P}$ is a neighborly consistent path system on G , then every selected path P_{uv} is an induced path of G .

Proof.

Suppose that two nonconsecutive vertices x, y of P_{uv} are adjacent in G . Neighborliness gives $P_{xy} = xy$, whereas consistency gives $P_{xy} = P_{uv}[x, y]$. The latter path has at least two edges, a contradiction. $\square$

This lemma will be used repeatedly to obtain finite lists of possible routes between branch vertices. In particular, it justifies the three-route description of a nonadjacent rim pair in W_5 .

Minors, blocks, and subdivision cores

A graph H is an **ordinary minor**, or simply a **minor**, of G if it can be obtained from G by deleting vertices, deleting edges, and contracting edges. After a contraction we delete loops and identify parallel copies, so the resulting minor is again simple. A graph H is a **topological minor** of G if G contains a subgraph isomorphic to a subdivision of H .

A **subdivision** of a graph K is obtained by replacing each edge $xy \in E(K)$ with an x - y path, with all replacement paths internally vertex-disjoint. The vertices inherited from K are the **core vertices**, and the replacement paths are the **core paths**. A core path of length one is **short**; a core path of length at least two, and its corresponding core edge, is **long**.

A **block** of a graph is a maximal 2-connected subgraph, a bridge together with its endpoints, or an isolated vertex. The **block-cut tree** of a connected graph is the bipartite tree whose nodes are its blocks and cut vertices, with incidence given by containment.

A vertex of degree at least three is a **branch vertex**. A path is **flat**, also called **suspended**, if every internal vertex has degree two in the ambient graph. A flat path is maximal if it is not properly contained in another flat path. Two distinct maximal flat paths are **parallel** if they have the same two branch endpoints. An edge xy is **compliant** if G contains a flat x - y path different from the edge xy .

To **suppress** a degree-two vertex v with distinct neighbors x, y is to delete v and replace xv, vy with an edge xy . Simultaneously suppressing all degree-two vertices replaces every maximal flat path by one edge. Before loops or parallel edges have been excluded, the resulting **suppressed core** is allowed to be a multigraph. Later structural lemmas verify when this core is simple.

Halos

Let x, y be distinct vertices of a graph G . An xy -**halo** is a subgraph

$$H = C \cup R_x \cup R_y$$

with the following properties:

1. C is a cycle;
2. R_x is an x - u path and R_y is a y - v path for distinct nonadjacent vertices $u, v \in V(C)$;
3. apart from their endpoints u, v , the two paths are disjoint from C , and they are vertex-disjoint from each other;
4. R_x or R_y may be a trivial path when $x = u$ or $y = v$.

This is the convention in the halo lemma of Chudnovsky, Cizma, and Linial [2, Lemma 2.3]. All later uses specify the ambient graph in which flatness and vertex degrees are measured.

Standard named graphs

For $n \geq 3$, W_n denotes the wheel consisting of a rim cycle

$$v_0 v_1 \cdots v_{n-1} v_0$$

and a hub h adjacent to every rim vertex. We write $W'_4 = K_5 - e$. When labels are needed, W'_4 has vertex set $\{A, B, C, X, Y\}$ and edge set

$$\{AB, AC, BC, AX, AY, BX, BY, CX, CY\}; \quad (2.4)$$

thus ABC is a triangle, X and Y are adjacent to each of A, B, C , and $XY \notin E(W'_4)$.

For positive integers a, b, c , the theta graph $\Theta_{a,b,c}$ consists of two poles s, t and three internally vertex-disjoint s - t paths of lengths a, b, c . The parameters may be reordered, and we normally assume $a \leq b \leq c$.

The six forbidden minors

We use the following fixed labelled descriptions.

1. $F_1 = K_{2,4}$, with bipartition $\{s, t\}$ and $\{x_1, x_2, x_3, x_4\}$.
2. $F_2 = C_3 \square K_2$, the triangular prism. Its vertices are $a_0, a_1, a_2, b_0, b_1, b_2$; the a_i 's and b_i 's each induce a triangle, and $a_i b_i$ is an edge for $i = 0, 1, 2$.
3. $F_3 = K_{3,3}$, with bipartition $\{a_1, a_2, a_3\}$ and $\{b_1, b_2, b_3\}$.
4. $F_4 = K_4^\Delta$. Start with K_4 on $\{a, b, c, d\}$, and replace each of ab, bc, ca by a path of length two with a distinct new internal vertex; leave ad, bd, cd as edges.
5. $F_5 = K_4^*$. Start with K_4 on $\{a, b, c, d\}$, and replace each of da, db, dc by a path of length two with a distinct new internal vertex; leave ab, bc, ca as edges.
6. $F_6 = \Theta_{3,3,3}$. Equivalently, its three pole-to-pole arms are $su_1 u_2 t$, $sv_1 v_2 t$, and $sz_1 z_2 t$.

Let

$$\mathcal{F} = \{F_1, F_2, F_3, F_4, F_5, F_6\}. \quad (2.5)$$

A graph is **six-free** if it has no member of $\mathcal{F}$ as an ordinary minor. The six graphs are not strictly metrizable by the explicit certificates of Chudnovsky, Cizma, and Linial [2, Figures 13–14 and Appendix A].

The six surviving subdivision families

The structural reduction leaves the following six classes. These definitions fix all labels used in the realization sections.

1. The theta family

For $2 \leq b \leq c$, write

$$\Theta_{2,b,c} = A \cup B \cup C,$$

where

$$A : s - a - t, \quad B : s = b_0, b_1, \dots, b_b = t, \quad C : s = c_0, c_1, \dots, c_c = t. \quad (2.6)$$

Here $b_0 = c_0 = s$, $b_b = c_c = t$, and the interiors of the three arms are pairwise disjoint.

2. Subdivisions of K_4 with at most two long edges

Let the core be K_4 on vertices $\{a, b, c, d\}$. For positive integers $\ell_{ab}, \ell_{ac}, \ell_{ad}, \ell_{bc}, \ell_{bd}, \ell_{cd}$, let

$$K_4(\ell_{ab}, \ell_{ac}, \ell_{ad}, \ell_{bc}, \ell_{bd}, \ell_{cd})$$

be the graph obtained by replacing each core edge xy by a path D_{xy} of length ℓ_{xy} , with the six paths internally disjoint. The family considered here is subject to

$$|\{xy \in E(K_4) : |E(D_{xy})| \geq 2\}| \leq 2. \quad (2.7)$$

The two-long cases separate into adjacent and opposite core edges; the zero-long and one-long cases are included.

3. The split K_4 family

For $p, r \geq 2$, $P_4(p, 2, r)$ is the subdivision of the K_4 on $\{a, b, c, d\}$ in which

$$D_{ab} : a = x_0 x_1 \cdots x_p = b, \quad D_{bc} : b = z_0 z_1 z_2 = c, \quad D_{cd} : c = y_0 y_1 \cdots y_r = d, \quad (2.8)$$

and ac, ad, bd remain single edges. Thus the three long core edges form the path $a-b-c-d$, and the middle core path has length exactly two. If $p = 1$ or $r = 1$, the graph belongs to the preceding at-most-two-long class rather than to the parameter domain in (2.8).

4–6. The U , rim–rim, and spoke–rim families

Use the labelled graph W'_4 from (2.4). For $m, n \geq 1$, replace AX and BY by internally disjoint paths

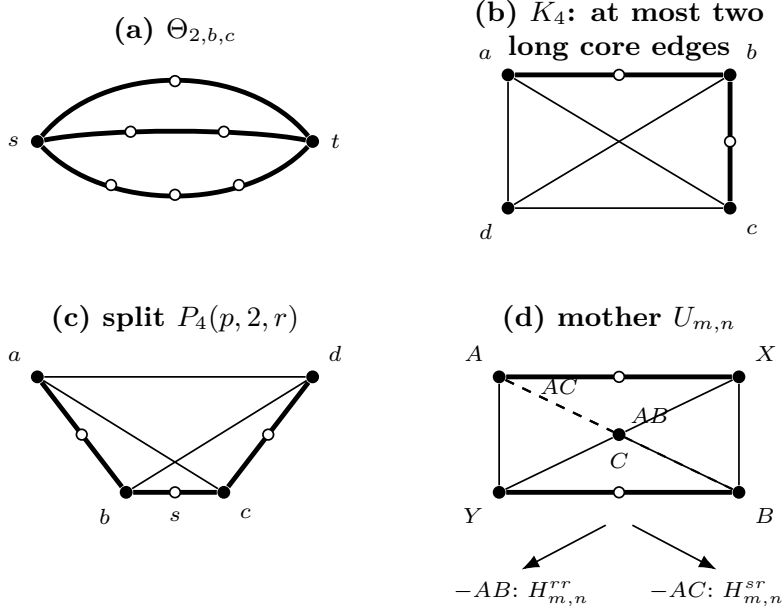
$$P : A = p_0 p_1 \cdots p_m = X, \quad Q : B = q_0 q_1 \cdots q_n = Y, \quad (2.9)$$

and keep every other listed core edge short. The resulting graph is $U_{m,n}$; explicitly, its seven short edges are

$$AB, AC, BC, BX, CX, AY, CY. \quad (2.10)$$

Define

$$H_{m,n}^{rr} = U_{m,n} - AB. \quad (2.11)$$



Heavy corridors are subdivided paths; white vertices are schematic and do not fix their lengths.

Figure 2: Schematic map of the six surviving subdivision families. Panel (b) shows the adjacent two-long-edge case; the opposite and boundary cases are included in the same class. In panel (d), deleting the dashed edge AB or AC from the mother graph gives the rim–rim or spoke–rim family, respectively.

Its short edges are

$$AC, BC, BX, CX, AY, CY.$$

In the wheel interpretation, C is the hub, the rim is $A - X - B - Y - A$, and $P = AX$ and $Q = BY$ replace opposite rim edges. This explains the superscript rr .

Finally, define

$$H_{m,n}^{sr} = U_{m,n} - AC. \quad (2.12)$$

Its short edges are

$$AB, BC, BX, CX, AY, CY.$$

Here B is the hub, the rim is $A - X - C - Y - A$, $P = AX$ replaces a rim edge, and $Q = BY$ replaces a spoke nonincident with that rim edge. This explains the superscript sr .

The value $m = 1$ or $n = 1$ means that the corresponding path in (2.9) is a single edge. Consequently

$$U_{1,1} = W'_4, \quad H_{1,1}^{rr} = H_{1,1}^{sr} = W_4, \quad (2.13)$$

and all zero-long and one-long boundary cases are included in the stated domains.

3 Structural reduction

Throughout this section, graphs are finite and simple. We work with connected graphs; equivalently, one may define strict metrizable componentwise and apply the argument to a non-strictly-metrizable component. Let

$$\mathcal{F} = \{F_1, \dots, F_6\}$$

be the six graphs in Figure 14 of Chudnovsky–Cizma–Linial, where

$$F_1 = K_{2,4}, \quad F_2 = C_3 \square K_2, \quad F_3 = K_{3,3},$$

F_4 is obtained from K_4 by subdividing the three edges of a triangle once, F_5 is obtained from K_4 by subdividing the three edges incident with one vertex once, and $F_6 = \Theta_{3,3,3}$. A graph is **six-free** if it contains no member of $\mathcal{F}$ as an ordinary minor.

We use the following notation. For a consistent path system $\mathcal{P} = \{P_{uv} : u, v \in V(G)\}$, let

$$\text{supp}(\mathcal{P}) = \bigcup_{u,v \in V(G)} E(P_{uv}).$$

The system is **neighborly** if $P_{uv} = uv$ for every edge $uv \in E(G)$. A path is **flat** if all its internal vertices have degree two in the ambient graph. A vertex of degree at least three is a **branch vertex**. Two maximal flat paths are **parallel** if they have the same two branch endpoints. An edge xy is **compliant** if there is a second flat x - y path.

We use several results of Chudnovsky, Cizma, and Linial [2]. Each imported statement is identified explicitly at the point where it is used.

3.1. Necessity

The class of strictly metrizable graphs is closed under ordinary minors [2, Theorem 1.1], and every F_i is not strictly metrizable by the explicit certificates in [2, Figures 13–14 and Appendix A]. Hence a strictly metrizable graph is six-free. It remains to prove the converse.

3.2. Blocks and support

We first record two elementary strict reductions. Their proofs are included because they are what allows all later family lemmas to be stated only for neighborly systems.

Lemma 3.1 (strict block reduction)

Let G be connected. Then G is strictly metrizable if and only if every block of G is strictly metrizable.

Proof.

Let B be a block and let $u, v \in V(B)$. A simple u - v path cannot leave B and later return to B : otherwise the block-cut tree would contain a cycle. Thus the restriction of a consistent system on G to pairs of vertices in B is a consistent system on B .

Suppose that every block is strictly metrizable. Strictly realize the restricted system on each block and use these weights on the corresponding edges of G . Any simple path between two vertices follows the unique chain of blocks in the block-cut tree. By consistency, the segment of the selected path inside each block is precisely the selected path between the two relevant cut vertices (or endpoints) in that block. Every competitor follows the same block chain and, unless it is the selected path, differs in at least one block. It is strictly longer in

every block in which it differs and has the same weight in the other blocks. Hence it is strictly longer globally.

Conversely, assume that G is strictly metrizable, fix a block B , and let $\mathcal{Q}_B$ be an arbitrary consistent system on B . On every other block choose any consistent system (for example, the unique paths of a fixed spanning tree of that block), and concatenate these local systems along the unique routes of the block-cut tree. More explicitly, the selected u - v path follows the unique block chain joining the blocks containing u and v , and in each block it uses the locally selected path between the two relevant cut vertices (with u or v replacing a cut vertex at an end). If x, y occur on this concatenation, its x - y subpath follows the corresponding consecutive subchain and, inside its first and last blocks, is locally selected by consistency; every intermediate block piece is unchanged. Hence that subpath is exactly the path constructed for x, y , so the global system $\mathcal{Q}$ is consistent. Its restriction to B is $\mathcal{Q}_B$. A strict realization of $\mathcal{Q}$ on G restricts to a strict realization of $\mathcal{Q}_B$. Since $\mathcal{Q}_B$ was arbitrary, B is strictly metrizable. $\square$

Lemma 3.2 (strict support reduction)

Let $\mathcal{P}$ be a consistent path system on a finite connected graph G , and let H be the spanning subgraph with $E(H) = \text{supp}(\mathcal{P})$. Then $\mathcal{P}$ is strictly realizable on G if and only if it is strictly realizable on H .

Proof.

Restriction proves the forward implication. Conversely, let $w_H : E(H) \rightarrow \mathbb{R}_{>0}$ strictly realize $\mathcal{P}$ on H , and choose

$$M > \sum_{e \in E(H)} w_H(e).$$

Keep the weights w_H on H and give every edge of $G - E(H)$ weight M . Every selected path is simple and contained in H , so its weight is at most $\sum_{e \in E(H)} w_H(e) < M$. A competitor using an edge outside H is therefore strictly longer, while a competitor contained in H is handled by the strict realization w_H . $\square$

Proposition 3.3 (a minimal counterexample is full-support neighborly)

If a six-free non-strictly-metrizable graph exists, then there is such a graph G satisfying all of the following:

1. G is 2-connected;
2. G has no compliant edge;
3. G carries a non-strictly-realizable consistent path system $\mathcal{P}$ with $\text{supp}(\mathcal{P}) = E(G)$; in particular, $\mathcal{P}$ is neighborly.

Proof.

Choose a six-free non-strictly-metrizable graph G minimal under ordinary minors. Six-freeness is itself minor-closed, so every proper minor of G is strictly metrizable by the choice of G . Notice that this conclusion uses only minimality among counterexamples, not minor closure of strict metrizability.

By Lemma 3.1, some block of a non-strictly-metrizable graph is not strictly metrizable. If G had more than one block, such a block would be a proper six-free minor, contrary to minimality. A one-vertex graph and K_2 are strictly metrizable, so G is 2-connected.

If e were compliant, then $G - e$ would be a proper six-free minor and hence strictly metrizable. Proposition 2.2 of [2] states that deleting a compliant edge preserves strict metrizability

in both directions; it would follow that G is strictly metrizable, a contradiction. Thus G has no compliant edge.

Choose a consistent system $\mathcal{P}$ on G which is not strictly realizable, and put $H = \text{supp}(\mathcal{P})$. The graph H is spanning and connected, because it contains the selected path P_{uv} for every pair u, v . If $H \neq G$, then H is a proper six-free edge-deletion minor and is strictly metrizable. Lemma 3.2 would then strictly realize $\mathcal{P}$ on G , a contradiction. Hence $H = G$.

Finally, if $xy \in E(G)$, then xy occurs as a one-edge subpath of some selected path. The subpath axiom for consistency gives $P_{xy} = xy$. Therefore $\mathcal{P}$ is neighborly. $\square$

3.3. Excluding two vertex-disjoint cycles

We reproduce the structural part of the two-cycle argument from [2, proof of Proposition 3.2], with the minor contractions incorporated into the proof.

Lemma 3.4 (two-cycle obstruction)

Let G be 2-connected and have no compliant edge. If G contains two vertex-disjoint cycles, then G contains $K_{2,4}$ or $C_3 \square K_2$ as a minor.

Proof.

Let C_1, C_2 be disjoint cycles. By the two-path form of Menger's theorem, there are vertex-disjoint C_1 - C_2 paths R_1, R_2 whose interiors lie outside $C_1 \cup C_2$. Write $R_1 = x_1 \cdots x_2$ and $R_2 = y_1 \cdots y_2$, where $x_i, y_i \in V(C_i)$ for $i = 1, 2$. Thus the two attachment vertices on each cycle are distinct. Put

$$H = C_1 \cup C_2 \cup R_1 \cup R_2.$$

There are three cases, according to whether x_i, y_i are adjacent on their cycle.

Neither pair is adjacent. Contract R_1 to identify x_1, x_2 to a pole x , and contract R_2 to identify y_1, y_2 to a pole y . The two x_1 - y_1 arcs of C_1 and the two x_2 - y_2 arcs of C_2 become four internally vertex-disjoint x - y paths. Each has an internal vertex because neither attachment pair was adjacent. Contracting each path to length two gives $K_{2,4}$.

Exactly one pair is adjacent. Assume that x_1, y_1 are not adjacent and that $x_2y_2 \in E(C_2)$. Write $P_2 = C_2 - x_2y_2$, an x_2 - y_2 path of length at least two. If $\{x_2, y_2\}$ separates the interior of P_2 from C_1 , then P_2 is not flat, for otherwise x_2y_2 would be compliant. The halo lemma [2, Lemma 2.3] gives an x_2y_2 -halo in that component. Its cycle has two nonadjacent attachment vertices. Because the halo lies in the component separated from C_1 , its cycle and attachment paths meet $C_1 \cup R_1 \cup R_2$ only at x_2, y_2 . Together with C_1, R_1, R_2 , they therefore give the preceding case.

Otherwise, choose a path Q , internally disjoint from H , from an internal vertex u of P_2 to $H - P_2$. By shortening Q , its other end lies either on the interior of R_1 or R_2 , at one of x_1, y_1 , or in $C_1 - \{x_1, y_1\}$. In the first situation, contract the segment of the connector between that end and the corresponding end on C_1 ; this reduces to the second situation. Suppose, by symmetry, that Q ends at x_1 . Contract R_2 , thereby identifying y_1, y_2 to a pole y . Between x_1 and y we now have four internally vertex-disjoint paths: the two arcs of C_1 , the path $R_1x_2y_2$, and the path QuP_2y_2 . All four have an internal vertex, so they contain a subdivision of $K_{2,4}$. If instead Q ends at an internal vertex v of C_1 , then the three vertices x_1, y_1, v split C_1 into three arcs, the three vertices x_2, y_2, u split C_2 into three arcs, and R_1, R_2, Q join the corresponding triples. Their union is a subdivision of the triangular prism $C_3 \square K_2$.

Both pairs are adjacent. Put $P_i = C_i - x_iy_i$. If $\{x_2, y_2\}$ separates the interior of P_2 from C_1 , then P_2 is non-flat, since otherwise x_2y_2 would be compliant. The halo lemma replaces C_2 by a cycle with nonadjacent attachments and reduces the problem to the preceding case; as

above, the separating component makes the replacement internally disjoint from $C_1 \cup R_1 \cup R_2$. Otherwise there is a path Q , internally disjoint from H , from an internal vertex u of P_2 to $H - C_2$. If the other end v of Q lies in the interior of R_1 , let

$$D = R_1[v, x_2] \cup x_2 y_2 \cup P_2[y_2, u] \cup Q.$$

Then D is a cycle disjoint from C_1 , and $R_1[x_1, v]$ and R_2 are vertex-disjoint paths joining C_1 to D . Their attachments x_1, y_1 on C_1 are adjacent, whereas their attachments v, y_2 on D are nonadjacent: each of the two v - y_2 arcs of D has at least two edges. This is the preceding case. The case in which v lies in the interior of R_2 is symmetric. If Q ends at an internal vertex of P_1 , then C_1, C_2, R_1, R_2, Q form a subdivision of the triangular prism. After interchanging the labels x_i, y_i if necessary, we may therefore assume that Q joins u to x_1 .

We now apply the same separation argument to P_1 . If $\{x_1, y_1\}$ separates the interior of P_1 from C_2 , then P_1 is non-flat, since otherwise the edge $x_1 y_1$ would be compliant. The halo lemma, applied in the component of $G - \{x_1, y_1\}$ containing the interior of P_1 , replaces C_1 by a cycle with nonadjacent attachments. That component is disjoint from $C_2 \cup R_1 \cup R_2 \cup Q$ apart from x_1, y_1 , so this reduces to the preceding case. We may thus assume that $\{x_1, y_1\}$ does not separate the interior of P_1 from C_2 . Hence there is a path Q' , internally disjoint from H , from an internal vertex u' of P_1 to $H - P_1$. If the other end v of Q' lies in the interior of R_1 , the cycle

$$D' = R_1[x_1, v] \cup Q' \cup P_1[u', y_1] \cup y_1 x_1$$

is disjoint from C_2 , and the connector paths $R_1[v, x_2]$ and R_2 have adjacent attachments x_2, y_2 on C_2 and nonadjacent attachments v, y_1 on D' , because both v - y_1 arcs of D' have at least two edges. Thus this again reduces to the preceding case; an endpoint in the interior of R_2 is symmetric. If the other end of Q' lies in the interior of P_2 , the triangular-prism subdivision is immediate. Consequently we may assume that Q' ends in $\{x_2, y_2\}$. If $Q' \cap Q \neq \emptyset$, let z be the first intersection encountered from u' along Q' . Then $Q'[u', z] \cup Q[z, u]$ is a path from the interior of P_1 to the interior of P_2 , internally disjoint from H , and hence gives the triangular-prism subdivision already treated. We may therefore assume that Q' is disjoint from Q . If Q' ends at x_2 , the cycles are split by x_1, y_1, u' and u, y_2, x_2 , and the three matching paths Q, R_2, Q' give a triangular-prism subdivision. If Q' ends at y_2 , then the following four paths between the poles x_1, y_2 are internally vertex-disjoint:

$$Q \cup P_2[u, y_2], \quad R_1 \cup x_2 y_2, \quad x_1 y_1 \cup R_2, \quad P_1[x_1, u'] \cup Q'.$$

Each contains an internal vertex, namely u, x_2, y_1, u' , respectively. They form a subdivision of $K_{2,4}$. This completes all cases. $\square$

Corollary 3.5

A six-free graph satisfying the conclusions of Proposition 3.3 has no two vertex-disjoint cycles.

Proof.

Both graphs supplied by Lemma 3.4 belong to $\mathcal{F}$. $\square$

3.4. Parallel flat paths

Suppressing all degree-two vertices can produce parallel edges. The parallel case must therefore be handled before the simple-graph theorem of Lov'asz is invoked.

Lemma 3.6 (parallel-flat-path reduction)

Let G be a 2-connected six-free graph with no compliant edge and no two vertex-disjoint cycles. If G contains two parallel maximal flat paths, then G is a subdivision of $K_{2,3}$. Moreover, all three arms of this subdivision have length at least two.

Proof.

Let x, y be the common branch endpoints of two parallel maximal flat paths. No member of this parallel family is the edge xy , since such an edge would be compliant. Thus every member has length at least two.

We first show that no second unordered pair of branch vertices can be joined by parallel maximal flat paths. Interiors of distinct maximal flat paths are disjoint unless the paths have the same branch endpoints: an internal vertex has degree two, so any overlap forces the maximal paths to coincide up to their branch endpoints. Hence two disjoint endpoint pairs would give two vertex-disjoint cycles, contrary to the hypothesis.

The only other possibility is that the endpoint pairs share exactly one vertex. Write them as $\{x, y\}$ and $\{x, z\}$. Since G is 2-connected, $G - x$ contains a $y-z$ path R . The interior of R is disjoint from the interiors of the four chosen flat paths. Indeed, a path avoiding x cannot enter the interior of an $x-y$ or $x-z$ flat path and later leave it, because every such internal vertex has degree two. Contract R , identifying y, z to a vertex r . The four chosen flat paths become four internally vertex-disjoint $x-r$ paths, each with an internal vertex. They contain a subdivision of $K_{2,4}$, a contradiction. Thus $\{x, y\}$ is the unique parallel endpoint pair.

Suppose first that x, y are the only branch vertices. By 2-connectivity, a maximal flat path cannot have both branch ends equal to x (or both equal to y): such a closed flat ear is attached to the rest of the graph only at that branch vertex, making it a cut vertex unless the whole graph is that cycle, contrary to the presence of the other branch vertex. Thus every degree-two vertex lies on a maximal flat $x-y$ path, and the graph is a subdivision of $K_{2,k}$, where $k = \deg(x) = \deg(y) \geq 3$. If $k \geq 4$, choosing four arms gives a $K_{2,4}$ minor. Hence six-freeness forces $k = 3$.

It remains to exclude a third branch vertex z . In a 2-connected graph there is an $x-y$ path through z : apply the two-fan form of Menger's theorem at z . Let S be the component of $G - \{x, y\}$ containing z . The induced graph on $S \cup \{x, y\}$ therefore contains a non-flat $x-y$ path of length at least two. The halo lemma [2, Lemma 2.3] yields an xy -halo in this induced graph.

The interiors of the original parallel flat paths lie in different components of $G - \{x, y\}$, so they are disjoint from the halo except at x, y . Contract the two halo attachment paths. The two arcs of the halo cycle then give two internally vertex-disjoint $x-y$ paths, each with an internal vertex because the attachment vertices are nonadjacent on the halo cycle. Together with two original parallel flat paths, they form a subdivision of $K_{2,4}$, a contradiction. Thus no third branch vertex exists, and G is a subdivision of $K_{2,3}$ with all arms of length at least two. $\square$

3.5. The simple suppressed core**Lemma 3.7 (the suppressed core is simple)**

Let G be 2-connected, have no compliant edge, and contain no parallel maximal flat paths. Suppressing every degree-two vertex produces a simple graph $\tilde{G}$ with minimum degree at least three. If G has no two vertex-disjoint cycles, neither does $\tilde{G}$.

Proof.

The graph G is not a cycle, because in a cycle every edge is compliant: the complementary

arc is a second flat path between its endpoints. Thus G has a branch vertex. After all degree-two vertices are suppressed, the remaining vertices are exactly the branch vertices, so $\delta(G) \geq 3$.

A loop of $\tilde{G}$ would come from a flat closed ear whose two ends are the same branch vertex x . If there are vertices outside that ear, then x is a cut vertex of G ; if there are not, G is a cycle. Both alternatives are impossible. Parallel edges of $\tilde{G}$ would come from parallel maximal flat paths in G , excluded by hypothesis. Hence $\tilde{G}$ is simple.

Finally, two vertex-disjoint cycles of $\tilde{G}$ lift, by replacing each suppressed edge with its flat path, to two vertex-disjoint cycles of G . $\square$

We now use the following graph-theoretic theorem.

Lemma 3.8 (Lov'asz)

If a simple graph H satisfies $\delta(H) \geq 3$ and has no two vertex-disjoint cycles, then

$$H \in \{K_5, W_n, K_{3,n}, K'_{3,n}, K''_{3,n}, K'''_{3,n}\},$$

where $K_{3,n}^{(i)}$ is obtained from $K_{3,n}$ by adding i edges inside its part of size three, for $i = 1, 2, 3$.

This is [2, Lemma 3.3]; see also the original theorem of Lov'asz [8].

Proposition 3.9 (core reduction)

Let G satisfy the conclusions of Proposition 3.3. Then exactly one of the following holds:

1. G is K_5 or W_5 ;
2. G is a subdivision of one of $K_{2,3}, K_4, W_4, W'_4$, where $W'_4 = K_5 - e$.

Proof.

By Corollary 3.5, G has no two vertex-disjoint cycles. If it contains parallel maximal flat paths, Lemma 3.6 gives a subdivision of $K_{2,3}$. Assume otherwise. By Lemma 3.7, suppressing the degree-two vertices gives a simple graph $\tilde{G}$ with minimum degree at least three and with no two vertex-disjoint cycles. Apply Lemma 3.8.

For $n \geq 3$, each of $K_{3,n}, K'_{3,n}, K''_{3,n}, K'''_{3,n}$ contains $K_{3,3} = F_3$ as a subgraph after deleting extra vertices and the added edges. The cases $n = 1, 2$ compatible with minimum degree at least three yield exactly K_4, W_4, W'_4 : for $n = 1$ all three internal edges are required and the graph is K_4 ; for $n = 2$, two internal edges give W_4 , while all three give $W'_4 = K_5 - e$.

For $n \geq 6$, the wheel W_n contains a subdivision of F_4 . Indeed, choose three rim vertices so that each of the three arcs between consecutive chosen vertices contains an internal vertex; together with the hub these vertices form a K_4 -subdivision whose rim triangle consists of three long paths. The smaller wheels give only $W_3 = K_4, W_4, W_5$. The remaining Lov'asz possibility is K_5 .

It remains to show that K_5 and W_5 cannot be properly subdivided. If a core edge uv of K_5 is replaced by a path of length at least two, use u, v as poles. The replacement path and the three routes through the other three core vertices form a subdivision of $K_{2,4} = F_1$.

Write the hub and rim of W_5 as $h, v_0, v_1, v_2, v_3, v_4$, with the rim indices cyclic. If a rim edge, say v_0v_1 , is long, then the branch vertices h, v_0, v_1, v_3 form a subdivision of F_4 : the three sides of the rim triangle are the long v_0v_1 path and the two rim arcs through v_2 and v_4 . If a spoke, say hv_0 , is long, then h, v_0, v_2, v_3 form a subdivision of F_5 : the three paths incident with v_0 are the long spoke and the two rim arcs through v_1 and v_4 . Thus every proper W_5 -subdivision contains F_4 or F_5 . Six-freeness forces both K_5 and W_5 to be unsubdivided. $\square$

Theorem 3.10 (minimal-counterexample reduction)

If a six-free graph is not strictly metrizable, then there is a six-free minor-minimal graph G carrying a full-support neighborly path system that is not strictly realizable, and G is either K_5 , W_5 , or a subdivision of one of

$$K_{2,3}, \quad K_4, \quad W_4, \quad W'_4 = K_5 - e.$$

Consequently, to prove the sufficiency direction it is enough to classify the six-free subdivisions of these four cores and strictly realize every neighborly consistent path system on each surviving family, together with the finite cores K_5, W_5 .

Proof.

Combine Proposition 3.3, Corollary 3.5, Lemma 3.6, and Proposition 3.9. $\square$

4 Core restrictions and finite cores

This section continues the global reduction. Thus G is a six-free subdivision of one of

$$K_{2,3}, \quad K_4, \quad W_4, \quad W'_4 = K_5 - e,$$

or else $G \in \{K_5, W_5\}$. In the subdivision cases, every core edge is replaced by a flat path. We call a core edge **long** if its replacement path has length at least two; a replacement path of length one is called **short**. Recall the six forbidden minors

$$F_1 = K_{2,4}, \quad F_2 = C_3 \square K_2, \quad F_3 = K_{3,3}, \quad F_4, \quad F_5, \quad F_6 = \Theta_{3,3,3},$$

where F_4 is the K_4 -subdivision in which the three edges of a triangle are subdivided, and F_5 is the K_4 -subdivision in which the three edges incident with one vertex are subdivided.

4.1. The $K_{2,3}$ core

Proposition 4.1

Every six-free subdivision of $K_{2,3}$ arising in the global reduction is of the form

$$\Theta_{2,b,c}, \quad 2 \leq b \leq c.$$

Proof.

Let x, y be the two degree-three vertices. The subdivision is the union of three internally vertex-disjoint x - y arms of lengths a, b, c . Each arm has length at least two: in the parallel-flat-path reduction there is no direct xy edge, since such an edge together with a second flat arm would be compliant. Equivalently, in the ordinary $K_{2,3}$ core each arm already contains its degree-two part vertex.

If $a, b, c \geq 3$, contract edges within each arm until every arm has length exactly three. The resulting minor is $\Theta_{3,3,3} = F_6$, contrary to six-freeness. Hence one arm has length two. Reordering the other two arms gives the stated form. $\square$

4.2. The K_4 core

Let $L \subseteq E(K_4)$ be the subgraph formed by the long core edges.

Lemma 4.2 (forbidden long-edge patterns in K_4)

The graph L contains none of the following:

1. a triangle;
2. a three-edge star;
3. a four-cycle.

Proof.

If the three edges of a core triangle are long, take all six replacement paths of the K_4 core. The three paths on that triangle have an internal vertex, so this subgraph is a subdivision of F_4 . Further subdivisions of the other three core edges are harmless. Thus a long triangle is forbidden.

Likewise, if the three core edges incident with one vertex are long, the six replacement paths form a subdivision of F_5 . Hence a long three-edge star is forbidden.

Finally suppose that

$$ab, bc, cd, da \in L.$$

If the diagonal ac were long, then abc would be a long triangle; if bd were long, then bcd would be a long triangle. Therefore both diagonals are short. Contract ac to one pole s and bd to a second pole t . The replacement paths of ab, bc, cd, da become four internally vertex-disjoint s - t paths, each with an internal vertex because all four core edges are long. They form a subdivision of $K_{2,4}$; contracting within the four arms gives $F_1 = K_{2,4}$ as a minor. Thus a long four-cycle is forbidden. $\square$

Proposition 4.3 (complete K_4 restriction)

For a six-free K_4 -subdivision, exactly one of the following holds:

1. at most two core edges are long;
2. exactly three core edges are long, they form a P_4 , and the replacement path of the middle edge has length exactly two.

Proof.

Here is the complete inspection. A three-edge subgraph of K_4 is a triangle, a three-edge star, or a P_4 . For four edges, if the two omitted edges are adjacent, the remaining graph contains a triangle; if they are disjoint, the remaining graph is a four-cycle. Every set of at least five edges contains a triangle. Lemma 4.2 excludes the triangle, star, and four-cycle patterns. It remains only to control the length of the middle edge in the P_4 case.

Label the three long core edges

$$ab, \quad bc, \quad cd,$$

in their path order. The other three core edges ac, ad, bd are short. If the replacement path of bc had length at least three, then between b and c there would be three internally vertex-disjoint paths:

1. the replacement path of bc , of length at least three;
2. the route through a , consisting of the long ba path followed by the short edge ac , of length at least three;
3. the route through d , consisting of the short edge bd followed by the long dc path, again of length at least three.

Contracting within these three routes produces $\Theta_{3,3,3} = F_6$. Hence the middle replacement path has length two. $\square$

Thus the second family is the split-core family

$$P_4(p, 2, r), \quad p, r \geq 2.$$

If $p = 1$ or $r = 1$, the corresponding side edge is no longer long and the graph belongs to the first case of Proposition 4.3. This is the required unit-side boundary convention.

4.3. The W_4 core

Write h for the hub and $v_0v_1v_2v_3v_0$ for the rim, with indices modulo four.

Lemma 4.4 (incident long edges in W_4 are forbidden)

No two incident core edges of a six-free W_4 -subdivision are long.

Proof.

There are three incidence types, with two subtypes when the two edges are spokes. We give an explicit forbidden subdivision or contraction in every case.

Two adjacent rim edges. Suppose v_0v_1 and v_1v_2 are long. Use h, v_0, v_1, v_2 as the branch vertices of a topological K_4 , and use the rim route $v_0v_3v_2$ for the v_0 - v_2 side. The three sides of the triangle $v_0v_1v_2$ are all long: two by hypothesis and the third because it uses two core edges. Hence this topological K_4 is a subdivision of F_4 .

A spoke and an incident rim edge. Suppose hv_0 and v_0v_1 are long. Again use h, v_0, v_1, v_2 as branch vertices, with $v_0v_3v_2$ as the v_0 - v_2 side. The three paths from v_0 to h, v_1, v_2 are all long. The other three sides are the replacement paths of hv_1, hv_2, v_1v_2 . This is a subdivision of F_5 .

Two spokes with adjacent rim endpoints. Suppose hv_0 and hv_1 are long. Contract the entire replacement path of the rim edge v_0v_1 , identifying v_0, v_1 to a pole z . Between h and z there are four internally vertex-disjoint paths:

$$hv_0, \quad hv_1, \quad hv_2v_1, \quad hv_3v_0.$$

The first two have internal vertices by hypothesis, and the last two use two distinct core edges and have internal branch vertices v_2, v_3 , respectively. Therefore these paths form a subdivision of $K_{2,4}$, and contraction within them gives F_1 .

Two spokes with opposite rim endpoints. Suppose hv_0 and hv_2 are long. Use h, v_0, v_1, v_2 as branch vertices and the rim route $v_0v_3v_2$ as the v_0 - v_2 side. The triangle with branch vertices h, v_0, v_2 has three long sides: the two given spokes and the two-edge rim route through v_3 . This gives a subdivision of F_4 .

These cases exhaust all pairs of incident edges of W_4 . $\square$

Proposition 4.5 (complete W_4 restriction)

The long core edges of a six-free W_4 -subdivision form a matching. A matching in W_4 has size at most two, and, up to automorphism, its two size-two possibilities are:

1. two opposite rim edges;
2. one spoke and one nonincident rim edge.

Proof.

The first assertion is Lemma 4.4. Since all spokes share the hub and the rim is a four-cycle, a matching has size at most two. A size-two matching either contains no spoke, in which case

it is a pair of opposite rim edges, or contains one spoke, in which case its other edge is a nonincident rim edge. The dihedral symmetries of the rim are transitive on each type. $\square$

Denote the two resulting families by

$$H_{m,n}^{rr} \quad \text{and} \quad H_{m,n}^{sr}, \quad m, n \geq 1,$$

where m, n are the lengths of the two designated replacement paths. In H^{rr} they replace opposite rim edges; in H^{sr} they replace a spoke and a nonincident rim edge. The value one means that the designated edge is short. Consequently:

- $H_{1,1}^{rr} = H_{1,1}^{sr} = W_4$;
- a single long rim edge occurs as a unit-parameter boundary of either family;
- a single long spoke occurs as a unit-parameter boundary of H^{sr} .

Thus the domains $m, n \geq 1$ cover every no-long, one-long, and two-long six-free W_4 subdivision.

4.4. The $W'_4 = K_5 - e$ core

Let A, B, C be the three degree-four vertices and X, Y the two degree-three vertices. Thus ABC is a triangle, each of X, Y is adjacent to all of A, B, C , and $XY \notin E(W'_4)$.

Lemma 4.6 (long high-high edges are forbidden)

No edge of the triangle ABC is long in a six-free subdivision of W'_4 .

Proof.

Suppose, by symmetry, that AB is long. Between A and B the following four routes are internally vertex-disjoint:

$$AB, \quad ACB, \quad AXB, \quad AYB.$$

The first route has an internal vertex because AB is long; each of the other three uses two core edges and has the distinct internal branch vertex C, X, Y , respectively. Thus the four routes form a subdivision of $K_{2,4}$, which contains F_1 as a minor after contractions within the routes. $\square$

Lemma 4.7 (incident long high-low edges are forbidden)

Among the six high-low edges, no two incident edges are long.

Proof.

There are two incidence types.

A common high endpoint. Suppose AX and AY are long. Use A, X, Y, B as the branch vertices of a topological K_4 , with the X - Y side represented by XCY . The triangle AXY has three long sides: AX, AY by hypothesis and XCY because it uses two core edges. The remaining sides are AB, BX, BY . Hence this is a subdivision of F_4 .

A common low endpoint. Suppose AX and BX are long. Use X, A, B, Y as branch vertices, with XCY as the X - Y side. The three sides incident with X , namely XA, XB, XCY , are all long; the other three sides are AB, AY, BY . Hence this is a subdivision of F_5 . $\square$

Proposition 4.8 (complete W'_4 restriction)

Every six-free subdivision of W'_4 has all three high-high edges short, and its long high-low edges form a matching of size at most two. Up to automorphism, it is obtained by replacing AX and BY by paths of lengths $m, n \geq 1$. This family is denoted by $U_{m,n}$.

Proof.

The first assertion is Lemma 4.6 and the matching assertion is Lemma 4.7. The high-low edges form $K_{3,2}$, whose matchings have size at most two. The automorphism group $S_3 \times S_2$, permuting the high vertices and interchanging X, Y , is transitive on one-edge matchings and on two-edge matchings. A two-edge representative is $\{AX, BY\}$. Allowing either replacement path to have length one also covers every one-edge matching and the empty matching. $\square$

In particular,

$$U_{1,1} = W'_4,$$

Moreover, $U_{1,n}$ for $n \geq 2$ and $U_{m,1}$ for $m \geq 2$ are precisely the one-long boundaries, while $U_{1,1} = W'_4$ is the zero-long case.

4.5. Summary of all surviving subdivision families

Combining Propositions 4.1, 4.3, 4.5, and 4.8, every subdivided core in a six-free minimal counterexample lies in one of the following six families:

$$\Theta_{2,b,c}, \quad K_4(\text{at most two long edges}), \quad P_4(p, 2, r), \quad H_{m,n}^{rr}, \quad H_{m,n}^{sr}, \quad U_{m,n},$$

with parameter domains

$$2 \leq b \leq c, \quad p, r \geq 2, \quad m, n \geq 1.$$

The length-one side boundaries of $P_4(p, 2, r)$ belong to the at-most-two-long K_4 family, while the unit parameters of H^{rr}, H^{sr}, U are part of those families themselves. Thus there is no uncovered no-long or one-long boundary.

4.6. The finite core K_5

Proposition 4.9

Every neighborly consistent path system on K_5 has a strict positive realization.

Proof.

Every pair of distinct vertices of K_5 is adjacent. Neighborliness therefore forces the selected path for every pair to be its single edge. Give every edge weight one. A selected path has weight one, whereas every different simple competitor contains at least two edges and has weight at least two. $\square$

4.7. The finite core W_5

Label the hub by 0 and the rim cyclically by 1, 2, 3, 4, 5, with indices on rim vertices read modulo five. A neighborly system selects the single edge for every adjacent pair. Moreover, every selected path is induced. Indeed, if a selected path had a chord xy , consistency would identify its x - y subpath with P_{xy} , whereas neighborliness would give $P_{xy} = xy$, a contradiction.

The induced i -($i+2$) paths are precisely the two rim arcs and the two-spoke path through the hub. Thus the only choices not already forced by neighborliness are the five nonadjacent rim pairs $(i, i+2)$. For such a pair write

$$\begin{aligned}
S_i &= i, (i+1), (i+2), \\
H_i &= i, 0, (i+2), \\
L_i &= i, (i-1), (i-2), (i+2),
\end{aligned}$$

for the short rim route, hub route, and long rim route, respectively. Encode the system by a cyclic word $\omega \in \{S, H, L\}^5$.

Lemma 4.10 (the exact consistency condition)

The word ω defines a neighborly consistent path system on W_5 if and only if, for every $i \in \mathbb{Z}_5$,

$$\omega_i = L \implies \omega_{i+2} = \omega_{i+3} = S. \quad (4.1)$$

Proof.

The paths S_i and H_i have length two, so all of their proper nontrivial subpaths are edges and are already forced by neighborliness. The path L_i has exactly two length-two subpaths whose endpoints are not adjacent. In the cyclic indexing, these are the short rim paths selected at positions $i+2$ and $i+3$. Thus consistency forces (4.1).

Conversely, if (4.1) holds, every subpath of every L_i is selected as prescribed, while the subpaths of S_i, H_i are selected edges. Hence the subpath characterization of consistency holds for the entire system. $\square$

Lemma 4.11 (57 words and 12 dihedral orbits)

There are exactly 57 admissible words satisfying (4.1). Under the dihedral group of the rim they form the following 12 orbits:

number of L 's	orbit representatives	
0	$HHHHH, HHHHS, HHHSS, HHSHS, HHSSS,$ $HSHSS, HSSSS, SSSSS,$	(4.2)
1	$HLHSS, HLSSS, LSSSS,$	
2	$LLSSS.$	

Proof.

If there is no L , every position is independently H or S , giving $2^5 = 32$ words. If there is exactly one L , its position can be chosen in five ways, two positions are forced to be S by (4.1), and the remaining two positions are independently H or S . This gives $5 \cdot 2^2 = 20$ words. Two L 's must be in adjacent cyclic positions; then the other three positions are forced to be S . There are five such words. Three L 's are impossible. Hence the total is

$$32 + 20 + 5 = 57.$$

For words without L , classify the binary H/S words by the number of S 's. With two S 's there are two types, adjacent and separated; with three S 's there are likewise two types, according as the two H 's are adjacent or separated. Every other number gives one type. This yields the eight representatives in the first row of (4.2).

With one L , rotate it to a fixed position. Reflection interchanges the two unforced positions, so their number of H 's, namely two, one, or zero, determines the three representatives in the second row. With two L 's, all admissible words are rotations or reflections of $LLSSS$. Thus there are $8 + 3 + 1 = 12$ dihedral orbits. $\square$

Proposition 4.12 (exact integer certificates for W_5)

Every neighborly consistent path system on W_5 has a strict positive integer realization.

Proof.

Use the edge order

$$01, 02, 03, 04, 05, 12, 15, 23, 34, 45. \quad (4.3)$$

For the 12 representatives in Lemma 4.11, take the following weight vectors. The last column is the minimum of $w(Q) - w(P_{uv})$ over all unordered vertex pairs u, v and all simple u - v paths $Q \neq P_{uv}$.

word	edge-weight vector in the order (4.3)	minimum margin
HHHHH	(3, 3, 3, 3, 3, 4, 4, 4, 4, 4)	2
HHHHS	(3, 4, 3, 3, 4, 3, 3, 5, 4, 5)	2
HHHSS	(4, 4, 3, 4, 4, 4, 2, 5, 5, 4)	2
HHS HS	(3, 4, 4, 3, 5, 3, 4, 6, 3, 4)	2
HHSSS	(4, 4, 4, 4, 5, 4, 3, 6, 4, 3)	2
HLHSS	(4, 5, 2, 5, 4, 3, 2, 5, 5, 3)	2
HLSSS	(4, 5, 3, 5, 5, 3, 3, 6, 4, 2)	2
HS HSS	(3, 5, 3, 5, 3, 4, 2, 4, 4, 4)	2
HSSSS	(3, 5, 3, 4, 4, 4, 3, 4, 3, 2)	2
LLSSS	(2, 2, 2, 2, 2, 1, 1, 3, 1, 1)	1
LSSSS	(2, 2, 2, 2, 2, 2, 1, 2, 1, 1)	1
SSSSS	(3, 3, 3, 3, 3, 2, 2, 2, 2, 2)	2

We explain why this table is an exact finite proof rather than numerical evidence. For a fixed row, the vector consists of positive integers. The wheel has exactly nine simple paths between the hub and a fixed rim vertex, 12 between a fixed adjacent rim pair, and 14 between a fixed nonadjacent rim pair. Hence, over all 15 unordered vertex pairs, there are

$$5 \cdot 9 + 5 \cdot 12 + 5 \cdot 14 = 175$$

simple paths, of which 15 are selected paths. Thus each row requires exactly 160 alternative-path comparisons. Each comparison is the subtraction of two sums of entries of the displayed integer vector. Direct enumeration of these finite sums gives the positive minimum in the last column. In particular, all 160 inequalities are strict in every row, with no rounding or linear-programming assumption. Since every displayed edge weight is positive, a non-simple competing walk can only become shorter when a closed subwalk is deleted. It is therefore enough to check the simple competitors counted above.

For illustration, in the first row every spoke has weight three and every rim edge has weight four. A selected hub route has weight six, while the short rim alternative has weight eight; an adjacent rim edge has weight four, while its two-spoke alternative has weight six. These comparisons already attain the displayed margin two, and every longer simple alternative has larger weight.

A rotation or reflection of the rim sends a path system to another system in the same orbit. Relabel the corresponding integer vector by the same graph automorphism. Path weights and all strict margins are preserved. Lemma 4.11 therefore extends the 12 rows to all 57 admissible words. $\square$

Complete exact certificate for Proposition 4.12

For completeness, we now write out the finite certificate that underlies the last column

of Proposition 4.12. This makes every strict inequality checkable from the manuscript alone. Fix the endpoint-pair order

$$\mathcal{O} = (01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 25, 34, 35, 45). \quad (4.4)$$

For a word ω , let p_ω be the vector of the 15 prescribed path lengths in the order $\mathcal{O}$, and let a_ω be the vector whose uv -coordinate is the minimum weight of a simple $u-v$ path different from the prescribed path. With the edge weights in Proposition 4.12, the two vectors are as follows.

ω	p_ω	a_ω	$\min(a_\omega - p_\omega)$
HHHHH	(3, 3, 3, 3, 3, 4, 6, 6, 4, 4, 6, 6, 4, 6, 4)	(7, 7, 7, 7, 7, 6, 8, 8, 6, 6, 8, 8, 6, 8, 6)	2
HHHHS	(3, 4, 3, 3, 4, 3, 6, 6, 3, 5, 7, 6, 4, 7, 5)	(7, 6, 7, 7, 6, 7, 8, 8, 7, 7, 9, 8, 6, 9, 7)	2
HHHSS	(4, 4, 3, 4, 4, 4, 7, 6, 2, 5, 8, 6, 5, 7, 4)	(6, 8, 9, 8, 6, 8, 9, 8, 8, 7, 10, 8, 7, 9, 8)	2
HHSHS	(3, 4, 4, 3, 5, 3, 7, 6, 4, 6, 7, 7, 3, 7, 4)	(7, 6, 6, 7, 7, 7, 9, 8, 8, 8, 9, 9, 7, 9, 8)	2
HHSSS	(4, 4, 4, 4, 5, 4, 8, 6, 3, 6, 8, 7, 4, 7, 3)	(8, 8, 8, 8, 7, 8, 10, 8, 9, 8, 10, 9, 8, 9, 9)	2
HLHSS	(4, 5, 2, 5, 4, 3, 6, 5, 2, 5, 8, 5, 5, 6, 3)	(6, 7, 10, 7, 6, 9, 8, 9, 8, 7, 10, 9, 7, 8, 9)	2
HLSSS	(4, 5, 3, 5, 5, 3, 7, 5, 3, 6, 8, 6, 4, 6, 2)	(8, 7, 9, 7, 7, 9, 9, 9, 9, 8, 10, 10, 8, 8, 10)	2
HSHS	(3, 5, 3, 5, 3, 4, 6, 6, 2, 4, 8, 6, 4, 6, 4)	(5, 7, 9, 7, 5, 8, 8, 8, 6, 8, 10, 8, 8, 8, 8)	2
HSSSS	(3, 5, 3, 4, 4, 4, 6, 5, 3, 4, 7, 7, 3, 5, 2)	(7, 7, 7, 6, 6, 8, 8, 7, 7, 8, 9, 9, 7, 7, 8)	2
LLSSS	(2, 2, 2, 2, 2, 1, 3, 2, 1, 3, 3, 2, 1, 2, 1)	(3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4)	1
LSSSS	(2, 2, 2, 2, 2, 2, 3, 2, 1, 2, 3, 3, 1, 2, 1)	(3, 4, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4)	1
SSSSS	(3, 3, 3, 3, 3, 2, 4, 4, 2, 2, 4, 4, 2, 4, 2)	(5, 5, 5, 5, 5, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6)	2

Here is a program-free way to check both the asserted minima and their exhaustiveness. Write a path as its vertex string. Up to a rotation or a reflection of the rim, the complete lists of simple paths for the three types of endpoint pairs are

$$\begin{array}{l|l} 01 & 01, 021, 023451, 0321, 03451, 04321, 0451, 054321, 051, \\ 12 & 102, 1032, 10432, 105432, 12, 1502, 15032, 150432, 15402, 154032, 154302, 15432, \\ 13 & 1023, 103, 1043, 10543, 1203, 12043, 120543, 123, 15023, 1503, 15043, 154023, 15403, 1543. \end{array} \quad (4.5)$$

For example, 023451 denotes $0-2-3-4-5-1$. The first line of (4.5) contains nine paths, the second 12, and the third 14. To see directly that no path is missing, a simple hub-rim path is either the spoke itself or first uses a spoke $0j$, with $j \neq i$, and then one of the two rim arcs from j to its endpoint i . This gives $1 + 4 \cdot 2 = 9$. A simple rim-rim path either avoids the hub, in which case it is one of the two rim arcs, or contains the hub exactly once. In the latter case it has a unique decomposition

$$(\text{rim segment}) - x - 0 - y - (\text{rim segment}),$$

where the two rim segments have disjoint vertex sets and $x \neq y$. Listing the possible ends x, y gives exactly the ten hub-using strings in the second line of (4.5) for adjacent endpoints and the twelve hub-using strings in the third line for nonadjacent endpoints. Thus (4.5), and its dihedral images and reversals, covers all 15 endpoint pairs and all 175 simple paths.

To check any entry of the static table, take the appropriate line of (4.5), relabel it to the endpoint pair in (4.4), add the edge weights from the row of Proposition 4.12, delete the one prescribed string, and take the least of the remaining integers. The result is the displayed coordinate of a_ω ; summing the prescribed string gives the corresponding coordinate of p_ω . Hence every coordinate satisfies $a_\omega > p_\omega$. This table, together with the explicit path catalogue (4.5), is the static certificate; it does not ask the reader to accept a computer enumeration.

The standalone verifier supplied with the supplementary material recomputes these finite additions using integer arithmetic only. It has no floating-point optimization dependency and is not a premise of the proof. The mathematical certificate is Lemma 4.11, the displayed integer vectors, the finite path catalogue (4.5), and the p_ω, a_ω table above.

4.8. Finite-core conclusion

Propositions 4.9 and 4.12 show that neither K_5 nor W_5 can be the full-support neighborly minimal counterexample furnished by the global reduction. Sections 5–9 establish strict neighborly realizability for the six displayed infinite families throughout their full parameter domains.

5 Strict realization tools and the theta input

This section records the two general strict-realization tools used throughout the remaining constructions and states the theta theorem used as an external input. The endpoint reduction keeps the endpoint edge in the competitor ledger, and the rationalization lemma preserves every strict margin.

5.1. Endpoint-chordless reduction

Let G be a finite simple graph. For distinct vertices u, v , a simple u - v path Q is **endpoint-chordless** if

- Q is induced in G when $uv \notin E(G)$; and
- Q is induced in $G - uv$ when $uv \in E(G)$.

Thus the only chord allowed in the second case is the endpoint edge uv itself.

Lemma 5.1 (strict endpoint-chord reduction)

Let $\mathcal{P}$ be a neighborly consistent path system on G , and let $w : E(G) \rightarrow \mathbb{R}_{>0}$. Suppose that, for every pair u, v ,

$$w(P_{uv}) < w(Q)$$

for every endpoint-chordless u - v path $Q \neq P_{uv}$. Then w strictly realizes $\mathcal{P}$.

Proof.

We first prove that every edge xy is strictly shorter than every other simple x - y path. Neighborliness gives $P_{xy} = xy$. Induct on the number of edges of a competing path R . If R is induced in $G - xy$, the hypothesis applies. Otherwise R has a chord $ab \neq xy$. The a - b segment $R[a, b]$ is a proper subpath of R , and induction applied to the edge ab gives

$$w(ab) < w(R[a, b]).$$

Replacing $R[a, b]$ by ab produces a shorter simple x - y path with fewer edges. A second application of induction gives the desired strict inequality for R .

Neighborliness and consistency also imply that every prescribed path is induced: a chord between two nonconsecutive vertices would make the selected path between its endpoints both that chord and the longer prescribed subpath. Now fix u, v and a simple competitor $R \neq P_{uv}$. Repeatedly replace a subpath spanning a chord by that chord. Each replacement strictly decreases weight by the preceding paragraph. The process ends either at P_{uv} , in which

case the replacements already prove the claim, or at a different endpoint-chordless path Q , in which case

$$w(P_{uv}) < w(Q) < w(R).$$

Positive weights allow every nonsimple walk to be shortened first by deleting closed subwalks. Thus P_{uv} is the unique shortest u - v path for every pair. $\square$

The distinction at an adjacent pair is essential: the edge $uv = P_{uv}$ must still be compared with induced detours in $G - uv$.

5.2. Rationalization

Lemma 5.2 (rational open-cone lemma)

If a finite path system has a positive real strict realization, then it has a positive rational strict realization.

Proof.

There are finitely many graph edges and finitely many simple paths. A strict realization is therefore a point of $\mathbb{R}^{E(G)}$ satisfying finitely many strict linear inequalities with integer coefficients: positivity of each edge and

$$w(Q) - w(P_{uv}) > 0$$

for every prescribed path and every different simple competitor. The minimum of these finitely many positive quantities is positive. Every sufficiently small perturbation preserves all inequalities, and rational vectors are dense in $\mathbb{R}^{E(G)}$. A rational vector in this open neighborhood is the required realization. $\square$

5.3. The theta theorem used here

The complete ordinary and strict classifications of theta graphs are proved in [10, Theorem 1.1]. We use only the following strict part as a cited external input and claim no novelty for the theta result in this article.

Theorem 5.3 (strict theta classification)

Let $1 \leq a \leq b \leq c$. The theta graph $\Theta_{a,b,c}$ is strictly metrizable if and only if $a \leq 2$. In particular, for every $2 \leq b \leq c$, every consistent path system on $\Theta_{2,b,c}$ has a positive rational strict realization.

The rational conclusion follows from the constructive proof in [10, Theorem 4.1 and Corollary 5.2], or directly from Lemma 5.2 after obtaining a real strict realization.

6 Strict realization of the K_4 and split-bridge families

Throughout this section, $\mathcal{P} = \{P_{uv}\}$ is a neighborly consistent path system. Every inequality constructed below is strict. We use Lemma 5.1 in the following form: it is enough to compare a prescribed path with every endpoint-chordless competitor, where a competitor for an edge uv is required to be induced in $G - uv$, rather than in G . Once a positive real strict realization has been obtained, Lemma 5.2 allows all weights to be chosen rational. In fact, the constructions below already take place in rational open intervals.

An accompanying manuscript classifies ordinary metrizable and the neighborly variant throughout the K_4 topological type and develops non-strict versions of the four-corner and split-bridge constructions [11]. The statements proved here are different and stronger in the direction needed for Theorem 1.1: Theorems 6.6 and 6.9 give *strict* realizations, but only for the K_4 -subdivision families that survive the six-minor reduction. We retain the argument, including all positive margins and endpoint ledgers, so that no strict conclusion is imported from the companion manuscript.

6.1. Order and margin tools

We first record the three order-theoretic tools used in both families. The monotone bipartite relations encoded below are the order-theoretic form of classical Ferrers, or difference, graphs; see [6, 7]. Our lemmas concern simultaneous strict affine representations and prescribed gaps, rather than enumerative properties of those graph classes.

Lemma 6.1 (four-corner representation)

Let $I = \{0, \dots, m\}$ and $J = \{0, \dots, n\}$, with their natural orders, and let

$$I \times J = R_{00} \dot{\cup} R_{01} \dot{\cup} R_{10} \dot{\cup} R_{11}.$$

Suppose that every $R_{\epsilon\eta}$ is closed toward its indicated corner. Thus, for example,

$$(i, j) \in R_{00}, \quad i' \leq i, \quad j' \leq j \quad \implies \quad (i', j') \in R_{00},$$

and the other three implications are obtained by reversing the appropriate orders. Then there are strictly increasing rational labels

$$X_0 < \dots < X_m, \quad Y_0 < \dots < Y_n$$

and rational constants $C_{00}, C_{01}, C_{10}, C_{11}$ for which the color of (i, j) is the unique minimizer of

$$C_{\epsilon\eta} + \sigma_\epsilon X_i + \sigma_\eta Y_j, \quad \sigma_0 = 1, \quad \sigma_1 = -1. \quad (6.1)$$

Proof.

In a fixed row, the two left-closed colors cannot both occur, and neither can the two right-closed colors. Hence each row consists of one left color followed by one right color. As the row index increases, the left color changes at most once, from 00 to 10, and the right color changes at most once, from 01 to 11.

If the first change occurs no later than the second, the nonempty row bands have types

$$00/01, \quad 10/01, \quad 10/11.$$

More explicitly, place the three row bands in $X < 0$, $0 < X < 1$, and $X > 1$, and place the three column bands in $Y < 0$, $0 < Y < 1$, and $Y > 1$. Within the central bands interleave the two orders so that color 10 occurs for $Y_j < X_i$ and color 01 for $X_i < Y_j$. The four forms may then be taken to be

$$X_i + Y_j, \quad X_i - Y_j, \quad -X_i + Y_j, \quad 2 - X_i - Y_j.$$

Indeed, comparison with the coordinate boundaries 0, 1, and then the comparison $Y_j < X_i$ in the central square, gives precisely the three displayed row bands. Thus the four unique-minimum regions are exactly the prescribed colors.

If the order of the two row changes is reversed, use the row bands

$$00/01, \quad 00/11, \quad 10/11$$

Place the row and column bands in $X < -1$, $-1 < X < 1$, $X > 1$ and $Y < -1$, $-1 < Y < 1$, $Y > 1$, respectively, and interleave the central labels at the boundary $X_i + Y_j = 0$. In this case use

$$X_i + Y_j, \quad X_i - Y_j + 2, \quad -X_i + Y_j + 2, \quad -X_i - Y_j.$$

Direct comparison gives the two outer bands and colors the central square by 00 when $X_i + Y_j < 0$ and by 11 when $X_i + Y_j > 0$. Again these are exactly the prescribed regions. Empty bands are simply omitted, and an unused boundary is placed in the adjacent open gap. Finally avoid the finitely many equality hyperplanes. A finite strict order has a rational linear extension, so all labels and constants may be chosen rational. $\square$

An immediate consequence is the following additive form. If $F \subseteq I \times J$ is a down-set, then there are strictly increasing rational labels x_i, y_j such that

$$(i, j) \in F \iff x_i + y_j < 0, \quad (6.2)$$

and every nonmember satisfies the reverse strict inequality.

Lemma 6.2 (two prescribed Ferrers gaps)

Let $F \subseteq I \times J$ be a down-set. Suppose that a row gap after k and a column gap after l must contain prescribed numbers a and b , respectively. There are increasing labels satisfying (6.2) and

$$x_k < a < x_{k+1}, \quad y_l < b < y_{l+1} \quad (6.3)$$

if and only if

$$\begin{array}{l|l} a + b < 0 & (k, l) \in F, \\ a + b = 0 & (k, l) \in F \text{ and } (k+1, l+1) \notin F, \\ a + b > 0 & (k+1, l+1) \notin F. \end{array} \quad (6.4)$$

There is also the following difference version. Let $Q \subseteq I \times J$ be southeast closed, and prescribe gaps

$$X_k < a < X_{k+1}, \quad z_l < c < z_{l+1}.$$

There are increasing labels with

$$(i, j) \in Q \iff X_i > z_j \quad (6.5)$$

if and only if

$$\begin{array}{l|l} a < c & (k, l+1) \notin Q, \\ a = c & (k, l+1) \notin Q \text{ and } (k+1, l) \in Q, \\ a > c & (k+1, l) \in Q. \end{array} \quad (6.6)$$

Proof.

For the additive statement, put $z_j = -y_j$. Order all x_i, z_j by their two internal orders and impose

$$x_i < z_j \quad \text{for } (i, j) \in F, \quad z_j < x_i \quad \text{for } (i, j) \notin F.$$

Downward closure is exactly the assertion that this finite comparison system is acyclic. Insert a in its row gap and $-b$ in the reversed column gap. Their possible relative orders are precisely the three alternatives in (6.4).

For the difference statement, order the symbols X_i, z_j internally and put $z_j < X_i$ for members of Q , and $X_i < z_j$ for nonmembers. Southeast closure again makes the comparison system acyclic. If $a < c$, the only possible obstruction is $z_{l+1} < X_k$, which is $(k, l+1) \in Q$. The case $a > c$ is symmetric, and equality requires both displayed comparisons. Choose rational values in the nonempty open gaps of a compatible finite linear extension. For the additive statement this is an order on the $x_i, z_j = -y_j$ symbols containing the fixed symbols $a, -b$; for the difference statement it is an order on the X_i, z_j symbols containing a, c . The labels can therefore be confined to arbitrarily small rational neighborhoods of their positions in any such compatible extension, but no common interval for the additive x - and y -families is asserted. $\square$

Lemma 6.3 (fixed-margin outward-safe extension)

Let

$$F \subseteq \{0, \dots, p\} \times \{p+1, \dots, m\}$$

be expanding:

$$(i, j) \in F, \quad i' \leq i, \quad j' \geq j \implies (i', j') \in F. \quad (6.7)$$

Assume every active row is nonempty. Suppose a finite family of strict auxiliary comparisons remains true when the active prefix $X_0, \dots, X_p$, with its internal order fixed, is moved sufficiently far to the left. If a prescribed portal prefix contains the active prefix, then for every sufficiently large rational $A > 0$ there are increasing rational labels and a rational threshold T , preserving all auxiliary comparisons and the portal cut, such that

$$(i, j) \in F \iff X_j - X_i > T, \quad T - \frac{X_m - X_0}{2} = A. \quad (6.8)$$

The symmetric assertion holds for an active suffix moved to the right.

Proof.

For an active row i , let

$$r_i = \min\{j : (i, j) \in F\}.$$

The numbers r_i are weakly increasing. Keep $X_{p+1} < \dots < X_m$ fixed and choose strictly increasing rational numbers H_i with

$$X_{r_i-1} < H_i < X_{r_i},$$

using $-\infty$ for the left endpoint when $r_i = p+1$. Put

$$T = 2A + X_m - H_0, \quad X_i = H_i - T \quad (0 \leq i \leq p).$$

Then

$$X_j - X_i > T \iff X_j > H_i \iff j \geq r_i,$$

and

$$T - \frac{X_m - X_0}{2} = \frac{T - X_m + H_0}{2} = A.$$

For large A , the translated prefix stays on its prescribed side of every portal and lies in the outward-safe region, so all auxiliary strict comparisons persist. $\square$

In the graphs considered below, consistency supplies the required outward safety. If a selected same-chain path detours outside the chain, every active left endpoint can occur only in a cross route entering through the left chain endpoint, and every active right endpoint only in a route entering through the right endpoint. Otherwise a cross route and the detour would prescribe two different subpaths between the same vertices. Thus all available affine forms involving an active left label have coefficient $+1$, and all those involving an active right label have coefficient -1 . Moving the blocks outward cannot reverse a selected comparison.

6.2. K_4 -subdivisions with at most two long core edges

An unsubdivided core edge will be called short, and a core edge replaced by a path of length greater than one will be called long. We prove all four possibilities: zero long edges, one long edge, and two opposite or adjacent long edges.

6.2.1. Zero and one long edge

For the zero-long case, give every edge of K_4 weight 1. Neighborliness prescribes each edge between its endpoints, and every different path has at least two edges. Hence every edge is uniquely shortest. Repeated chord elimination, as in Lemma 5.1, then makes every prescribed path uniquely shortest.

Now let the only possibly long chain be

$$P : A = p_0, \dots, p_m = B$$

and let its total weight be M . Write

$$w(AC) = u, \quad w(BC) = v, \quad w(AD) = r, \quad w(BD) = s, \quad w(CD) = t.$$

The two portal cuts on P are

$$C_0 = v - u, \quad D_0 = s - r. \tag{6.9}$$

When P is represented by centered labels

$$-M = X_0 < \dots < X_m = M, \quad w(p_{i-1}p_i) = \frac{X_i - X_{i-1}}{2},$$

the route from p_i to C exits at A precisely when $X_i < C_0$, and the corresponding statement for D uses D_0 . Neighborliness at both ends makes the cuts internal:

$$-M < C_0, D_0 < M. \tag{6.10}$$

Consistency makes every same-chain detour relation expanding, and all detours use the same exterior A - B connector. Choose a rational number h on the side of

$$\frac{D_0 - C_0}{2}$$

that makes the prescribed connector the shorter one, and then put

$$r = L, \quad u = L + h, \quad v = L + h + C_0, \quad s = L + D_0, \quad t = L. \quad (6.11)$$

Indeed, the two exterior connector totals are

$$u + v = 2L + 2h + C_0, \quad r + s = 2L + D_0,$$

so their strict order is exactly the chosen side of the displayed cut. For sufficiently large rational L , all five short-edge weights are positive and both triangles ACD and BCD satisfy all strict triangle inequalities.

If the prescribed A - B path is an exterior connector, apply Lemma 6.3 with its total as the prescribed margin. This realizes every same-chain detour, keeps both portal gaps, and makes the selected connector strictly shorter than the direct chain. The unused connector is still strictly longer by its preceding order. If $P_{AB} = P$, consistency makes the external same-chain relation empty; choose M above $\max\{|C_0|, |D_0|\}$ and below both exterior connector totals, which is possible after increasing L .

The endpoint-chordless competitors for the five short edges are exactly

$$\begin{array}{l|ll} AC & A - P - B - C & A - D - C \\ BC & B - P - A - C & B - D - C \\ AD & A - P - B - D & A - C - D \\ BD & B - P - A - D & B - C - D \\ CD & C - A - D & C - B - D. \end{array} \quad (6.12)$$

The first column is strict by (6.10), and the second by the two strict triangles. For a chain edge, every endpoint-chordless detour is the complementary chain segment plus one of the two exterior connectors; the same-chain relation handles the selected connector and its strict endpoint order handles the other. Lemma 5.1 completes the one-long case.

6.2.2. Two opposite long edges

Let the opposite chains be

$$P : A = p_0, \dots, p_m = B, \quad Q : C = q_0, \dots, q_n = D,$$

with totals M and N . Denote the four bridges by

$$AC = q_{00}, \quad AD = q_{01}, \quad BC = q_{10}, \quad BD = q_{11}. \quad (6.13)$$

For p_i, q_j , an induced cross-chain path uses exactly one bridge. Consistency makes the resulting four-color table corner closed. Apply Lemma 6.1 and give the chain edges the half-potential increments. After centering the two label sequences, the doubled route lengths, with a common term deleted, have the form

$$X_i + Y_j + 2q_{00}, \quad X_i - Y_j + 2q_{01}, \quad -X_i + Y_j + 2q_{10}, \quad -X_i - Y_j + 2q_{11}. \quad (6.14)$$

Adding the same constant to the four affine constants does not change their unique minimizers; hence the $q_{e\eta}$ may first be chosen positive. Boundary rows and columns in the same table give all portal cuts. Explicitly,

$$\begin{aligned} C_P &= q_{10} - q_{00}, & D_P &= q_{11} - q_{01}, \\ A_Q &= q_{01} - q_{00}, & B_Q &= q_{11} - q_{10}, \end{aligned} \quad (6.15)$$

and neighborliness places C_P, D_P strictly in the centered P -interval and A_Q, B_Q strictly in the centered Q -interval.

Let F_P and F_Q be the same-chain detour relations on P and Q . They are expanding. Moreover, F_P is nonempty if and only if $P_{AB} \neq P$. Indeed, an internal detour contains one of the two exterior A - B connectors as its A - B subpath, and consistency then prescribes that connector for P_{AB} ; conversely the endpoint pair itself belongs to F_P when $P_{AB} \neq P$. The same statement holds for Q . The two connector weights for P are $q_{00} + q_{10}$ and $q_{01} + q_{11}$, and those for Q are $q_{00} + q_{01}$ and $q_{10} + q_{11}$.

We record the gauge freedom in the four-corner representation. Its only invariant under translations of the two potential sequences and addition of a common route constant is the alternating difference

$$d = q_{00} + q_{11} - q_{01} - q_{10}. \quad (6.16)$$

Indeed, if $X'_i = X_i + a$, $Y'_j = Y_j + b$, and all four doubled route forms are to increase by the same number c , set

$$\begin{aligned} 2q'_{00} &= 2q_{00} + c - a - b, & 2q'_{01} &= 2q_{01} + c - a + b, \\ 2q'_{10} &= 2q_{10} + c + a - b, & 2q'_{11} &= 2q_{11} + c + a + b. \end{aligned}$$

Then all four forms in (6.14) change by the common constant c . Conversely, any two bridge matrices with the same value of d are related in this way. Hence all cross-route winners and all portal comparisons are unchanged.

Relabel the ends of P and Q , independently, so that an active P -relation uses the column-zero connector and an active Q -relation uses the row-zero connector. If a relation is empty, make the corresponding choice arbitrarily. Choose a rational $x > 0$, then choose rational R, S sufficiently large, and use the preceding gauge freedom to arrange

$$q_{00} = x, \quad q_{01} = R - x, \quad q_{10} = S - x, \quad q_{11} = R + S + d - 3x. \quad (6.17)$$

Thus the selected Q -connector, when active, has weight $R = q_{00} + q_{01}$, and the selected P -connector has weight $S = q_{00} + q_{10}$. The other row and column sums are

$$q_{10} + q_{11} = R + 2S + d - 4x, \quad q_{01} + q_{11} = 2R + S + d - 4x.$$

Taking R, S large makes all four bridges positive and makes each unused connector strictly longer than the selected one. If F_P is empty, also require both column sums to exceed the current total M ; if F_Q is empty, require both row sums to exceed the current total N .

Before fitting the active detour relations, add a common rational $L > 0$ to all four bridge weights. This adds the same constant $2L$ to each doubled form in (6.14), preserves every portal difference and every row/column connector order, and replaces the selected connector weights by $S + 2L$ and $R + 2L$. Choose L so large that every three-bridge inequality in the following complete short-edge ledger is strict:

edge	portal detours	three-bridge detour
q_{00}	$M + q_{10} \quad N + q_{01}$	$q_{01} + q_{11} + q_{10}$
q_{01}	$M + q_{11} \quad N + q_{00}$	$q_{00} + q_{10} + q_{11}$
q_{10}	$M + q_{00} \quad N + q_{11}$	$q_{11} + q_{01} + q_{00}$
q_{11}	$M + q_{01} \quad N + q_{10}$	$q_{10} + q_{00} + q_{01}$

(6.18)

The first two columns are strict boundary cells of the four-corner table, and the last column is strict by the choice of L .

Now apply Lemma 6.3 to the prefix of nonempty active rows of F_P , if it is active, with prescribed margin $S + 2L$. It gives a threshold T_P satisfying

$$T_P - M = S + 2L,$$

which is exactly the comparison between a direct P -segment and its complementary segment followed by the selected exterior connector. Apply Lemma 6.3 symmetrically to the corresponding active prefix of F_Q , with margin $R + 2L$. Consistency gives the outward-safety hypotheses described after Lemma 6.3: an active left endpoint occurs only in cross routes entering at the left end of its chain, and an active right endpoint only in routes entering at the right end. Hence the extensions preserve every cross winner and every portal cut. If both relations are active, perform the extensions successively. On the intersection of two active blocks the selected corner form is strengthened in both outward directions, and neither selected physical connector contains an edge of the other long chain, so the first fitted relation is unchanged.

For a chain edge, every endpoint-chordless detour consists of the complementary chain segment followed by one of the two row/column connectors. Lemma 6.3 handles the selected connector, and the unused connector is strictly longer by the displayed row/column-sum inequalities. If the endpoint relation is empty, both connectors exceed the chain total, so every chain subpath remains direct. Lemma 5.1 completes the opposite case.

6.2.3. Two adjacent long edges

Let the two chains share the endpoint z :

$$A : z = a_0, \dots, a_m = u, \quad B : z = b_0, \dots, b_n = v,$$

let x be the fourth kernel vertex, and retain the four short edges

$$zx, \quad ux, \quad vx, \quad uv.$$

We first construct one system of chain labels that simultaneously realizes all cross-chain paths, both x -entry cuts, both same-chain relations, and the strict triangle margins at u, v, x .

Lemma 6.4 (adjacent four-route normal form)

There are strictly increasing rational labels

$$\xi_0 < \dots < \xi_m, \quad \eta_0 < \dots < \eta_n$$

and rational numbers

$$\Delta < h < 0 \tag{6.19}$$

with the following properties. Every selected a_i - b_j path is one of

$$\begin{array}{l|l} S & a_i A z B b_j \\ T & a_i A u - v B b_j \\ U & a_i A z - x - v B b_j \\ V & a_i A u - x - z B b_j, \end{array} \tag{6.20}$$

and its letter is the unique minimizer of

$$L_S = \xi_i + \eta_j, \quad L_U = \xi_i - \eta_j, \quad L_V = -\xi_i + \eta_j, \quad L_T = 2\Delta - \xi_i - \eta_j. \tag{6.21}$$

Moreover, the two prefixes of vertices reached from x through z are

$$D_A = \{i : \xi_i < h\}, \quad D_B = \{j : \eta_j < h\}. \quad (6.22)$$

Proof.

Every selected path is induced. Delete from a selected cross-chain path its initial segment in A and terminal segment in B . An induced path in the four-vertex kernel can only have one of the four forms in (6.20). The route through $u - x - v$ has the chord uv , the route through $z - x - u - v$ has the chord xv , and a longer kernel walk either repeats a vertex or contains one of these chords.

An induced path from x to a_i enters A at z or at u . If it enters at z for index i , consistency gives the same entrance for every smaller index. Hence D_A is a prefix. Neighborliness gives $0 \in D_A$ and $m \notin D_A$. The statement for D_B is symmetric. Taking x -to-chain subpaths in (6.20) gives

$$U(i, j) \implies i \in D_A, j \notin D_B, \quad V(i, j) \implies i \notin D_A, j \in D_B. \quad (6.23)$$

Thus no row contains both U and V , and taking subpaths gives corner closure.

Apply the band decomposition in Lemma 6.1 with $(00, 01, 10, 11) = (S, U, V, T)$. In its first band order the possible nonempty row bands are

$$S/U, \quad V/U, \quad V/T.$$

A genuinely nonconstant middle row would contain both V and U , contrary to (6.23). Every surviving constant U -row or V -row may be absorbed into the adjacent outer band. The reverse band order is already

$$S/U, \quad S/T, \quad V/T.$$

Thus, after the harmless absorption of constant rows and the symmetric absorption of constant columns, the only nonconstant outer bands are S/U and V/T , and the remaining central rectangle uses only S, T .

The outer cuts are common across all relevant rows and columns. Indeed, if two rows containing U selected U and S , respectively, in the same column, the two selected paths would give different prescribed z - b_j subpaths. The V -rows and the corresponding column statements are symmetric. Delete these constant-color outer rows and columns. Only S, T remain in the central rectangle, and the S -cells form a down-set. By (6.2), write

$$S(i, j) \iff r_i + s_j < 0 \quad (6.24)$$

with both label sequences strictly increasing. The two prefixes in (6.22) determine open gaps in the two sequences. Replacing

$$r_i \mapsto r_i + \lambda, \quad s_j \mapsto s_j - \lambda$$

preserves every sum. Choose λ so that the two open gaps overlap, and choose a common rational point g in their intersection. Fix a rational $\Delta < 0$, take a sufficiently small rational $\varepsilon > 0$, and put on the central rows and columns

$$\xi_i = \frac{\Delta}{2} + \varepsilon r_i, \quad \eta_j = \frac{\Delta}{2} + \varepsilon s_j, \quad h = \frac{\Delta}{2} + \varepsilon g. \quad (6.25)$$

All these values lie in $(\Delta, 0)$. Place the outer U -rows and V -columns below Δ , and the opposite outer rows and columns above 0, preserving their internal orders. Direct comparison of (6.21) gives

	$\eta < \Delta$	$\Delta < \eta < 0$	$\eta > 0$
$\xi < \Delta$	S	S	U
$\Delta < \xi < 0$	S	S/T	T
$\xi > 0$	V	T	T

(6.26)

This recovers every outer rectangle, the central staircase, and the two entry prefixes. Empty outer or central bands are omitted, with their unused thresholds placed in adjacent open gaps. $\square$

The midpoint inequalities implicit in the short-edge triangles require a stronger normalization than merely placing the two midpoints below $h/2$.

Lemma 6.5 (strengthened midpoint normalization)

The representation in Lemma 6.4 may be chosen simultaneously with both same-chain detour relations so that

$$\xi_0 + \xi_m < \Delta, \quad \eta_0 + \eta_m < \Delta. \quad (6.27)$$

All cross-route winners, both x -entry cuts, and both same-chain relations remain strict.

Proof.

We first prove a quantitative centering move. Let

$$F \subseteq \{0, \dots, p\} \times \{p+1, \dots, m\}$$

be an expanding same-chain detour relation, and suppose its physical threshold has the form

$$\tau = C - \xi_0.$$

For every nonempty right column j , put

$$q_j = \max\{i \leq p : (i, j) \in F\}.$$

Choose rational slopes

$$\lambda_0 > \lambda_1 > \dots > \lambda_p > 0, \quad \lambda_{p+1} = 0,$$

and set

$$\xi_i(T) = \bar{\xi}_i - \lambda_i T \quad (0 \leq i \leq p), \quad \tau(T) = C - \xi_0(T). \quad (6.28)$$

For each value q occurring among the q_j , choose one common rational slope

$$\lambda_q > \gamma(q) > \lambda_{q+1}.$$

Choose the $\gamma(q)$ strictly decreasing as q increases. All columns at the same level use this common slope and are separated only by bounded, strictly increasing rational constants. Put

$$H_j(T) = -\gamma(q_j)T + O(1), \quad \xi_j(T) = H_j(T) + \tau(T). \quad (6.29)$$

Since the levels q_j are nondecreasing with j , these choices preserve the right-label order for large T . Empty right columns form an initial block; leave them fixed below C . For a nonempty column,

$$\begin{aligned}\xi_j(T) - \xi_i(T) > \tau(T) &\iff H_j(T) > \xi_i(T) \\ &\iff i \leq q_j.\end{aligned}\tag{6.30}$$

For an empty column, $\xi_j < C$ and $\xi_i \geq \xi_0$, so it remains a nonmember.

No same-side relation is created. If $i < k \leq p$, the coefficient of T in

$$\xi_k - \xi_i - \tau$$

is $\lambda_i - \lambda_k - \lambda_0 < 0$. For two nonempty right columns $j < k$, the corresponding coefficient is

$$\gamma(q_j) - \gamma(q_k) - \lambda_0 < 0.$$

Two empty columns have bounded difference while τ tends to infinity. For one empty and one nonempty column, the leading term after subtracting τ is $-\gamma(q_k)T < 0$. Thus every same-side pair remains direct. If the endpoint pair belongs to F , the last column is nonempty and

$$\xi_m = H_m + \tau, \quad \xi_0 + \xi_m = C + H_m \longrightarrow -\infty.\tag{6.31}$$

The outward-safety observation following Lemma 6.3 shows that all existing cross comparisons and the entry cut persist.

We now distinguish the endpoint configurations.

Configuration I: neither endpoint path uses the other chain. If $P_{zu} = A$, consistency makes every A -subpath direct. The A -left boundary block contains only S/U forms. Move only ξ_0 to the left. This preserves the internal order and the entry cut, and strengthens every selected S/U boundary comparison. The endpoint S -winner also gives

$$\xi_m < 0, \quad \xi_m < \Delta - \eta_0,$$

so all A -subpaths remain shorter than both unused exterior connectors. Consequently $\xi_0 + \xi_m$ can be made arbitrarily negative. If $P_{zu} = zxu$, apply (6.28)–(6.31) with $C = 0$. Treat B symmetrically. The two moves may be made successively: at an intersection of their active sides the four possibilities are

left–left	S
left–right	U
right–left	V
right–right	T ,

and movement in the indicated outward directions only strengthens the selected form.

Configuration II: the A -endpoint path uses B . Assume

$$P_{zu} = zBvu, \quad P_{zv} = B.\tag{6.32}$$

Consistency excludes U, V throughout the cross table: a U -path would force $P_{zv} = z xv$, and a V -path would force $P_{zu} = z xu$. Thus the table is an S/T Ferrers table. Before applying the scaling in (6.25), write

$$S(i, j) \iff r_i + s_j < 0, \quad D_A = \{i : r_i < g\}, \quad D_B = \{j : s_j < g\}.\tag{6.33}$$

Let

$$I = \{i : T(i, 0)\}.$$

The set I is a suffix and every row in I is entirely T . Moreover, $I \cap D_A = \emptyset$: if $i \in D_A$, the selected path $P_{xa_i} = xzAa_i$ forces the boundary cell $(i, 0)$ to be S .

Choose a rational number

$$t_0 > \max\{0, s_0 + s_n\}$$

and perform the selective shift

$$s'_0 = s_0 - t_0, \quad r'_i = r_i + t_0 \quad (i \in I), \quad (6.34)$$

leaving all other labels fixed. Both internal orders persist because I is a suffix and s_0 is the first B -label. In column 0, an unchanged S -row becomes more strongly S , while an altered T -row has unchanged sum. In every other column, an altered row was already entirely T , and its sum only increases. Hence the complete S/T table is unchanged. The two entry cuts also persist: every moved A -row lies outside D_A , while lowering $s_0 \in D_B$ keeps it inside D_B .

Now fix $\Delta < 0$, take a sufficiently small rational $\varepsilon > 0$, and set

$$\xi_i = \frac{\Delta}{2} + \varepsilon r'_i, \quad \eta_j = \frac{\Delta}{2} + \varepsilon s'_j, \quad h = \frac{\Delta}{2} + \varepsilon g. \quad (6.35)$$

Then

$$\eta_0 + \eta_n = \Delta + \varepsilon(s'_0 + s_n) < \Delta. \quad (6.36)$$

Apply the centering move to the A -relation with

$$C = \Delta - \eta_0.$$

The boundary cell identifies exactly the empty/nonempty cut needed in (6.29):

$$S(j, 0) < T(j, 0) \iff \xi_j + \eta_0 < \Delta \iff \xi_j < C. \quad (6.37)$$

Choose the centering parameter so that $H_m < \eta_0$. Equations (6.31) and (6.37) then give

$$\xi_0 + \xi_m = \Delta - \eta_0 + H_m < \Delta. \quad (6.38)$$

The B -endpoint path is direct, so its same-chain relation is empty. The active left A -rows are S -rows and the active right rows are T -rows; the centering move therefore cannot reintroduce U, V .

Configuration III. If the B -endpoint path uses A and uv , interchange A and B in Configuration II. The two cross-endpoint alternatives cannot occur simultaneously, since either one contains the other endpoint path as a prescribed subpath and forces that path to be direct. These three configurations exhaust all possibilities and prove (6.27). All parameter choices lie in rational open intervals. $\square$

We can now assign the short-edge weights. Put

$$\mu_A = \frac{\xi_0 + \xi_m}{2}, \quad \mu_B = \frac{\eta_0 + \eta_n}{2},$$

give the chain edges their half-potential increments, and define

$$\begin{aligned}
w(zx) &= -\frac{h}{2}, & w(ux) &= \frac{h}{2} - \mu_A, \\
w(vx) &= \frac{h}{2} - \mu_B, & w(uv) &= \Delta - \mu_A - \mu_B.
\end{aligned} \tag{6.39}$$

Equations (6.19) and (6.27) make all four weights positive. After deletion of a common term, the doubled cross-route lengths are exactly (6.21), so all cross-chain paths are uniquely shortest. The two triangle slacks not already encoded by the entry cuts are

$$\begin{aligned}
w(uv) + w(vx) - w(ux) &= \Delta - 2\mu_B > 0, \\
w(uv) + w(ux) - w(vx) &= \Delta - 2\mu_A > 0.
\end{aligned} \tag{6.40}$$

The complete endpoint-chordless short-edge bases are

$$\begin{array}{l|ll}
zx & A + ux, & B + vx \\
ux & A + zx, & uv + vx \\
vx & B + zx, & uv + ux \\
uv & A + B, & ux + vx.
\end{array} \tag{6.41}$$

For zx , the two excesses are $h - \xi_0$ and $h - \eta_0$. For ux, vx , the entrance excesses are $\xi_m - h, \eta_n - h$, and the remaining excesses are those in (6.40). For uv , the route through z is controlled by the corner comparison $T(m, n) < S(m, n)$, while the route through x has excess $h - \Delta > 0$. Every chain-edge base is covered by the strict same-chain relation and the strict order between its two endpoint connectors. Lemma 5.1 now applies.

Theorem 6.6 (the K_4 realization theorem)

Let G be a subdivision of K_4 in which at most two core edges are long. Then every neighborly consistent path system on G has a positive rational strict realization. This includes zero long edges, one long edge, two opposite long edges, and two adjacent long edges.

The non-strict precursor is [11, Theorem 6.3]; for neighborly systems, the claim here is the strict strengthening with the endpoint margins recorded above.

Proof.

The zero-long and one-long cases were proved in Section 6.2.1, the opposite case in Section 6.2.2, and the adjacent case in Section 6.2.3. If one of the two chains in the adjacent or opposite description has length one, neighborliness forces its endpoint path to be that edge; its internal order and same-chain detour relation disappear, and the formulas reduce to the one-long or zero-long case. Lemma 5.2 gives positive rational weights without changing any strict comparison. $\square$

6.3. The split family $P_4(p, 2, r)$

Let

$$A : a_0, \dots, a_p, \quad B : b_0, \dots, b_r$$

be disjoint chains. Add the three ordinary bridges

$$a_0b_0, \quad a_0b_r, \quad a_pb_r$$

and the split bridge $a_p - s - b_0$. Write

$$\begin{aligned}
w(a_0b_0) &= q_{00}, & w(a_0b_r) &= q_{01}, & w(a_pb_r) &= q_{11}, \\
\alpha &= w(a_ps), & \beta &= w(sb_0), & q_{10} &= \alpha + \beta, & \delta &= \alpha - \beta.
\end{aligned} \tag{6.42}$$

The main construction treats $p, r \geq 2$. Unit-chain boundaries are handled at the end.

Lemma 6.7 (route and mode normal forms)

For $p, r \geq 2$, every selected endpoint-chordless path between the two chains is one of the four one-bridge routes, denoted 00, 01, Q , 11, or, when it is chordless, the complementary route

$$W(i, j) = a_i A a_p - b_r - a_0 - b_0 B b_j. \tag{6.43}$$

Here Q is the one-bridge route using $a_p - s - b_0$. A Q -route and a W -route cannot occur in the same consistent system. Thus every system belongs to exactly one of three modes: Q -mode, W -mode, or N -mode, in which neither route is selected.

The six possible induced paths from s to the two chains are

$$\begin{array}{ll}
R_A \mid s - a_p A a_i & L_B \mid s - b_0 B b_j \\
L_A \mid s - b_0 - a_0 A a_i & R_B \mid s - a_p - b_r B b_j \\
D_A \mid s - a_p - b_r - a_0 A a_i & D_B \mid s - b_0 - a_0 - b_r B b_j.
\end{array} \tag{6.44}$$

On A , the non- R_A indices form a prefix and use only L_A or only D_A . On B , the L_B indices form a prefix, after which only R_B or only D_B occurs. The two diagonal types D_A, D_B cannot coexist.

The cross tables have the following normal forms.

1. In W -mode, the W -cells and the 01-cells are opposite rectangles; the remaining cells form an additive 00/11 staircase.
2. In Q -mode, the Q -cells and 01-cells are opposite rectangles with an additive 00/11 staircase between them, or the central rectangle is a southeast difference staircase. The latter has the form

$$Q(i, j) \iff X_i - Y_j > q_{10} - q_{01}. \tag{6.45}$$

3. In N -mode, the 01-cells form a rectangle and all remaining cells form an additive 00/11 staircase.

All staircases and all entrance prefixes admit simultaneous strict rational representations.

Proof.

Delete the initial A -segment and terminal B -segment of an induced cross-chain path. The remaining walk lies in the five-vertex kernel obtained by subdividing the fourth bridge at s . If it crosses the A/B partition once, it uses one of

$$a_0b_0, \quad a_0b_r, \quad a_ps b_0, \quad a_pb_r.$$

If it crosses three times, the only possibility without an ordinary-bridge chord is the complementary cycle arc $a_p - b_r - a_0 - b_0$, which is (6.43). Every other kernel walk repeats a vertex or has one of the three ordinary bridges as a chord. This proves the route exhaustion.

Every Q -route contains the prescribed subpath $P_{a_pb_0} = a_p - s - b_0$, whereas every W -route contains $P_{a_pb_0} = a_p - b_r - a_0 - b_0$. Consistency excludes their coexistence.

For an s -rooted path, the first edge is sa_p or sb_0 . Continuing through the kernel without a chord gives exactly the six paths in (6.44). Taking subpaths proves the prefix statements. A non- R_A prefix cannot contain both L_A, D_A , since those two routes prescribe different $s-a_0$

subpaths; the B -statement is symmetric. A D_A -path gives $P_{sb_r} = s - a_p - b_r$, while a D_B -path gives $P_{sb_r} = s - b_0 - a_0 - b_r$, so the two diagonal types cannot coexist. If E_A is the non- R_A prefix and E_B the L_B prefix, the two s -root subpaths of a Q -cell give

$$Q(i, j) \implies i \notin E_A, \quad j \in E_B. \quad (6.46)$$

We next prove the three cross-table assertions.

The W -mode. Let F_A, F_B be the two same-chain detour relations, and put

$$R_A^* = \{i : (0, i) \in F_A\}, \quad L_B^* = \{j : (j, r) \in F_B\}.$$

A $W(i, j)$ -path contains both endpoint-detour subpaths, so its cell lies in $R_A^* \times L_B^*$. Conversely, within this product the colors 00, 01 conflict with the A -detour, the colors 01, 11 conflict with the B -detour, and Q is excluded. Hence

$$c(i, j) = W \iff (i, j) \in R_A^* \times L_B^*. \quad (6.47)$$

Let R_{01} be the rows in which 01 occurs and C_{01} the corresponding columns. Corner closure makes R_{01} a prefix and C_{01} a suffix. A cell in their product cannot be 00 or 11, because it would conflict with an 01-cell in the same row or column, and it cannot be W . Thus

$$c(i, j) = 01 \iff (i, j) \in R_{01} \times C_{01}. \quad (6.48)$$

After deleting these opposite rectangles, only 00, 11 remain, and subpath consistency makes the 00-cells southwest closed.

The Q -mode. Before choosing edge weights, apply Lemma 6.1 to the four one-bridge color classes. It gives increasing labels and affine constants $C_{00}, C_{01}, C_{10}, C_{11}$. Let

$$\begin{aligned} \mu_A &= \frac{X_0 + X_p}{2}, & \mu_B &= \frac{Y_0 + Y_r}{2}, \\ x_i &= X_i - \mu_A, & y_j &= Y_j - \mu_B, \end{aligned}$$

and absorb centering into

$$\begin{aligned} \tilde{C}_{00} &= C_{00} + \mu_A + \mu_B, & \tilde{C}_{01} &= C_{01} + \mu_A - \mu_B, \\ \tilde{C}_{10} &= C_{10} - \mu_A + \mu_B, & \tilde{C}_{11} &= C_{11} - \mu_A - \mu_B. \end{aligned} \quad (6.49)$$

Choose a common rational K sufficiently large and set

$$q_{\epsilon\eta} = \frac{\tilde{C}_{\epsilon\eta} + K}{2} > 0. \quad (6.50)$$

If the chain increments are half-potential differences, the four actual one-bridge lengths have the common term $(M_A + M_B)/2$, where

$$M_A = \frac{X_p - X_0}{2}, \quad M_B = \frac{Y_r - Y_0}{2}.$$

After deleting that term and multiplying lengths by 2, the four affine forms are

$$x_i + y_j + 2q_{00}, \quad x_i - y_j + 2q_{01}, \quad -x_i + y_j + 2q_{10}, \quad -x_i - y_j + 2q_{11}. \quad (6.51)$$

For the remainder of this lemma we rename the centered coordinates x_i, y_j as X_i, Y_j . Thus $X_0 = -M_A, X_p = M_A, Y_0 = -M_B, Y_r = M_B$. All thresholds in (6.53)–(6.54) refer to these centered coordinates, not to the uncentered labels used before (6.49).

The factor 2 in every bridge constant is forced by the half-potential chain convention. The centering corrections cancel in the alternating sum, so

$$\begin{aligned} d &= q_{00} + q_{11} - q_{01} - q_{10} \\ &= \frac{C_{00} + C_{11} - C_{01} - C_{10}}{2}. \end{aligned} \quad (6.52)$$

Thus the sign of d records which of the two combinatorial band orders was produced by Lemma 6.1; it is not assumed before the representation is constructed.

Put

$$\begin{aligned} r_0 &= q_{10} - q_{00}, & r_1 &= q_{11} - q_{01}, \\ c_0 &= q_{01} - q_{00}, & c_1 &= q_{11} - q_{10}. \end{aligned} \quad (6.53)$$

Then $r_1 - r_0 = c_1 - c_0 = d$. Comparing the Q -form in (6.51) with the other three gives

$$Q(i, j) \iff X_i > r_0, \quad Y_j < c_1, \quad X_i - Y_j > q_{10} - q_{01}. \quad (6.54)$$

If $d \leq 0$, the first two inequalities strictly imply the third, so the Q -cells form a rectangle; the symmetric calculation gives the opposite 01-rectangle, and the remaining 00/11 cells form an additive staircase. If $d > 0$, the third inequality cuts the central rectangle into the difference staircase (6.45). This proves the Q -mode normal form in the correct logical order.

The N -mode. Both Q, W are absent, so the 10-class of the ordinary four-corner table is empty. In the first band order

$$00/01, \quad 10/01, \quad 10/11,$$

the middle band, when nonempty, is entirely 01, and the last band is entirely 11. The 01-part of the first band is the same column suffix. If the left column band is nonempty, the middle row band must be empty, or its left cells would have color 10. Hence the 01-cells form a row-prefix by column-suffix rectangle. In the reverse band order

$$00/01, \quad 00/11, \quad 10/11,$$

the last band is entirely 11, and 01 again appears only as one rectangle. In either order, cells southwest of that rectangle are 00, cells northeast are 11, and the remaining 00/11 cells form an additive staircase.

Finally, Lemma 6.2 inserts the entrance gaps simultaneously with each additive or difference staircase. For an additive staircase, if the gaps follow row k and column l , the down-set gives exactly the three alternatives in (6.4). Choose the sum of the two targets negative, zero, or positive accordingly. For a difference staircase, (6.46) supplies $(k, l+1) \notin Q$ when the first target must precede the second, while a diagonal entrance supplies $(k+1, l) \in Q$ in the reverse orientation. When both hold, the two open gaps overlap. Lemma 6.2 therefore supplies all labels and cuts rationally and strictly. $\square$

Lemma 6.8 (strict scalar calibration)

Assume $p, r \geq 2$. There are strictly increasing rational labels

$$X_0 < \cdots < X_p, \quad Y_0 < \cdots < Y_r,$$

positive rational bridge weights as in (6.42), and rational same-chain thresholds with the following simultaneous properties:

1. in every cross-chain cell, the prescribed route is strictly shorter than every other available endpoint-chordless cross route;
2. all six s -root routes in (6.44) have exactly their prescribed entrance prefixes;
3. both same-chain detour relations are represented strictly with the same labels;
4. every endpoint-chordless competitor for a graph edge is strictly longer than that edge.

Proof.

Put

$$\begin{aligned}
M_A &= \frac{X_p - X_0}{2}, & M_B &= \frac{Y_r - Y_0}{2}, \\
\mu_A &= \frac{X_0 + X_p}{2}, & \mu_B &= \frac{Y_0 + Y_r}{2}, \\
\kappa_A &= q_{00} - q_{01} - q_{11}, & \kappa_B &= q_{00} + q_{01} - q_{11}.
\end{aligned} \tag{6.55}$$

We carry out the three constructions in the centered gauge. Namely, after each finite order representation is chosen, translate its two label sequences and absorb the translations into the four affine constants as in (6.49). Hence from this point onward $\mu_A = \mu_B = 0$, $X_0 = -M_A$, $X_p = M_A$, and $Y_0 = -M_B$, $Y_r = M_B$. The symbols μ_A, μ_B retained in (6.56)–(6.57) only display the translation-invariant form of those calculations. In particular, every later occurrence of X_i, Y_j , including (6.61), (6.78), and (6.87), denotes a centered coordinate.

The actual lengths of the six s -root routes are

$$\begin{array}{l|l}
R_A & \alpha + (X_p - X_i)/2 \\
L_A & \beta + q_{00} + (X_i - X_0)/2 \\
D_A & \alpha + q_{11} + q_{01} + (X_i - X_0)/2
\end{array}
\quad
\begin{array}{l|l}
L_B & \beta + (Y_j - Y_0)/2 \\
R_B & \alpha + q_{11} + (Y_r - Y_j)/2 \\
D_B & \beta + q_{00} + q_{01} + (Y_r - Y_j)/2.
\end{array} \tag{6.56}$$

Let

$$I_A = (X_k - \mu_A, X_{k+1} - \mu_A), \quad I_B = (Y_l - \mu_B, Y_{l+1} - \mu_B)$$

be the nontrivial centered entrance gaps when they exist. Pairwise comparison in (6.56) gives the following complete list:

type	open interval for δ	endpoint or fixed-gap condition
$A : L$	$(q_{00} + I_A) \cap (\kappa_A, \infty)$	
$A : D$	$(-\infty, \kappa_A)$	$-q_{01} - q_{11} \in I_A$
$A : R$	$(-\infty, q_{00} - M_A)$	$q_{01} + q_{11} > M_A$
$B : R$	$(-q_{11} + I_B) \cap (-\infty, \kappa_B)$	
$B : D$	(κ_B, ∞)	$q_{00} + q_{01} \in I_B$
$B : L$	$(M_B - q_{11}, \infty)$	$q_{00} + q_{01} > M_B$.

For example,

$$L_A < R_A \iff X_i - \mu_A < \delta - q_{00},$$

and $L_A < D_A$ is equivalent to $\delta > \kappa_A$, giving the first row. The other five rows follow symmetrically. Since

$$\alpha = \frac{q_{10} + \delta}{2}, \quad \beta = \frac{q_{10} - \delta}{2},$$

all s -root routes and both split-edge positivities are realized exactly when

$$\delta \in \mathcal{J}_A \cap \mathcal{J}_B \cap (-q_{10}, q_{10}), \quad (6.58)$$

where $\mathcal{J}_A, \mathcal{J}_B$ are the intervals selected from (6.57). We construct a rational point in this intersection separately in the three modes.

The W-mode.

Let T_A, T_B be the same-chain detour thresholds. By Lemma 6.3, prescribe the same sufficiently large rational margin

$$T_A - M_A = T_B - M_B = L.$$

Fix a small rational q with $0 < q < L$, and set

$$q_{01} = q, \quad q_{00} = q_{11} = L - q, \quad q_{10} > 2L - q. \quad (6.59)$$

The selected endpoint connectors are

$$E_A^1 = q_{01} + q_{11} = L, \quad E_B^0 = q_{00} + q_{01} = L, \quad (6.60)$$

whereas the unused connectors contain q_{10} .

To realize all entrance pairs, first use the reference value $\hat{q}_{10} = 2L - q$ and put

$$u_i = X_i - L, \quad v_j = Y_j + L.$$

After deleting a common term, the four available route forms are

route	00	01	11	W
form	$u_i + v_j$	$4q + u_i - v_j$	$-u_i - v_j$	$-u_i + v_j$.

(6.61)

Thus the two opposite rectangles and the central additive staircase are represented strictly. The only possible entrance pairs are

$$(A : L, B : R), \quad (A : D, B : R), \quad (A : L, B : D). \quad (6.62)$$

For an ordinary entrance cut, define the two centered gap targets directly by

$$g_A = \delta - q_{00} = -L + (\delta + q), \quad g_B = \delta + q_{11} = L + (\delta - q).$$

Substitution in (6.57) gives

pair	g_A	g_B	condition on δ
$(A : L, B : R)$	$-L + (\delta + q)$	$L + (\delta - q)$	$-q < \delta < q$
$(A : D, B : R)$	$-L$	$L + (\delta - q)$	$\delta < -q$
$(A : L, B : D)$	$-L + (\delta + q)$	L	$\delta > q$.

(6.63)

The two gap targets have sums

$$2\delta, \quad \delta - q < 0, \quad \delta + q > 0, \quad (6.64)$$

respectively. For the first pair, choose $\delta = -q/2, 0, q/2$ according as the additive corner data require a negative, zero, or positive target sum in Lemma 6.2. For the second pair, the diagonal subpath forces the negative alternative; take $\delta = -2q$. For the third, take $\delta = 2q$. Each choice lies strictly in (6.63). If a cut is empty or full, delete its absent gap condition; the

same values satisfy the remaining strict interval, and Lemma 6.3 places the inactive labels on the proper sides.

Finally choose the actual q_{10} larger than

$$2L - q, \quad |\delta|,$$

and every fixed route bound not involving q_{10} . Increasing q_{10} affects only the unavailable Q -route and the unused connectors. It leaves (6.61), the gap orders, and the two selected connectors unchanged. Hence (6.58) holds throughout W -mode.

The Q -mode.

Use the Hadamard coordinates

$$\begin{aligned} d &= q_{00} + q_{11} - q_{01} - q_{10}, \\ C &= q_{01} + q_{11} - q_{00} - q_{10}, \\ R &= q_{10} + q_{11} - q_{00} - q_{01}, \\ \Lambda &= q_{00} + q_{01} + q_{10} + q_{11}. \end{aligned} \tag{6.65}$$

Their inverse is

$$\begin{aligned} q_{00} &= (\Lambda + d - C - R)/4, & q_{01} &= (\Lambda - d + C - R)/4, \\ q_{10} &= (\Lambda - d - C + R)/4, & q_{11} &= (\Lambda + d + C + R)/4. \end{aligned} \tag{6.66}$$

The coordinate d selects the additive or difference central staircase; C, R control the two possible diagonal entrance signs; and Λ supplies common bridge positivity. From (6.57), the three nondegenerate entrance intervals are

$$\begin{array}{l|l} (A : L, B : R) & \max\{-q_{10}, -q_{10} - C\} < \delta < \min\{q_{10}, q_{10} - R\} \\ (A : D, B : R) & -q_{10} < \delta < -q_{10} - C \\ (A : L, B : D) & q_{10} - R < \delta < q_{10}. \end{array} \tag{6.67}$$

Thus an A -diagonal requires and is permitted by $C < 0$, and a B -diagonal requires and is permitted by $R > 0$. The two diagonals cannot coexist.

We verify that the gap orientation, these signs, and bridge positivity can be chosen simultaneously. Write

$$t = q_{10}, \quad a = q_{00}, \quad b = q_{11}, \quad P = t + d = a + b - q_{01}. \tag{6.68}$$

Then

$$\begin{aligned} r_0 &= t - a, & r_1 &= P - a, \\ c_0 &= b - P, & c_1 &= b - t. \end{aligned} \tag{6.69}$$

For the ordinary pair $(A : L, B : R)$, the two gap targets are

$$g_A = \delta - a, \quad g_B = \delta + b,$$

and

$$\begin{aligned} g_A - r_0 &= \delta - t, & g_A - r_1 &= \delta - P, \\ g_B - c_0 &= \delta + P, & g_B - c_1 &= \delta + t. \end{aligned} \tag{6.70}$$

First suppose the central staircase is additive. Since all color comparisons are strict, perturb within their common open cone so that $d < 0$, hence $P < t$. Choose $0 < \varepsilon < t/3$. The four possible relative positions of the two gaps have the following witnesses:

A-gap	B-gap	(P, δ)
before r_0, r_1	between c_0, c_1	$(0, -\varepsilon)$
before r_0, r_1	after c_0, c_1	$(2\varepsilon, 0)$
between r_0, r_1	between c_0, c_1	$(-2\varepsilon, 0)$
between r_0, r_1	after c_0, c_1	$(0, \varepsilon)$.

(6.71)

Substitution in (6.70) gives, respectively, the required four sign patterns

$$(-, -, -, +), \quad (-, -, +, +), \quad (-, +, -, +), \quad (-, +, +, +).$$

They are precisely the alternatives allowed by (6.4), and every row has $|\delta| < t$.

If the central staircase is a difference staircase, then $d > 0$ and $P > t$. Within $|\delta| < t$, (6.70) forces the A -gap before both row thresholds and the B -gap after both column thresholds. This is exactly the order forced by (6.46) at the two boundaries of the difference rectangle. Take $\delta = 0$ and use the difference part of Lemma 6.2. The two other formal gap orientations would require $\delta < -t$ or $\delta > t$, and therefore cannot arise from positive split edges in a consistent Q -mode system.

The diagonal signs are independent of these choices. Equation (6.68) gives

$$C = 2b - (P + t), \quad R = P + t - 2a. \quad (6.72)$$

Put $H = (P + t)/2 > 0$, after a common positive rescaling if necessary. With no diagonal entrance, choose $a, b > H$; then $C > 0, R < 0$, and (6.67) reduces to $-t < \delta < t$. With an A -diagonal, choose a small rational $\eta > 0$, set $b = H - \eta$, take $a > H$, and set

$$\delta = -t + \eta.$$

Then $C = -2\eta < 0, R < 0$, and $-t < \delta < -t + 2\eta$. With a B -diagonal, set $a = H - \eta$, take $b > H$, and set

$$\delta = t - \eta.$$

Then $R = 2\eta > 0, C > 0$, and $t - 2\eta < \delta < t$. Taking

$$0 < \eta < \min\{H, t/4\}$$

and choosing the variable on the other side of H sufficiently close to H also makes

$$q_{01} = a + b - P > 0. \quad (6.73)$$

The fixed diagonal gap $-q_{01} - q_{11}$, or $q_{00} + q_{01}$, and the remaining ordinary gap fall into the corresponding alternative of Lemma 6.2. The subpath condition (6.46) excludes the incompatible alternative. Thus the gap interval remains nonempty after the signs of C, R have been fixed.

It remains to make all bridges positive and to include empty and full entrance prefixes. Increase Λ in (6.66), equivalently apply the common lift

$$q_{\varepsilon\eta}^\lambda = q_{\varepsilon\eta} + \lambda, \quad t^\lambda = t + \lambda, \quad \lambda \geq 0. \quad (6.74)$$

Then

$$d^\lambda = d, \quad C^\lambda = C, \quad R^\lambda = R, \quad \kappa_A^\lambda = \kappa_A - \lambda, \quad \kappa_B^\lambda = \kappa_B + \lambda. \quad (6.75)$$

The complete list of possible boundary pairs and the required split balance is

entrance pair	balance after the common lift
$(A : L, B : R)$	$\delta^\lambda = \delta$
$(A : D, B : R)$	$\delta^\lambda = \delta - \lambda$
$(A : L, B : D)$	$\delta^\lambda = \delta + \lambda$
$(A : R, B : L)$	$\delta^\lambda = \delta$
$(A : R, B : R)$	$\delta^\lambda = \delta - \lambda$
$(A : L, B : L)$	$\delta^\lambda = \delta + \lambda$
$(A : D, B : L)$	$\delta^\lambda = \delta - \lambda$
$(A : R, B : D)$	$\delta^\lambda = \delta + \lambda$

(6.76)

We verify every row. The ordinary gap targets and diagonal fixed gaps are

$$g_A = \delta - q_{00}, \quad g_B = \delta + q_{11}, \quad h_A = -q_{01} - q_{11}, \quad h_B = q_{00} + q_{01}. \quad (6.77)$$

For $(A : L, B : R)$, the two ordinary targets move by $-\lambda, +\lambda$. For each fixed λ , rerun the rational order insertion in Lemma 6.2. In the additive case their sum is unchanged; in the difference case the required order only becomes stronger.

For $(A : D, B : R)$, taking $\delta^\lambda = \delta - \lambda$ gives

$$\delta^\lambda < \kappa_A^\lambda \iff \delta < \kappa_A, \quad g_B^\lambda = g_B.$$

The fixed A -gap moves to $h_A - 2\lambda$, which Lemma 6.3 inserts on the outward-safe side. The case $(A : L, B : D)$ is symmetric: g_A is fixed, $\delta^\lambda > \kappa_B^\lambda$ is equivalent to $\delta > \kappa_B$, and the B -gap moves to $h_B + 2\lambda$.

For $(A : R, B : L)$, keeping δ fixed increases both direct endpoint slacks. For $(A : R, B : R)$, taking $\delta^\lambda = \delta - \lambda$ fixes the B -gap and gives

$$q_{00}^\lambda - M_A - \delta^\lambda = q_{00} - M_A - \delta + 2\lambda > 0.$$

The case $(A : L, B : L)$ is symmetric. Finally, $(A : D, B : L)$ satisfies

$$\begin{aligned} \delta^\lambda < \kappa_A^\lambda &\iff \delta < \kappa_A, \\ \delta^\lambda > M_B - q_{11}^\lambda &\iff \delta > M_B - q_{11}, \end{aligned}$$

and the symmetric pair $(A : R, B : D)$ satisfies

$$\begin{aligned} \delta^\lambda < q_{00}^\lambda - M_A &\iff \delta < q_{00} - M_A, \\ \delta^\lambda > \kappa_B^\lambda &\iff \delta > \kappa_B. \end{aligned}$$

These last two rows are essential when an ordinary cut is empty or full but a diagonal entrance remains.

Split-edge positivity persists in every row. If $\delta^\lambda = \delta - \lambda$, then

$$t^\lambda + \delta^\lambda = t + \delta > 0, \quad t^\lambda - \delta^\lambda = t - \delta + 2\lambda > 0,$$

and the plus case is symmetric. If δ is fixed, both numerators increase. Every one-bridge route receives the same increment λ , so the cross table is unchanged. The unused W -margin, the ordinary triangle slacks, and every missing midpoint margin increase strictly. Taking λ large makes all four bridge weights positive. This proves (6.58) for every Q -mode entrance pair.

The N -mode.

Put

$$q_{00} = q_{11} = u, \quad q_{01} = v, \quad \rho = u - v.$$

Direct comparison of the four one-bridge forms in (6.51), with the Q -form unavailable, gives

$$\begin{aligned} 01 \text{ is minimal} &\iff X_i < \rho, \quad Y_j > -\rho, \\ 00 \text{ is minimal outside this rectangle} &\iff X_i + Y_j < 0. \end{aligned} \tag{6.78}$$

Thus (6.57) specializes to

type	$\mathcal{J}$	endpoint or fixed-gap condition
$A : L$	$(u + I_A) \cap (-v, \infty)$	$-(u + v) \in I_A, \quad M_A > u + v$ $M_A < u + v$
$A : D$	$(-\infty, -v)$	
$A : R$	$(-\infty, u - M_A)$	
$B : R$	$(-u + I_B) \cap (-\infty, v)$	$u + v \in I_B, \quad M_B > u + v$ $M_B < u + v$
$B : D$	(v, ∞)	
$B : L$	$(M_B - u, \infty)$	

(6.79)

For an ordinary entrance gap, write

$$g_A = \delta - u, \quad g_B = \delta + u. \tag{6.80}$$

The four possible positions of these gaps relative to the two rectangle boundaries in (6.78) are realized by the following finite list:

	g_A relative to ρ	g_B relative to $-\rho$	(u, v, δ)
1	$g_A < \rho$	$g_B < -\rho$	$(5/2, 9/2, -2)$
2	$g_A < \rho$	$g_B > -\rho$	$(1, 1, 0)$
3	$g_A > \rho$	$g_B < -\rho$	$(1, 3, 0)$
4	$g_A > \rho$	$g_B > -\rho$	$(5/2, 9/2, 2)$.

(6.81)

Indeed, the corresponding values of (ρ, g_A, g_B) are

$$(-2, -9/2, 1/2), \quad (0, -1, 1), \quad (-2, -1, 1), \quad (-2, -1/2, 9/2), \tag{6.82}$$

which give exactly the four strict sign patterns in (6.81). The first and fourth triples are obtained from $(1, 3, -2)$ and $(1, 3, 2)$, respectively, by the finite common lift

$$(u, v) \mapsto (u + 3/2, v + 3/2).$$

This leaves ρ fixed, moves g_A down by $3/2$, and moves g_B up by $3/2$. Moreover,

$$g_A + g_B = 2\delta, \tag{6.83}$$

so the negative, zero, or positive alternative required by the additive part of Lemma 6.2 is unchanged. After choosing the appropriate row of (6.81), apply the finite rational order insertion of Lemma 6.2 anew and place every inactive symbol in one of the resulting nonempty open intervals. No deformation of an already fixed label set is used.

A single diagonal entrance is realized by

$$(u, v, \delta) = (3, 1, -2) \quad (A : D), \quad (u, v, \delta) = (3, 1, 2) \quad (B : D). \quad (6.84)$$

The first triple satisfies $\delta < -v$ and fixes the centered A -gap at $-(u+v) = -4$; the second satisfies $\delta > v$ and fixes the centered B -gap at $u+v = 4$. Lemma 6.3 inserts the fixed gap on the outward-safe side, while Lemma 6.2 simultaneously inserts an ordinary gap on the other chain when one is present.

There are exactly eight admissible entrance pairs, since $(A : D, B : D)$ is excluded by Lemma 6.7. They are realized as follows.

1. For $(A : L, B : R)$, choose the row of (6.81) determined by the positions of I_A, I_B relative to $\rho, -\rho$, and insert $g_A \in I_A, g_B \in I_B$. Every row also satisfies $-v < \delta < v$, so both intervals in (6.79) contain δ .
2. For $(A : D, B : R)$, use $(3, 1, -2)$, take $M_A > 4$, fix the A -gap at -4 , and insert $g_B = 1$. Then $\delta = -2 \in (-\infty, -1) \cap (-u + I_B)$.
3. For $(A : L, B : D)$, use $(3, 1, 2)$, take $M_B > 4$, fix the B -gap at 4 , and insert $g_A = -1$.
4. For $(A : R, B : L)$, use $(1, 1, 0)$ and choose $0 < M_A, M_B < 1$. Then $M_B - u < 0 < u - M_A$.
5. For $(A : R, B : R)$, choose the row of (6.81) required by the B -gap. Since every row has $\delta < u$, first choose $0 < M_A < \min\{u+v, u-\delta\}$; then $\delta < u - M_A$.
6. For $(A : L, B : L)$, choose the row required by the A -gap. Since every row has $\delta > -u$, first choose $0 < M_B < \min\{u+v, u+\delta\}$; then $M_B - u < \delta$.
7. For $(A : D, B : L)$, use $(3, 1, -2)$, take $M_A > 4$, and choose $0 < M_B < 1$. Then the fixed A -gap exists and $M_B - u < -2 < -v$.
8. For $(A : R, B : D)$, use $(3, 1, 2)$, take $M_B > 4$, and choose $0 < M_A < 1$. Then $v < 2 < u - M_A$.

We now impose the physical same-chain comparisons, writing T_A, T_B for their thresholds and putting $E = u+v$. In N -mode an active endpoint detour cannot use the split path, since that would select a Q -subpath. Its selected exterior connector is therefore the one avoiding q_{10} , of weight E .

Here the prescribed margin E need not be left to the *sufficiently large* quantifier in Lemma 6.3. For an active left block, insert the fixed symbol E into the row-0 boundary gap of its expanding relation in the finite order construction. In the notation of the proof of Lemma 6.3, take $A = E$ and $H_0 = E$. Then

$$T = 2E + X_m - H_0 = X_m + E, \quad X_0 = H_0 - T = -X_m,$$

and every other active-prefix label satisfies $X_i = H_i - T < -E$, since $H_i < X_m$. Thus the labels are already centered and $T - (X_m - X_0)/2 = E$. The reflected construction for an active right block inserts $-E$ in its terminal-column gap and puts all of that block strictly to the right of E . Every marker actually used in the N -mode order insertion— $\rho, -\rho$ and any present ordinary gap target g_A, g_B —lies strictly in $(-E, E)$; a present diagonal marker is exactly $-E$ or E . Hence these two specialized constructions preserve all marker orders and all outward-safe comparisons while giving

$$T_A - M_A = E \quad \text{or} \quad T_B - M_B = E,$$

as appropriate. On a side whose endpoint path is direct, choose instead $M_A < E$ or $M_B < E$ during the finite open-gap insertion. The direct chain is then shorter than the connector of weight E , and also shorter than the other connector, whose weight is $u + q_{10} > E$ by (6.86). The conditions $M_A > E$ and $M_B > E$ in the diagonal rows of (6.79) cause no conflict: the corresponding diagonal entrance already forces that side's endpoint detour to be active.

The small half-spans in the fourth through eighth cases do not conflict with a same-chain fixed margin. Type $A : R$ prescribes the path $s - a_p A a_0$; consistency therefore forces the direct path $P_{a_0 a_p} = A$, so no A -endpoint detour is active. Symmetrically, type $B : L$ forces the direct path on B . These are precisely the two types for which an upper bound on a half-span is used. Choose the small half-span by a positive rational rescaling before the finite order insertion. On a side with an active detour, apply the specialized Lemma 6.3 construction above afterwards. Its finite auxiliary family consists of (i) all strict cross-winner comparisons in (6.78), (ii) the ordinary entrance memberships $g_A \in I_A$, $g_B \in I_B$ and the diagonal memberships $-(u+v) \in I_A$, $u+v \in I_B$ that are present, and (iii) every label-marker order against $\rho, -\rho, g_A, g_B, \pm(u+v)$ used in the finite order insertion. Place the active prefix beyond the nearest fixed marker on its left (or, symmetrically, the active suffix beyond the nearest fixed marker on its right) before applying the lemma. Consistency gives the outward-safety property recorded after Lemma 6.3, so the extension moves that block only farther outward and preserves every listed strict comparison.

All six triples in (6.81) and (6.84) satisfy

$$u > |\delta|.$$

Consequently,

$$|\delta| < \min\{M_A + u, M_B + u, 2u + v\}. \quad (6.85)$$

After the finite open-gap insertion, choose

$$q_{10} > \max\{|\delta|, v, \text{ the total weight already assigned outside the split path}\}. \quad (6.86)$$

Then Q cannot be selected, and every unused connector containing q_{10} is strictly too long. The three possible excesses of the W -route over the selected N -mode expression are

$$2(u + v - X_i), \quad 2(2u - X_i + Y_j), \quad 2(u + v + Y_j). \quad (6.87)$$

The first expression is the excess over a selected 00-route. That route contains the direct $a_i - a_0$ subpath; consistency selects this subpath, and the strict same-chain comparison with the complementary connector of weight $q_{01} + q_{11} = u + v$ is exactly $X_i < u + v$. Similarly, the third expression is the excess over a selected 11-route, whose direct $b_j - b_r$ subpath gives $Y_j > -(u+v)$. The middle expression occurs in the 01-rectangle, where (6.78) gives $X_i < u - v$ and $Y_j > v - u$, whence $2u - X_i + Y_j > 2v > 0$. Thus every expression in (6.87) is positive. Together with $q_{10} > v, |\delta|$, this makes both W and Q strictly longer than the selected route in every cell. This proves (6.58) in N -mode.

The endpoint ledger.

It remains to verify the edge cases required by the endpoint-chord reduction in Lemma 5.1. For each ordinary bridge, the following table lists every endpoint-chordless alternative in the graph with that bridge deleted:

edge	alternative lengths		
q_{00}	$M_A + q_{10}$	$q_{01} + M_B$	$q_{01} + q_{11} + q_{10}$
q_{01}	$M_A + q_{11}$	$q_{00} + M_B$	$q_{00} + q_{10} + q_{11}$
q_{11}	$M_A + q_{01}$	$q_{10} + M_B$	$q_{10} + q_{00} + q_{01}$

(6.88)

The first two columns are the corresponding corner comparisons already made strict by the cross-table construction. The third column requires

$$\begin{aligned}
q_{00} &< q_{01} + q_{11} + q_{10}, \\
q_{01} &< q_{00} + q_{10} + q_{11}, \\
q_{11} &< q_{10} + q_{00} + q_{01}.
\end{aligned}
\tag{6.89}$$

In Q -mode the common lift (6.74) increases the right side minus the left side in each line by 2λ . In W -mode, (6.59) and a sufficiently large L give all three inequalities. In N -mode they follow from (6.86).

For the edge $a_p s$ of weight α , the three alternatives through sb_0 have lengths

$$\beta + M_A + q_{00}, \quad \beta + M_B + q_{11}, \quad \beta + q_{00} + q_{01} + q_{11}.$$

For the edge sb_0 of weight β , the three symmetric alternatives through $a_p s$ have the same three added terms. Hence all six split-edge bases are exactly

$$|\delta| < \min\{M_A + q_{00}, M_B + q_{11}, q_{00} + q_{01} + q_{11}\}. \tag{6.90}$$

In Q -mode, the selected corner $(p, 0)$ gives

$$q_{10} < M_A + q_{00}, \quad q_{10} < M_B + q_{11}, \quad q_{10} < q_{00} + q_{01} + q_{11}; \tag{6.91}$$

together with $|\delta| < q_{10}$, this proves (6.90). In W -mode the three quantities on the right of (6.90) grow with L , while δ stays in the bounded intervals used in (6.63). In N -mode, (6.90) is (6.85).

Finally, the two exterior connector totals for each chain are

$$\begin{aligned}
E_A^0 &= q_{00} + q_{10}, & E_A^1 &= q_{01} + q_{11}, \\
E_B^0 &= q_{00} + q_{01}, & E_B^1 &= q_{10} + q_{11}.
\end{aligned}
\tag{6.92}$$

If an endpoint path detours, Lemma 6.3 makes its selected connector strictly shorter than both the direct chain and the unused connector. If the endpoint path is direct, both connectors are placed strictly beyond the chain total. The fixed-margin relations then compare every internal chain edge with both exterior endpoint-chordless routes. The cross-table and s -root constructions already compare every remaining competitor. Thus (6.88)–(6.92) complete the endpoint ledger, and all four assertions of the lemma follow. $\square$

Theorem 6.9 (the split-family realization theorem)

For every $p, r \geq 1$, every neighborly consistent path system on $P_4(p, 2, r)$ has a positive rational realization in which every prescribed path is the unique shortest path.

For $p, r \geq 2$, the corresponding neighborly non-strict statement is [11, Theorem 6.7]; the unit-boundary cases are covered by [11, Theorem 6.3]. The proof below uses the strict scalar calibration and complete endpoint ledger established in Lemma 6.8.

Proof.

Suppose first that $p, r \geq 2$. Lemma 6.7 exhausts all endpoint-chordless paths: they are cross-chain routes, one of the six s -root routes, same-chain paths, or competitors for graph edges. Lemma 6.8 makes the prescribed member strictly shorter in every one of these comparisons. Lemma 5.1, applied with edge competitors taken in $G - e$, therefore proves that every prescribed path is uniquely shortest. All parameters were chosen in rational open intervals, so every edge weight is positive and rational.

If $p = 1$ or $r = 1$, the graph itself is a subdivision of K_4 with at most two adjacent long core edges: the subdivided edge $a_p - s - b_0$ and, when present, the other nonunit chain. Theorem 6.6 applies. When both $p = r = 1$, only the edge subdivided at s is long, so this is the one-long case of that theorem. Notice also that a diagonal entrance on a unit chain is impossible: neighborliness prescribes its single edge, while a diagonal route would prescribe a different path between the same two vertices. Thus no entrance case is lost at either unit boundary. $\square$

7 The rim–rim family $H_{m,n}^{rr}$

We give a direct, non-computational construction for the only branch class in which all six triangle expansions can fail, followed by the expansion and lifting arguments for every remaining branch class.

Theorem 7.1

For every $m, n \geq 1$, every neighborly consistent path system on $H_{m,n}^{rr}$ has a positive rational strict realization.

The proof occupies Subsections 7.1–7.12. We first isolate the unique branch class not reducible by triangle expansion, construct it directly, and then give a finite expansion-and-lifting table for every remaining class.

7.1. Setup and the exceptional branch class

Let

$$P : A = p_0 p_1 \cdots p_m = X, \quad Q : B = q_0 q_1 \cdots q_n = Y, \quad m, n \geq 1$$

and let the six short core edges of $H_{m,n}^{rr} = U_{m,n} - AB$ be

$$AC, BC, BX, CX, AY, CY.$$

Inducedness leaves exactly three routes for each of the four nonadjacent branch pairs. We label a route by the long chain or the C -route that it uses:

pair	C	P	Q
AB	ACB	$APXB$	$AYQ^{-1}B$
AX	ACX	P	$AYQ^{-1}BX$
BY	BCY	$BXP^{-1}AY$	Q
XY	XCX	$XP^{-1}AY$	$XBQY$.

Taking the indicated branch-to-branch subpaths gives all nontrivial compatibility implications:

$$\begin{aligned}
AB = P &\implies AX = P, & AB = Q &\implies BY = Q, \\
AX = Q &\implies AB = BY = XY = Q, & BY = P &\implies AB = AX = XY = P, \\
XY = P &\implies AX = P, & XY = Q &\implies BY = Q.
\end{aligned}$$

Conversely, any two named routes not excluded by these implications meet in only their common prescribed branch subpaths, so there is no further branch-level compatibility condition. The catalogue therefore has twenty types. If $AB = C$, then $AX \in \{C, P\}$ and $BY \in \{C, Q\}$; the choices $XY = C, P, Q$ contribute 4, 2, 2 types, respectively. If $AB = P$, then $AX = P$, and the three possibilities for BY contribute 2, 1, 3 choices for XY , for a total of six. The case $AB = Q$ is symmetric and contributes another six. When $m = 1$ or $n = 1$, neighborliness forces the corresponding endpoint path to be the chain edge, so the unavailable alternatives are simply deleted from this same catalogue.

We prove the following branch lemma.

Lemma 7.2 (exceptional $C/P/Q/C$ class). Every neighborly consistent path system satisfying

$$P_{AB} = ACB, \quad P_{AX} = P, \quad P_{BY} = Q, \quad P_{XY} = XCY \quad (7.1)$$

has a positive rational strict realization.

This is precisely the branch class in which no edge of either core triangle admits the direct triangle expansion proved below.

7.2. Portal prefixes and the central merged order

For a vertex of P , record through which endpoint its prescribed path to a core target leaves P . Let

$$I_B, I_C, I_Y$$

be the prefixes which leave through A on their paths to B, C, Y , respectively. Similarly let

$$J_A, J_C, J_X$$

be the prefixes of Q , from its B -end, which leave through B on their paths to A, C, X . Consistency gives

$$I_B \subseteq I_C \subseteq I_Y, \quad J_A \subseteq J_C \subseteq J_X. \quad (7.2)$$

For example, a path from p_i to B which uses ACB contains the prescribed p_i -to- C subpath through A , so $I_B \subseteq I_C$. If $p_i \in I_C$ but $p_i \notin I_Y$, the prescribed p_i -to- Y path through XCY would contain a p_i -to- C subpath through X , a contradiction. The other two inclusions are symmetric. Each set is a prefix because any earlier chain vertex occurs as a subpath endpoint on the route through the same end.

Put

$$P^\circ = I_Y \setminus I_B, \quad Q^\circ = J_X \setminus J_A.$$

For $p_i \in P^\circ, q_j \in Q^\circ$, the prescribed cross path cannot use ACB or XCY : either choice would contradict one of the four defining portal memberships. Thus it uses exactly one of AY and XB . Define an order relation by

$$p_i \prec q_j \iff P_{p_i q_j} \text{ uses } AY,$$

and otherwise put $q_j \prec p_i$. Retain the natural orders on P° and Q° .

This is a merged total order. Indeed, if $i' \leq i, j' \geq j$, and the (i, j) -cell uses AY , then its selected path contains $p_{i'}$ and $q_{j'}$, and consistency forces the (i', j') -cell to use the same AY route. The complementary closure holds for XB . These are exactly the Ferrers conditions for merging two ordered chains.

7.3. The two-anchor embedding

Insert an abstract anchor T into the merged order after every vertex of $I_C \setminus I_B$ and before every vertex of $I_Y \setminus I_C$. This is possible because the first block precedes the second on P . Insert another anchor S after $J_C \setminus J_A$ and before $J_X \setminus J_C$. Equivalently, insert each anchor into its cut interval of the already merged total order. The resulting order is acyclic; if the two cut intervals overlap, choose either order for T, S .

If $T \prec S$, put

$$t = 1/3, \quad s = 2/3;$$

if $S \prec T$, put

$$t = 2/3, \quad s = 1/3.$$

Embed all remaining elements strictly in $(0, 1)$, respecting the augmented order, with T at t and S at $s = 1 - t$. Extend the labels along the whole chains so that

$$-2 = x_0 < x_1 < \dots < x_m = 2, \quad -2 = y_0 < y_1 < \dots < y_n = 2, \quad (7.3)$$

and

$$\begin{array}{c|cccc} P & I_B & I_C \setminus I_B & I_Y \setminus I_C & P \setminus I_Y \\ & (-2, 0) & (0, t) & (t, 1) & (1, 2) \\ Q & J_A & J_C \setminus J_A & J_X \setminus J_C & Q \setminus J_X \\ & (-2, 0) & (0, s) & (s, 1) & (1, 2). \end{array} \quad (7.4)$$

Empty blocks cause no problem. Give each chain edge half the difference of its endpoint labels. Both chain totals are therefore 2.

The six portal thresholds are now

$$\beta = \alpha = 0, \quad \gamma = t, \quad \nu = s, \quad \eta = \xi = 1. \quad (7.5)$$

In particular

$$\nu - \alpha = \eta - \gamma = s, \quad \xi - \nu = \gamma - \beta = t, \quad (7.6)$$

and the central comparison is exactly

$$x_i - \eta < y_j - \xi \iff x_i < y_j, \quad (7.7)$$

which is the merged order constructed above. This proves the simultaneous six-gap/Ferrers embedding, including coincident or empty portal blocks.

7.4. Explicit core weights

Write b, c, d, e, f, g for the weights of AC, BC, BX, CX, AY, CY . Set

$$b = c = 2, \quad d = f = 4, \quad e = 2 + t, \quad g = 2 + s. \quad (7.8)$$

Equivalently, these are the portal equations

$$d = b + c + \beta, \quad e = b + \gamma, \quad g = c + \nu, \quad f = b + c + \alpha. \quad (7.9)$$

All weights are positive. The four branch paths in (7.1) have the following lengths:

pair	selected	other induced branch routes
AB	$ACB = 4$	$APXB = 6, AYQB = 6$
AX	$P = 2$	$ACX = 4 + t, AYQBX = 10$
BY	$Q = 2$	$BCY = 4 + s, BXPAY = 10$
XY	$XCY = 5$	$XPAY = 6, XBQY = 6.$

(7.10)

Thus all four branch prescriptions have a uniform positive margin.

7.5. Portal and cross pairs

For p_i, q_j , delete the common endpoint-to-portal constants. The four canonical cross-route lengths are

ACB	$(x_i + y_j)/2 + 4$
AY	$(x_i - y_j)/2 + 4$
XB	$(-x_i + y_j)/2 + 4$
XCY	$(-x_i - y_j)/2 + 5.$

(7.11)

After subtracting (7.4) and multiplying by 2, these are

$$x_i + y_j, \quad x_i - y_j, \quad -x_i + y_j, \quad -x_i - y_j + 2. \quad (7.12)$$

Consequently:

- ACB beats its adjacent routes exactly when $x_i < 0$ and $y_j < 0$;
- AY beats them exactly when $y_j > 0$ and $x_i < 1$;
- XB beats them exactly when $x_i > 0$ and $y_j < 1$;
- XCY beats them exactly when $x_i > 1$ and $y_j > 1$;
- in the sole overlap $0 < x_i, y_j < 1$, $AY < XB$ if and only if $x_i < y_j$.

The portal memberships in (7.4), consistency, and (7.7) therefore select exactly the prescribed canonical route in every cross cell, strictly.

The same calculation gives the six one-chain portal cuts. For example, the two shortest p_i -to- C routes have lengths

$$\frac{x_i + 2}{2} + 2, \quad \frac{2 - x_i}{2} + (2 + t),$$

so the first is shorter exactly when $x_i < t$. The other five cuts are 0, 1 on P and 0, s , 1 on Q , as prescribed by (7.4). Every third portal route contains one of the strictly longer branch/core detours in the next section and is therefore strictly longer.

Since $P_{AX} = P$ and $P_{BY} = Q$, consistency forces every same-chain path to be its direct chain segment. Any complementary P -detour contains an external A -to- X connector of length at least

$$b + e = 4 + t > 2 = w(P),$$

and the alternative connector through Y, Q, B is longer still. Thus every direct P -segment is uniquely shorter than its complements. The argument for Q is symmetric, using $c + g = 4 + s > 2$.

7.6. Endpoint-chordless edge ledger

The complete short-core ledger is

edge	first competitor	second competitor	
$AC = b = 2$	$A - P - X - C = 4 + t$	$A - Y - C = 6 + s$	(7.13)
$BC = c = 2$	$B - Q - Y - C = 4 + s$	$B - X - C = 6 + t$	
$BX = d = 4$	$B - Q - Y - A - P - X = 8$	$B - C - X = 4 + t$	
$CX = e = 2 + t$	$C - A - P - X = 4$	$C - B - X = 6$	
$AY = f = 4$	$A - P - X - B - Q - Y = 8$	$A - C - Y = 4 + s$	
$CY = g = 2 + s$	$C - A - Y = 6$	$C - B - Q - Y = 4.$	

Every displayed comparison is strict because $0 < t, s < 1$. The two endpoint-chordless competitors of a P -chain edge use, respectively, the connectors $A-C-X$ and $A-Y-Q-B-X$; those of a Q -chain edge are symmetric. Their strictness follows from the connector comparison in the previous section.

For completeness, every endpoint-chordless path in this graph belongs to one of the following classes:

1. a direct same-chain segment or one of its two connector complements;
2. one of the six portal pairs just considered;
3. one of the four canonical cross routes in (7.11);
4. one of the four branch pairs in (7.10);
5. one of the edge competitors in (7.13).

This follows by recording the first endpoint through which a path leaves P and the last endpoint through which it enters Q . A second passage through a chain, or a noncanonical passage through either core triangle, creates an endpoint chord.

Lemma 5.1 now applies: replace any chorded competitor subpath by its chord edge and iterate. Positivity deletes any resulting closed walk. Since every endpoint-chordless competitor is strictly longer, the weights (7.3), (7.8) strictly realize the prescribed path system. This proves the exceptional $C/P/Q/C$ lemma.

7.7. Exact scalar region for the whole main C sector

The same anchored normalization (7.3)–(7.6) works for all four main C-sector branch choices

$$AB = C, \quad XY = C, \quad AX \in \{C, P\}, \quad BY \in \{C, Q\}.$$

Let $s = 1 - t$ and retain

$$d = f = b + c, \quad e = b + t, \quad g = c + s. \quad (7.14)$$

The complete scalar conditions coming from the core-edge ledger are simply

$$b > s/2, \quad c > t/2. \quad (7.15)$$

The branch endpoints add

$$\begin{array}{ll} AX = C & \left| \begin{array}{l} s/2 < b < 1 - t/2 \\ b > 1 - t/2 \end{array} \right. & \begin{array}{l} BY = C \\ BY = Q \end{array} & \left| \begin{array}{l} t/2 < c < 1 - s/2 \\ c > 1 - s/2 \end{array} \right. \end{array} \quad (7.16)$$

All four regions are nonempty. One explicit choice is

$$b = \begin{cases} 3s/4, & AX = C, \\ 2, & AX = P, \end{cases} \quad c = \begin{cases} 3t/4, & BY = C, \\ 2, & BY = Q. \end{cases} \quad (7.17)$$

Equations (7.15)–(7.16) include all branch and core-edge inequalities; in particular, they show directly that no simultaneous large-parameter choice of b, c is required.

It remains to check that a same-chain complement system can be installed without moving a portal or changing a cross cell. Suppose first that $AX = C$, so the selected endpoint connector is $A - C - X$, of total weight

$$R_P = b + e = 2b + t < 2.$$

If a selected p_i -to- p_j path uses this connector, with $i < j$, then its p_i -to- X subpath is $p_i \cdots A - C - X$. Were $P_{p_i B}$ to leave through X , its prescribed path would contain $p_i \cdots A - C - X - B$, which has the chord CB . Hence $p_i \in I_B$. At the other end, $P_{Ap_j} = A - C - X \cdots p_j$. Were $P_{p_j Y}$ to leave through A , the resulting prescribed path would contain $p_j \cdots X - C - A - Y$, which has the chord CY . Hence $p_j \notin I_Y$. Thus every active left endpoint lies strictly to the left of all three P-portal cuts, and every active right endpoint lies strictly to their right.

The standard strict cycle-potential lemma now applies with its middle labels fixed, but the chain endpoints must remain pinned. Put $\delta_P = 2 - R_P > 0$. Consistency makes the connector-selected pairs a Ferrers relation between an active prefix L and an active suffix U . Write

$$U = \{u_1 < \cdots < u_r = X\}$$

and choose

$$0 < h_1 < \cdots < h_r = \delta_P.$$

For each nonendpoint $p_i \in L$, its selected columns form a terminal interval $\{u_k, \dots, u_r\}$. Choose a_i in the open interval (h_{k-1}, h_k) , where $h_0 = 0$, preserving the strict order of the left labels; set $a_A = 0$. Put

$$b_{u_j} = \delta_P - h_j.$$

Ferrers nesting makes these choices simultaneous, and it gives the strict equivalence

$$(p_i, u_j) \text{ uses the connector} \iff a_i < h_j \iff a_i + b_{u_j} < \delta_P.$$

In particular $b_X = 0$. Place the active prefix labels at $-2 + a_i$ and the active suffix labels at $2 - b_j$. Thus $x_A = -2$ and $x_X = 2$: the chain total remains exactly 2. Place every inactive vertex in the open middle interval $(-2 + \delta_P, 2 - \delta_P)$, respecting the chain order and retaining all previously fixed central labels. This retention is possible because $0 < \delta_P < 1$, so the entire

anchored interval $[0, 1]$ lies strictly inside that middle interval; inactive vertices in the outer portal blocks still have nonempty intervals $(-2 + \delta_P, 0)$ and $(1, 2 - \delta_P)$. Then

$$x_j - x_i > 2 + R_P$$

holds exactly for the connector-selected pairs, which is precisely the strict comparison between the connector complement and the direct P -segment. Every pair involving an inactive vertex has coordinate difference strictly less than $2 + R_P = 4 - \delta_P$, so its direct segment remains strictly selected. With the explicit value $b = 3s/4$ in (7.17),

$$R_P = \frac{3}{2}s + t = \frac{3}{2} - \frac{1}{2}t \in (1, 2).$$

Consequently the nonendpoint active prefix stays in $(-2, -1)$, the nonendpoint active suffix stays in $(1, 2)$, while the endpoints remain at $-2, 2$; every prescribed middle label from the two-anchor construction can remain fixed.

The Q -chain statement is symmetric, with $y_B = -2$ and $y_Y = 2$. If $BY = C$, a complement path $q_i \cdots B - C - Y \cdots q_j$ forces $q_i \in J_A$: otherwise its path to A through Y would contain the chord CA . It also forces $q_j \notin J_X$: otherwise its path to X through B would contain the chord CX . With $c = 3t/4$,

$$R_Q = c + g = 2c + s = 1 + \frac{1}{2}t \in (1, 2),$$

so the same fixed-middle Ferrers construction works.

These two outward moves occur in disjoint extreme blocks. Cross cells having an extreme P - or Q -coordinate are already forced by the signs $x < 0, x > 1, y < 0, y > 1$; they do not use the central diagonal comparison. Thus the moves preserve all six portal memberships and every cross choice. They also preserve both chain totals $M = N = 2$. The twelve short-edge comparisons depend only on those totals and on the core weights, so their strict margins are unchanged. This closes all four main C-sector types, including the case in which both complement systems are active.

7.8. Boundary C-sector types reduce to adjacent-long K_4

Suppose $AB = C$ and $XY \neq C$. Then edge CY is legally expandable through A . To see this, orient any selected path containing CY from the C -side to the Y -side.

It cannot continue from C through X , because its X -to- Y subpath would be XCY , contrary to $XY \neq C$. Hence its C -side can only continue through B and possibly along Q . For any such vertex z , the selected z -to- Y subpath goes through $B - C - Y$. A z -to- A path through $Y - A$ would contain the incompatible direct- Q z -to- Y subpath; therefore it exits through B , and $P_{AB} = ACB$ forces it through $C - A$.

On the Y -side, any vertex z lies on Q and its selected C -to- z subpath goes through $C - Y$. If P_{Az} exited Q through B , then $P_{AB} = ACB$ would force its z -to- C subpath through $B - C$, contrary to the former subpath. Thus P_{Az} goes through $A - Y$. These are exactly the two side conditions for replacing every occurrence of CY by $C - A - Y$ consistently.

After deleting CY , suppress the degree-two vertex Y . The result is a K_4 subdivision on branch vertices A, B, C, X , with adjacent long edges

$$AX = P, \quad AB = A - Y - Q^{-1} - B$$

(one of them may have its minimal allowed length). Hence Theorem 6.6 strictly realizes the expanded system.

7.9. Triangle expansion and strict lifting

Let uvw be a triangle and $e = uv$. Replace every selected occurrence of uv by $u - w - v$. The resulting family on $G - e$ is consistent if and only if, for every selected path oriented through $u - v$,

- every old vertex on its u -side has its selected path to w through $u - w$; and
- every old vertex on its v -side has its selected path to w through $v - w$.

Necessity follows by taking subpaths incident with the inserted w . Sufficiency follows because all old-old subpaths are unchanged and these are the only new subpaths. Original inducedness ensures that w did not already lie on a path containing uv .

The same side conditions also preserve inducedness. If the inserted middle vertex had an additional neighbor on either side of the transformed path, neighborliness would prescribe that chord edge, contradicting the required side path through uw or vw .

Suppose the expanded system is strictly realized by w_0 on $G - e$. Set

$$w(e) = w_0(uw) + w_0(vw) - \varepsilon \quad (7.18)$$

and keep all other weights. Expand every competitor which uses e into a positive walk in $G - e$, and delete positive closed subwalks until a simple path remains. If the result is not the selected expanded path, strict realization by w_0 gives a positive margin; if it is the selected path, every deleted nonempty closed subwalk itself supplies a positive margin. Thus the expanded walk is strictly longer whenever the original competitor is distinct. If selected path and competitor both use e , their lengths both decrease by ε ; if both avoid it, neither changes; and if only the selected path uses e , its margin improves. For the remaining finite collection, in which only the competitor uses e , take ε below the minimum positive expanded-walk margin and below $w_0(uw) + w_0(vw)$. Then every comparison remains strict and all weights stay positive. Thus (7.18) strictly realizes the original compressed system.

7.10. Fixed expansions for the P and Q sectors

Only three compatibility observations are needed.

AC expansion rule. If $P_{AB} \neq ACB$ and $P_{XY} \neq XBQY$, then AC is legally expandable through Y .

To prove it, orient a selected path containing AC from its A -side to its C -side. It avoids Y . Its C -side cannot continue through B , since the resulting A - B subpath would be ACB . Hence it can only continue through X and possibly into P . Every P -vertex on the A -side must reach Y through AY : an exit through XCY would give the wrong subpath to C , while an exit through $XBQY$ would give the forbidden X - Y subpath. Every P -vertex on the C -side must reach Y through CY : an exit through A gives the wrong A -subpath, and an exit through $XBQY$ is again forbidden. These are exactly the two side conditions of the expansion criterion in Subsection 7.9.

CX expansion rule. If $P_{AB} = APXB$ and $P_{XY} \neq XCY$, then CX is legally expandable through B .

Here $P_{AB} = APXB$ forces $P_{AX} = P$. A selected path containing CX cannot continue from C through A , since it would prescribe $AX = ACX$, and cannot continue through Y , since it would prescribe $XY = XCY$. Thus C is its C -side endpoint. Its X -side can only

continue into P , and every P -vertex reaches B through XB as a subpath of $APXB$. Hence the expansion side conditions hold.

CY expansion rule. If $P_{AB} = AYQ^{-1}B$ and $P_{XY} \neq XCY$, then CY is legally expandable through A .

The first branch path forces $P_{BY} = Q$. A path containing CY cannot continue from C through B , since it would prescribe $BY = BCY$, and cannot continue through X , since it would prescribe $XY = XCY$. Thus C is its C -side endpoint. Its Y -side can only continue into Q , and every Q -vertex reaches A through YA as a subpath of $AYQ^{-1}B$. Again the side conditions hold.

These rules give a fixed expansion for every one of the twelve P- and Q-sector branch types:

AB	AX	BY	XY	fixed legal expansion
P	P	C	C	$AC \mapsto AYC$
P	P	C	P	$CX \mapsto CBX$
P	P	P	P	$CX \mapsto CBX$
P	P	Q	C	$AC \mapsto AYC$
P	P	Q	P	$CX \mapsto CBX$
P	P	Q	Q	$CX \mapsto CBX$
Q	C	Q	C	$AC \mapsto AYC$
Q	C	Q	Q	$CY \mapsto CAY$
Q	P	Q	C	$AC \mapsto AYC$
Q	P	Q	P	$CY \mapsto CAY$
Q	P	Q	Q	$CY \mapsto CAY$
Q	Q	Q	Q	$CY \mapsto CAY$

(7.19)

No finite enumeration is used in this table.

7.11. The edge-deleted topological types

Every expansion in (7.19), and the C-boundary expansion in Subsection 7.8, lands in the already closed adjacent-two-chain K_4 family:

deleted	suppressed vertex	K_4 branch vertices	two long edge paths
AC	A	B, C, X, Y	$BY = Q, \quad XY = X - P^{-1} - A - Y$
CX	X	A, B, C, Y	$AB = A - P - X - B, \quad BY = Q$
CY	Y	A, B, C, X	$AB = A - Y - Q^{-1} - B, \quad AX = P$

(7.20)

The two displayed paths in each row share one branch vertex. Apply the one-long-edge part of Theorem 6.6 exactly when one of those displayed paths is unsubdivided (in the first two rows this occurs when $n = 1$, and in the third when $m = 1$); otherwise apply its adjacent-two-long-edge part. Neighborliness forces the endpoint path of any unit chain to be its edge. Empty middle portal blocks and same-chain relations simply disappear when $m = 1$ or $n = 1$. None of the twelve rows requires an opposite-long or split-bridge family. (Deleting AY or BX would produce the split-bridge core, but the fixed table never needs those deletions.)

7.12. Proof of Theorem 7.1

The branch catalogue has eight C-sector, six P-sector and six Q-sector types.

- If $AB = C$ and $XY = C$, Subsections 7.2–7.7 give a simultaneous two-anchor construction, the complete scalar region, and the fixed-middle outward extensions. This covers all four main C types, including $C/P/Q/C$.
- If $AB = C$ and $XY \neq C$, Subsection 7.8 expands CY through A , applies the adjacent or one-long K_4 lemma, and lifts the strict metric by Subsection 7.9. This covers the other four C types.
- The twelve P/Q types use the fixed table (7.19), then (7.20), the adjacent or one-long K_4 lemma, and the strict lifting lemma.

The one-long and adjacent-two-chain K_4 realization results invoked here are contained in Theorem 6.6, and strict lifting is proved in Subsection 7.9. These twenty cases exhaust the catalogue. Rationality follows from Lemma 5.2. $\square$

8 The spoke–rim family $H_{m,n}^{sr}$

This section proves the spoke–rim realization theorem needed for the forbidden-minor characterization.

Theorem 8.1

For every $m, n \geq 1$, every neighborly consistent path system on $H_{m,n}^{sr}$ admits positive rational edge weights for which every prescribed path is the unique shortest path between its endpoints.

The proof is uniform in m and n . Its two parts are a legal triangle expansion for sixteen branch interfaces and a strict long-ear construction for the remaining eighteen interfaces.

8.1. Earlier tools

We use the following results from the preceding sections.

Corollary 8.2 (the two K_4 -subdivision families)

Every neighborly consistent path system on either of the following graphs has a positive rational strict realization:

1. a subdivision of K_4 with at most two long core edges, including the zero-long, one-long, adjacent-two-long, and opposite-two-long cases;
2. the split graph $P_4(p, 2, r)$, for $p, r \geq 2$.

Proof.

The two assertions are Theorems 6.6 and 6.9, respectively. $\square$

Lemma 8.3 (fixed-margin outward-safe extension)

Let

$$F \subseteq \{0, \dots, p\} \times \{p+1, \dots, m\}$$

be a nonempty expanding relation, with every active row nonempty. Suppose that every comparison in a fixed finite auxiliary family remains true when the active prefix, with its internal order fixed, is translated sufficiently far to the left; suppose also that every prescribed

portal prefix contains the active left block and avoids the active right block. A sufficient condition used below is that every available form contains an active-left label with coefficient $+x_i$ and an active-right label with coefficient $-x_j$, with the prescribed inequality pointing in the outward direction. Then, for every sufficiently large rational connector weight $D > 0$, there are rational labels

$$-M = x_0 < x_1 < \cdots < x_m = M$$

with $M > D$, preserving all previously prescribed strict orders and portal cuts, such that

$$(i, j) \in F \iff x_j - x_i > M + D \quad (8.1)$$

holds strictly in both directions. The symmetric suffix statement holds as well. Here M is an output of the construction; it is not an arbitrarily preassigned half-span.

Proof.

Apply Lemma 6.3 with its margin parameter $A = D$. It supplies increasing rational labels and a threshold T , preserves all outward-safe auxiliary comparisons and portal cuts, and gives

$$T - \frac{x_m - x_0}{2} = D.$$

Put $M = (x_m - x_0)/2$. Since $F \neq \emptyset$ and is expanding, $(0, m) \in F$, so $x_m - x_0 > T = M + D$. Hence $M > D$, and the threshold is $T = M + D$, as required. Translate the complete chain label set and all of its portal symbols by a common constant to obtain $x_0 = -M, x_m = M$. Differences and cut orders are unchanged; the remaining absolute scalar parameters are chosen only after this centering. $\square$

8.2. The graph and its scalar data

Write the two possibly subdivided paths as

$$P : A = p_0 p_1 \cdots p_m = X, \quad Q : B = q_0 q_1 \cdots q_n = Y. \quad (8.2)$$

The six remaining edges are

$$AB, \quad BC, \quad BX, \quad CX, \quad AY, \quad CY. \quad (8.3)$$

Thus $H_{m,n}^{sr}$ is obtained from the graph $U_{m,n}$ defined in Section 2 by deleting AC . Put

$$\begin{aligned} a &= w(AB), & c &= w(BC), & d &= w(BX), \\ e &= w(CX), & f &= w(AY), & g &= w(CY), \end{aligned} \quad (8.4)$$

and set $t = 2a$. Denote the total weight of P by M , and use centered labels

$$-M = x_0 < x_1 < \cdots < x_m = M, \quad w(p_{i-1} p_i) = \frac{x_i - x_{i-1}}{2}. \quad (8.5)$$

The portal scalars used below are

$$\beta = d - a, \quad \alpha = f - a, \quad \nu = g - c, \quad (8.6)$$

$$\xi = e + g - d, \quad \eta = e + g - f, \quad \delta = \xi - \alpha = \eta - \beta. \quad (8.7)$$

Finally define

$$\gamma = e - w(P_{AC}). \quad (8.8)$$

The prescribed A -to- C route is one of three possibilities, and

$$w(P_{AC}) = \begin{cases} a + c, & P_{AC} = ABC, \\ f + g, & P_{AC} = AYC, \\ M + e, & P_{AC} = APXC. \end{cases} \quad (8.9)$$

We write, for example, $XY = XCY$ to mean that XCY is the prescribed X -to- Y path.

8.3. Three path-system lemmas

Lemma 8.4 (inducedness)

Every prescribed path in a neighborly consistent path system is induced.

Proof.

If two nonconsecutive vertices u, v of a prescribed path were adjacent, neighborliness would give $P_{uv} = uv$, whereas consistency would give the longer u -to- v subpath. $\square$

For adjacent endpoints u, v , call a different u -to- v path endpoint-chordless if it is induced in $G - uv$. For nonadjacent endpoints, endpoint-chordless means induced in G .

Lemma 8.5 (endpoint-chord reduction)

Let w be a positive edge weighting of a neighborly consistent path system. If every prescribed path is strictly shorter than every distinct endpoint-chordless competitor, then the weighting strictly realizes the whole system.

Proof.

We first prove that every edge uv is strictly shorter than every different u -to- v path. Induct on the number of edges of the competitor. An endpoint-chordless competitor is covered by the hypothesis. Otherwise it contains an internal chord $xy \neq uv$. Its proper x -to- y subpath has fewer edges, so by induction it is strictly longer than the edge xy . Replacing that subpath by xy , and deleting any closed subwalk thereby created, strictly shortens the competitor. The induction terminates at an endpoint-chordless path.

Now let $R \neq P_{uv}$ be an arbitrary simple competitor. Repeatedly replace a subpath cut off by an internal chord with that chord. The edge case just proved makes every replacement strict. If the terminal path is an endpoint-chordless competitor distinct from P_{uv} , use the hypothesis. If it is P_{uv} , at least one replacement was made because $R \neq P_{uv}$, and the accumulated inequality is already strict. $\square$

Lemma 8.6 (legal triangle expansion and strict lifting)

Let uvw be a triangle. Uniformly replace every prescribed occurrence of uv by uvw . The resulting paths form a consistent path system on $G - uv$ if and only if, for every selected occurrence

$$R = R_u u, v R_v,$$

the following side conditions hold:

$$\begin{aligned} z \in R_u &\implies P_{zw} \text{ reaches } w \text{ through } uw, \\ z \in R_v &\implies P_{zw} \text{ reaches } w \text{ through } vw. \end{aligned} \quad (8.10)$$

If the expanded system has a positive rational strict realization w_0 , then the original system has one.

Proof.

By Lemma 8.4, a prescribed path containing uv cannot contain w elsewhere. Hence expansion preserves simplicity. Old-old subpaths transform as before; the only new subpaths have one endpoint at the inserted copy of w , and their consistency is exactly (8.10). This proves the first assertion.

For the lifting assertion, keep the old weights and put

$$w_\varepsilon(uv) = w_0(uw) + w_0(wv) - \varepsilon. \quad (8.11)$$

Expand the occurrence of uv , if any, in a competitor R , and call the resulting walk $E(R)$. For a prescribed path S ,

$$\ell_{w_\varepsilon}(R) - \ell_{w_\varepsilon}(S) = \ell_{w_0}(E(R)) - \ell_{w_0}(E(S)) - \varepsilon(\mathbf{1}_{uv \in R} - \mathbf{1}_{uv \in S}). \quad (8.12)$$

Delete positive closed subwalks from $E(R)$. Strict realization by w_0 leaves a nonnegative difference, with equality only if the resulting simple path is $E(S)$. If no closed subwalk was deleted and equality holds, injectivity of expansion gives $R = S$, except for the impossible case in which a prescribed path avoids uv but contains uvw , which would have chord uv . Thus every case in which the epsilon term can decrease a margin has a positive w_0 -margin. There are finitely many simple competitors, so a positive rational ε , smaller than all these margins and than $w_0(uw) + w_0(wv)$, preserves strictness and positivity. $\square$

8.4. Exhaustion of the 34 branch interfaces

Lemma 8.7 (branch-route catalogue)

The four branch paths have the following possibilities:

$$\begin{array}{l|l} AC & ABC, AYC, APXC, \\ AX & ABX, P, AYCX, \\ BY & BAY, BCY, Q, \\ XY & XBQY, XCY, XPAY, XBAY. \end{array} \quad (8.13)$$

Among these four choices, consistency imposes exactly the following implications:

$$\begin{array}{l|l} AC = APXC & AX = P, \\ XY = XBQY & BY = Q, \\ XY = XPAY & AX = P, \\ XY = XBAY & AX = ABX \text{ and } BY = BAY, \\ AX = AYCX & XY = XCY, \\ AC = ABC & AX \neq AYCX. \end{array} \quad (8.14)$$

Consequently every neighborly consistent system has one of 34 branch signatures.

Proof.

The internal vertices of P and Q have degree two. A branch path that enters one of these chains must either leave at the same end, in which case it contains a forbidden repeated vertex or a chord, or traverse the unique indicated segment. Applying Lemma 8.4 to the finite kernel gives exactly the routes in (8.13).

It remains to justify both the implications and their completeness. Consistency of two prescribed paths is equivalent to agreement on every nonempty common subpath. The only

pairs of branch paths that can have a disconnected intersection, or can prescribe different routes on a common pair of branch vertices, are listed below.

pair	incompatible choices	forced condition
(AC, AX)	$AC = APXC, AX \neq P;$ $AC = ABC, AX = AYC X;$	$AC = APXC \Rightarrow AX = P;$ $AC = ABC \Rightarrow AX \neq AYC X;$
(AX, XY)	$XY = XPAY, AX \neq P;$ $XY = XBAY, AX \neq ABX;$	$XY = XPAY \Rightarrow AX = P;$ $XY = XBAY \Rightarrow AX = ABX;$
(BY, XY)	$AX = AYC X, XY \neq XCY;$ $XY = XBQY, BY \neq Q;$ $XY = XBAY, BY \neq BAY;$	$AX = AYC X \Rightarrow XY = XCY;$ $XY = XBQY \Rightarrow BY = Q;$ $XY = XBAY \Rightarrow BY = BAY.$

(8.15)

For example, $AC = APXC$ contains the full A -to- X chain, so its prescribed A -to- X subpath must be P . If $AC = ABC$ and $AX = AYC X$, the two paths meet at A and C but do not share the same A -to- C subpath. The remaining entries follow identically by reading the named route as a prescribed subpath.

For the other three unordered pairs among AC, AX, BY, XY , every intersection is either a single vertex or one of the named common paths $AB, AY, BC, CY, BX, CX, P, Q$, and the routes agree on it. After the conditions in (8.15) are imposed, the same is true for the three pairs shown in the table. Hence there is no further branch-level consistency condition. Interior entrance cuts may remove a branch signature from a particular system, but cannot create a new one.

Counting the signatures satisfying (8.14) gives

XY	compatible choices	#
$XC Y$	$AC = ABC : AX \in \{ABX, P\}, BY \in \{BAY, BCY, Q\};$ $AC = AYC : AX \in \{ABX, P, AYC X\}, BY \in \{BAY, BCY, Q\};$ $AC = APXC : AX = P, BY \in \{BAY, BCY, Q\}$	$6 + 9 + 3$
$XPAY$	$AC \in \{ABC, AYC, APXC\}, AX = P, BY \in \{BAY, BCY, Q\}$	9
$XBQY$	$AC \in \{ABC, AYC\} : AX \in \{ABX, P\}, BY = Q;$ $AC = APXC : AX = P, BY = Q$	$2 + 2 + 1$
$XBAY$	$AC \in \{ABC, AYC\}, AX = ABX, BY = BAY$	2.

(8.16)

Thus there are $18 + 9 + 5 + 2 = 34$ signatures. $\square$

Lemma 8.8 (the 16 + 18 partition)

The 34 signatures are the disjoint union of the following eighteen direct interfaces:

XY	direct interfaces	#
$XC Y$	$AC = ABC, AX \in \{ABX, P\}, BY \in \{BAY, BCY, Q\}$	6
$XPAY$	$AC \in \{ABC, AYC, APXC\}, AX = P, BY \in \{BAY, BCY, Q\}$	9
$XBQY$	$AC = ABC, AX \in \{ABX, P\}, BY = Q$	2
$XBAY$	$AC = ABC, AX = ABX, BY = BAY$	1,

(8.17)

and the following sixteen expansion interfaces:

XY	AC	AX	BY	$\#$
XCY	AYC	$ABX, P, AYCX$	BAY, BCY, Q	9
XCY	$APXC$	P	BAY, BCY, Q	3
$XBQY$	AYC	ABX, P	Q	2
$XBQY$	$APXC$	P	Q	1
$XBAY$	AYC	ABX	BAY	1.

(8.18)

Proof.

Partition the rows of (8.16) according as $AC = ABC$, except that all $XPAY$ rows are placed in the direct class. Reading the four XY blocks gives (8.17) and (8.18), whose sizes are respectively $6 + 9 + 2 + 1 = 18$ and $9 + 3 + 2 + 1 + 1 = 16$. The displayed conditions are disjoint and their union is (8.16). $\square$

8.5. The sixteen expansion interfaces

Proposition 8.9 (legal expansion of BC)

For every interface in (8.18), uniformly replacing

$$BC \mapsto BXC \tag{8.19}$$

gives a neighborly consistent path system on $H_{m,n}^{sr} - BC$.

Proof.

By Lemma 8.6, it is enough to check every selected occurrence of BC :

$$\begin{aligned} z \text{ on its } B\text{-side} &\implies P_{zX} \text{ reaches } X \text{ through } BX, \\ z \text{ on its } C\text{-side} &\implies P_{zX} \text{ reaches } X \text{ through } CX. \end{aligned} \tag{8.20}$$

Throughout (8.18), $AC \neq ABC$. Orient a selected occurrence of BC from B to C . The vertex X cannot occur elsewhere on this path: near B or C , one of BX, CX is a chord, and the same chord argument applies at every more distant occurrence.

Apart from B , a vertex on the B -side must be an internal Q -vertex whose segment to B is the direct Q -prefix. Indeed, A would make ABC the selected A -to- C subpath; a P -interior vertex can avoid X only by passing through A ; and Y gives the chord CY . A Q -vertex cannot approach B through Y for the same reason. Hence every such vertex satisfies

$$P_{q_iC} = q_iQBC. \tag{8.21}$$

Apart from C , every C -side vertex lies on CYQ . An occurrence of A , or of a P -interior vertex, creates the chord AB . Thus

$$P_{Bq_i} = BCYQq_i. \tag{8.22}$$

If $BY \neq BCY$, the C -side stops at C , since otherwise Y would force the selected subpath BCY .

There are three possible XY routes in (8.18).

1. If $XY = XCY$, a B -side q_i cannot reach X through Y : consistency with $P_{YX} = YCX$ would change its q_i -to- C subpath in (8.21). It therefore exits through B and uses BX . A C -side q_i cannot exit through B by (8.22), and hence uses YCX .

2. If $XY = XBQY$, then $BY = Q$. The C -side condition is vacuous beyond C , and every Q -vertex reaches X through the selected subpath of $XBQY$, hence through BX .
3. If $XY = XBAY$, then $BY = BAY \neq BCY$, so again the C -side stops at C . A B -side q_i cannot reach X through $YABX$, since this would contradict the direct prefix in (8.21); it uses BX .

This proves (8.20) for every occurrence, including arbitrary interior portal cuts and arbitrary same-chain choices. $\square$

Corollary 8.10

Every interface in (8.18) has a positive rational strict realization.

Proof.

After deleting BC , the vertex C has degree two on XCY . With

$$a_0 = A, \quad a_p = X, \quad s = C, \quad b_0 = Y, \quad b_r = B,$$

the graph is $P_4(m, 2, n)$, with kernel path $A - X - Y - B$. Apply Corollary 8.2 according to

(m, n)	case
$m, n \geq 2$	split $P_4(m, 2, n)$,
$m = 1 < n$	adjacent at-most-two-long K_4 ,
$n = 1 < m$	adjacent at-most-two-long K_4 ,
$m = n = 1$	one-long K_4 .

(8.23)

Lemma 8.6 then restores BC while preserving all strict inequalities. $\square$

8.6. A strict long-ear extension theorem

The next lemma is stated for an arbitrary finite core. This form is useful because the core remaining after Q is removed still contains the whole chain P . A terminal-potential extension for non-strict metrizable appears in the K_4 -subdivision companion manuscript [11]; Lemma 8.11 supplies the positive margins and unique-geodesic hypotheses required here.

Lemma 8.11 (strict long-ear extension)

Let J be a finite graph with terminals s, t , and attach a chain

$$C : s = z_0^*, z_1^*, \dots, z_q^* = t. \tag{8.24}$$

Let a consistent path system on $J \cup C$ be given.

If $P_{st} \neq C$, suppose that a positive rational metric w strictly realizes the restriction to J . If $P_{st} = C$, contract C to a proxy edge st , write $K = J + st$, and suppose that w strictly realizes the contracted system on K , with st its unique s -to- t geodesic. In the relevant core put

$$\psi(v) = d_w(t, v) - d_w(s, v). \tag{8.25}$$

For every old vertex v , consistency gives an actual entrance cut k_v : the prescribed z_i^* -to- v path enters J through s for $i \leq k_v$, and through t for $i > k_v$. Suppose that

$$k_u < k_v \implies \psi(u) < \psi(v). \tag{8.26}$$

Then C has positive rational edge weights that strictly realize all old-ear and ear-ear paths while preserving every prescribed old core path. In the proxy case, C replaces the edge st .

Proof.

We first justify the cut notation. If two ear vertices z_i^*, z_k^* , with $i < k$, enter a prescribed path to v through different terminals, consistency on their common subpaths prevents a later switch back to the first terminal. Thus the indices entering through s form a prefix, possibly empty or full, and k_v is well-defined.

Suppose first that $P_{st} \neq C$. Write

$$D = d_w(s, t), \quad p = k_t, \quad h = k_s.$$

The prescribed paths to the two terminals, and then consistency with their subpaths to an arbitrary old vertex, give

$$0 \leq p \leq k_v \leq h \leq q - 1 \quad (8.27)$$

for every old vertex v . The triangle inequality gives $-D \leq \psi(v) \leq D$.

Choose rational right labels

$$z_{p+1} < z_{p+2} < \cdots < z_q$$

so that every potential block with cut $k > p$ lies in the open gap (z_k, z_{k+1}) . For the cut- p block choose only its upper endpoint at this stage. Require

$$-D < z_{p+1}, \quad z_q > D, \quad (8.28)$$

and, if $h > p$, require $z_h < D < z_{h+1}$. If $h = p$, require $D < z_{p+1}$. Hypothesis (8.26) is exactly the condition needed to place the distinct cut blocks in this order; potentials with the same cut may be ordered arbitrarily inside the same open gap.

Let F consist of the pairs $i < j$ whose prescribed z_i^* -to- z_j^* path leaves the ear through s , crosses the core from s to t , and returns through t . Consistency gives

$$F \subseteq \{0, \dots, p\} \times \{h + 1, \dots, q\}. \quad (8.29)$$

It also makes F expanding: extending the left endpoint toward s , or the right endpoint toward t , preserves the selected core-crossing subpath. Every row $0 \leq i \leq p$ is nonempty because the prescribed path from z_i^* to t enters through s . If

$$r_i = \min\{j : (i, j) \in F\},$$

then

$$r_0 = h + 1, \quad r_0 \leq r_1 \leq \cdots \leq r_p. \quad (8.30)$$

If $h > p$, choose increasing rational numbers $H_0, \dots, H_p$ with

$$H_0 = D, \quad z_{r_i-1} < H_i < z_{r_i}. \quad (8.31)$$

If $h = p$, set $H_0 = D$; for $i > 0$ with $r_i = p + 1$, choose $D < H_i < z_{p+1}$, and use (8.31) for the remaining indices. Repeated values of r_i cause no problem because all the intervals are open. Put

$$T = D + z_q, \quad z_i = H_i - T \quad (0 \leq i \leq p). \quad (8.32)$$

Then $z_0 = -z_q$, the labels are strictly increasing, $z_p < -D < z_{p+1}$, and

$$(i, j) \in F \iff z_j - z_i > T. \quad (8.33)$$

Give the ear edge $z_{i-1}^* z_i^*$ weight

$$\frac{z_i - z_{i-1}}{2}. \quad (8.34)$$

For an old vertex v , the cost through s minus the cost through t from z_i^* equals $z_i - \psi(v)$. Hence the route enters through s precisely when $z_i < \psi(v)$, so all actual entrance cuts are strict.

For $i \leq p < h + 1 \leq j$, the direct ear segment has weight $(z_j - z_i)/2$, while the route through J has weight

$$\frac{z_i - z_0}{2} + D + \frac{z_q - z_j}{2}.$$

The latter minus the former is $T - (z_j - z_i)$, proving the desired choice by (8.33). If a pair in the rectangle (8.29) is not in F , the open choice of H_i gives the strict reverse inequality. If $i > p$, then $z_i > -D$, and hence $z_j - z_i < z_q + D = T$. If $j \leq h$, then $z_j < D$, and $z_j - z_i < D + z_q = T$. Thus every pair outside the rectangle also selects its direct segment strictly. A competitor that leaves and returns through the same terminal contains a positive closed subwalk and is longer still.

The total ear weight is $z_q > D$. Because every internal ear vertex has degree two, an old-old simple path using an ear edge must enter at one terminal and leave at the other, and therefore traverses the full ear. Replacing it by the old s -to- t geodesic strictly shortens it. Thus all old core paths are preserved.

Now suppose $P_{st} = C$. Let the proxy edge have weight L in the strict metric on K . The two extreme cut blocks satisfy

$$k_v = -1 \implies \psi(v) = -L, \quad k_v = q \implies \psi(v) = L. \quad (8.35)$$

Choose

$$-L = z_0 < z_1 < \dots < z_q = L \quad (8.36)$$

with every nonextreme potential block in its prescribed open cut gap, and again use (8.34). The chain has total weight L , so old walks using the whole chain correspond weight-for-weight to walks using the proxy. The strict old metric is therefore preserved. The equalities in (8.35) are the two forced extreme cuts; all nonextreme entrance comparisons are strict.

No ear pair can prescribe a core detour, since that would give an s -to- t prescribed subpath different from C . Every proxy-avoiding s -to- t route is strictly longer than L , because the proxy is the unique s -to- t geodesic. Replacing any direct subsegment of C by a core route therefore increases its weight. All choices lie in rational open intervals. $\square$

8.7. Removing Q : terminal potentials and actual cuts

We now consider the eighteen interfaces in (8.17). Delete the internal vertices of Q . If $P_{BY} \neq Q$, no prescribed path between two remaining vertices can use Q , because it would

contain the whole subpath BQY and consistency would force $P_{BY} = Q$. Denote the remaining graph by G_P ; this is the non-proxy case.

If $P_{BY} = Q$, contract Q to a proxy edge BY and write

$$G_P^+ = G_P + BY. \quad (8.37)$$

The contracted paths are again neighborly and consistent.

For a core vertex v , let $\kappa(v)$ be its actual entrance cut on the original Q : the vertices on the B -side enter through B , and those on the Y -side enter through Y . In a core metric define

$$\psi(v) = d(Y, v) - d(B, v). \quad (8.38)$$

Because the six short edges are prescribed,

$$\psi(A) = \alpha, \quad \psi(C) = \nu, \quad (8.39)$$

and

$$\psi(B) = D, \quad \psi(Y) = -D, \quad (8.40)$$

where $D = d(B, Y)$ in the non-proxy case and $D = w(BY)$ in the proxy case.

Lemma 8.12 (four clamp identities)

For the labels in (8.5), the terminal potential along P has one of the following forms.

1. If $XY = XCY$ and $\beta < \eta$, then

$$\psi(p_i) = \begin{cases} \alpha, & x_i \leq \beta, \\ \alpha + x_i - \beta, & \beta \leq x_i \leq \eta, \\ \xi, & x_i \geq \eta. \end{cases} \quad (8.41)$$

If $\eta < \beta$, then

$$\psi(p_i) = \begin{cases} \alpha, & x_i \leq \eta, \\ \alpha + \eta - x_i, & \eta \leq x_i \leq \beta, \\ \xi, & x_i \geq \beta. \end{cases} \quad (8.42)$$

Here $\xi - \alpha = \eta - \beta$. If $\beta = \eta$, the two plateaux coincide.

2. If $XY = XPAY$, then

$$\psi(p_i) = \begin{cases} \alpha, & x_i \leq \beta, \\ \alpha + x_i - \beta, & x_i \geq \beta, \end{cases} \quad \psi(X) = M + \alpha - \beta. \quad (8.43)$$

3. If $XY = XBQY$, let $L = w(BY)$ and put

$$R_0 = L + \beta - \alpha.$$

Then

$$\psi(p_i) = \begin{cases} \alpha, & x_i \leq \beta, \\ \alpha + x_i - \beta, & \beta \leq x_i \leq R_0, \\ L, & x_i \geq R_0. \end{cases} \quad (8.44)$$

The right plateau is a full-cut block and $\psi(X) = L = \psi(B)$.

4. If $XY = XBAY$, put

$$D = t + \alpha, \quad H = t + \beta.$$

Then

$$\psi(p_i) = \begin{cases} \alpha, & x_i \leq \beta, \\ \alpha + x_i - \beta, & \beta \leq x_i \leq H, \\ D, & x_i \geq H. \end{cases} \quad (8.45)$$

Thus $\psi(X) = D = \psi(B)$.

Proof.

In the main sector,

$$d(B, p_i) = \min \left\{ \frac{M + x_i}{2} + a, \frac{M - x_i}{2} + d \right\},$$

whose switch is $x_i = \beta$, while

$$d(Y, p_i) = \min \left\{ \frac{M + x_i}{2} + f, \frac{M - x_i}{2} + e + g \right\},$$

whose switch is $x_i = \eta$. Subtracting on the three resulting intervals gives (8.41) and (8.42). In the via- P sector, consistency with $XPAY$ retains only the first Y -expression, giving (8.43). In the via- Q sector, the second Y -expression is $(M - x_i)/2 + d + L$, whose switch with the left expression is R_0 . In the special sector it is $(M - x_i)/2 + d + a + f$, whose switch is $H = t + \beta$. $\square$

Lemma 8.13 (actual-cut order)

On a constant piece of (8.41)–(8.45),

$$\kappa(p_i) = \kappa(A) \quad \text{or} \quad \kappa(p_i) = \kappa(X), \quad (8.46)$$

according as it is the left or right plateau. On a middle piece the cuts change in the same direction as the displayed potential. Consequently, the scalar data can and will be chosen so that

$$\kappa(u) < \kappa(v) \implies \psi(u) < \psi(v) \quad (8.47)$$

for all core vertices. Vertices with the same cut need only lie in the same open potential gap; their potentials need not be equal.

Proof.

Suppose that the selected B -to- p_i and Y -to- p_i paths merge at the same endpoint $E \in \{A, X\}$ and then share EPp_i . For every q_j , consistency makes the q_j -to- E subpath of $P_{q_j p_i}$ equal to $P_{q_j E}$. Thus p_i and E have the same entrance decision for every q_j , proving (8.46).

On a middle piece the terminal paths enter P from opposite ends. Take $i < i'$. In an increasing sector, if $\kappa(p_i) > \kappa(p_{i'})$, choose q_j strictly between the two cuts. The selected path from q_j , through X , to p_i contains $p_{i'}$, but its q_j -to- $p_{i'}$ subpath then uses the opposite entrance from the prescribed one. This contradicts consistency. Reversing the directions gives the decreasing case (8.42). Hence cuts and potentials have the same strict direction on every middle piece.

The same separating-vertex argument covers a transition between a plateau and an adjacent middle piece. For example, in an increasing sector let p_i lie on the left plateau and let $p_{i'}$, with $i < i'$, lie on the middle piece. If $\kappa(p_i) > \kappa(p_{i'})$, choose q_j with $\kappa(p_{i'}) < j \leq \kappa(p_i)$. Then $P_{q_j p_{i'}}$ enters the core through Y , reaches P through A , and contains p_i . Its q_j -to- p_i subpath therefore enters the core through Y , whereas $P_{q_j p_i}$ enters through B , a contradiction. The middle-right transition is analogous, and reversing B and Y gives the decreasing sector. Thus no reversal occurs at a plateau-middle boundary.

At the endpoints this yields

sector	forced cut relation
main	$\kappa(A) < \kappa(X) \Rightarrow \beta < \eta, \quad \kappa(X) < \kappa(A) \Rightarrow \eta < \beta,$
via- P	$\kappa(A) \leq \kappa(X),$
via- Q and special	$\kappa(X) = \kappa(B).$

(8.48)

In the main sector XCY also gives $\kappa(C) \leq \kappa(X)$. Distinct cut blocks can be assigned disjoint rational open intervals in their order, and equal-cut vertices can be finely ordered in one interval. The affine pieces in (8.41)–(8.45) can then be inverted inside those intervals: the main sector uses the two switches β, η , the via- P sector uses β , and the via- Q and special sectors use R_0 and H , respectively. The remaining P -to- C switch creates no extra cycle. If $AC = ABC$, consistency gives $\gamma < \beta$ after separating coincident gaps. In the main sector it also gives $\gamma < \eta$; in the via- P , via- Q , and special sectors $\gamma < \eta$ follows, respectively, from $\eta > M > \gamma$, from $\xi > L > |\alpha|$ and $\eta - \beta = \xi - \alpha$, and from $\delta > t > 0$. If $AC = AYC$, place γ and β in their prescribed, possibly equal, portal gaps; by compressing the finitely many intervening gaps their signed difference can be made arbitrarily small without changing its prescribed order. If $AC = APXC$, (8.8)–(8.9) force $\gamma = -M$. The scalar construction in the next subsection makes these choices inside rational open intervals and simultaneously enforces positivity and all route inequalities. $\square$

8.8. Strict realization of the eighteen direct cores

Proposition 8.14

For every interface in (8.17), the restricted path system on G_P , or on G_P^+ in the proxy case, has a positive rational strict realization such that (8.47) holds.

Proof.

We first fix the required order type on P , then solve for the six short-edge weights, and finally perform the numerical same-chain extension inside each of the six required open cones.

8.8.1. The chain order and same- P detours

Every set of P -vertices that exits a prescribed path through one end of P is a prefix or suffix, so all portal thresholds lie in the same linear label order. To realize the actual Q -cut preorder, leave a rational open interval between every two distinct cut blocks and place equal-cut vertices in any fine order inside their common interval. In the main sector choose α and ξ in the intervals assigned to A and X , impose $\eta - \beta = \xi - \alpha$, and invert the appropriate middle part of (8.41) or (8.42) to obtain the labels x_i . In the via- P sector choose $\alpha < \psi(X)$,

put $\beta = M + \alpha - \psi(X)$, and invert (8.43). In the via- Q and special sectors use the switch points $R_0 = L + \beta - \alpha$ and $H = t + \beta$, respectively. Place $\nu = \psi(C)$ in the interval assigned to $\kappa(C)$, and place the independent P -to- C threshold γ in its prescribed portal gap. Lemma 8.13 shows that these requirements have no cyclic order. All choices are made in open rational intervals; the macro constructions in Subsection 8.8.3 leave those intervals free.

If $AX = P$, every prescribed subpath of P is direct and there is no same- P detour relation.

If $AX = ABX$, the pairs that detour through the exterior connector ABX form an expanding relation. Its active left block is contained in every relevant prefix, and its active right block is disjoint from every such prefix. Indeed, if an active left endpoint p_i used a route first leaving toward X , that route would pass through the matching active right endpoint p_j and contain the direct segment $p_i P p_j$, contradicting the prescribed detour. Thus active left labels occur only with coefficient $+x_i$; symmetrically, active right labels occur only with coefficient $-x_j$. Lemma 8.3 applies with

$$K_P^B = w(ABX) = t + \beta. \quad (8.49)$$

The other exterior connector has weight

$$K_P^Y = w(AYCX) = t + 2\alpha + \eta. \quad (8.50)$$

The inequalities below make it strictly longer whenever it is not selected. All scalar symbols in what follows refer to the centered gauge. The numerical use of Lemma 8.3 is deferred until the macro parameters have been fixed in Witnesses 1, 5, and 6. In those three constructions $\beta = K$ is chosen after all parameters that bound it from below, so we first fix the centered targets t, K , put $D = t + K$, and use the common translation in Lemma 8.3 so that the centered portal symbol is $\beta = K$. The short-edge weights are reconstructed from these final centered values. Thus $D = t + \beta$ can be made sufficiently large for Lemma 8.3. The active left and right endpoints lie on terminal-potential plateaux; translating an active tail therefore preserves the actual-cut order. The lemma then supplies, rather than presupposes, a half-span $M > D$. The remaining span requirements are automatic: in Witness 1, $D - \eta = t - \delta > 0$; in Witness 5, $D - R_0 = t - L + \alpha = T_0 > 0$; and Witness 6 requires only $M > D$.

8.8.2. Scalar reconstruction

The equations below are obtained by solving (8.6)–(8.9); they define the six short-edge weights after $\alpha, \beta, \nu, \xi, \eta, \gamma, t, M$, and h where needed, have been chosen. In every sector we impose

$$M > 0, \quad -M < \beta < M, \quad M + \eta > 0, \quad \nu < \xi. \quad (8.51)$$

If $AC = ABC$, then

$$\begin{aligned} d &= a + \beta, & f &= a + \alpha, \\ 2g &= \alpha + \eta + \nu - \gamma, & 2c &= \alpha + \eta - \nu - \gamma, \\ 2e &= t + \alpha + \eta - \nu + \gamma. \end{aligned} \quad (8.52)$$

Choose the open gaps so that

$$\gamma < \beta, \quad \gamma < \eta, \quad \nu < \xi, \quad \alpha + \nu > 0, \quad 2\alpha + \eta - \gamma > 0, \quad M > \gamma - \alpha - \nu, \quad M + \gamma > 0, \quad (8.53)$$

and then take

$$t > \Lambda_{ABC}(M) := \max\{0, -2\beta, -2\alpha, -\alpha - \eta + \nu - \gamma, -\alpha - \nu, -\beta - \gamma, \nu - \alpha, -\alpha - M + \nu - \gamma\}. \quad (8.54)$$

If $AC = AYC$, then

$$2g = \eta - \gamma, \quad 2c = \eta - \gamma - 2\nu, \quad 2e = t + 2\alpha + \eta + \gamma. \quad (8.55)$$

Put

$$T = \beta - \gamma - \alpha - \nu, \quad \Omega = (\beta - \gamma) + (\xi - \nu). \quad (8.56)$$

Require

$$\begin{aligned} \alpha + \nu < 0, \quad T > 0, \quad \Omega > 0, \quad \eta - \gamma > 0, \quad \eta - \gamma - 2\nu > 0, \\ M > \gamma, \quad M + \gamma > 0, \quad \eta - \gamma - \alpha - \nu > 0, \end{aligned} \quad (8.57)$$

and

$$t > \Lambda_{AYC}(M) := \max\{0, -2\beta, -2\alpha, -2\alpha - \eta - \gamma, -\alpha - \nu, -\alpha - \beta - \nu - \gamma, \nu - \alpha, -2\alpha - M - \gamma\}. \quad (8.58)$$

If $AC = APXC$, then $\gamma = -M$. Choose $h \in \mathbb{Q}$ and put

$$g = a + h, \quad e = \alpha + \eta - h, \quad c = a + h - \nu. \quad (8.59)$$

First require

$$2h < \min\{2\alpha + 2\eta, \alpha + \beta + \eta + \nu, 2\alpha + \eta + M\}, \quad (8.60)$$

and then

$$\begin{aligned} t > \Lambda_{PXC}(M, h) := \max\{0, -2\beta, -2\alpha, -2h, 2\nu - 2h, \nu - \alpha - 2h, \\ -\alpha - \nu, \xi + \nu - 2h, \eta - M - 2h, \alpha + \nu - 2h, \\ \nu - \alpha, M + \alpha + \eta + \nu - 2h, M + \eta - 2h\}. \end{aligned} \quad (8.61)$$

The two exterior B -to- Y connectors have total weights

$$K_Q^A = t + \alpha \quad (8.62)$$

and

$$K_Q^C = \begin{cases} \alpha + \eta - \gamma, & AC = ABC, \\ \eta - \gamma - \nu, & AC = AYC, \\ t + 2h - \nu, & AC = APXC. \end{cases} \quad (8.63)$$

The endpoint choices amount to the following additional strict inequalities:

interface	conditions
main ABC	$M > \eta, \quad t > \delta;$
$AX = ABX$	$M > t + \beta;$
$AX = P$	$M < \min\{t + \beta, t + 2\alpha + \eta\};$
$BY = BAY$	$t < \eta - \gamma;$
$BY = BCY$	$t > \eta - \gamma;$
$BY = Q$	$\max_{v \neq B, Y} \psi(v) < L < \min\{t + \alpha, \alpha + \eta - \gamma\};$
via- P	$\eta > M, \quad t > M - \beta, \quad t > M - 2\alpha - \eta;$
$ABC, BY = BAY/BCY$	$t < \eta - \gamma \text{ / } t > \eta - \gamma;$
$AYC, BY = BAY/BCY$	$t < \delta + T \text{ / } t > \delta + T;$
$APXC, BY = BAY/BCY$	$2h > \alpha + \nu \text{ / } 2h < \alpha + \nu;$
$BY = Q$	$\max_{v \neq B, Y} \psi(v) < L < \min\{K_Q^A, K_Q^C\};$
via- Q, ABC	$\xi > L, \quad M > L + \beta - \alpha, \quad t > L - \alpha;$
$AX = ABX$	$M > t + \beta;$
$AX = P$	$L + \beta - \alpha < M < t + \beta < t + 2\alpha + \eta;$
special ABC	$t < \delta, \quad M > t + \beta, \quad BY = BAY.$

(8.64)

The following identities record the positive widths between bounds that will be used:

$$(\eta - \gamma) - \delta = \beta - \gamma, \quad (8.65)$$

$$\begin{aligned} (\delta + T) - \delta &= T, & (\delta + T) - (\nu - \alpha) &= \eta - \gamma - 2\nu, \\ (\delta + T) - (-2\alpha) &= \Omega, & (\delta + T) - (-\alpha - \nu) &= \eta - \gamma, \end{aligned} \quad (8.66)$$

$$\begin{aligned} (\eta - \gamma) - (M - \beta) &= (\eta - M) + (\beta - \gamma), \\ (\delta + T) - (M - \beta) &= (\eta - M) + T, \\ (t + \beta) - (L + \beta - \alpha) &= t + \alpha - L, \\ (\alpha + \eta - \gamma) - L &= (\xi - L) + (\beta - \gamma). \end{aligned} \quad (8.67)$$

8.8.3. The eighteen-to-six witness map

The following table is the complete assignment of the eighteen interfaces to the six scalar constructions below.

witness	XY	AC	AX	BY	#
1	XCY	ABC	ABX, P	BAY, BCY, Q	6
2	$XPAY$	ABC	P	BAY, BCY, Q	3
3	$XPAY$	AYC	P	BAY, BCY, Q	3
4	$XPAY$	$APXC$	P	BAY, BCY, Q	3
5	$XBQY$	ABC	ABX, P	Q	2
6	$XBAY$	ABC	ABX	BAY	1.

(8.68)

The rows are disjoint and contain $6 + 3 + 3 + 3 + 2 + 1 = 18$ interfaces, exactly the list (8.17).

We use the following elementary finite-dominance convention. After some rational parameters have been fixed, writing $R \gg S$ means: choose the rational R larger than the maximum of the finite list of affine lower bounds on R displayed in (8.51)–(8.64). Such a choice exists because there is no upper bound on R at that stage. Thus every hierarchy below is a finite nested choice of rational numbers, not a limiting argument.

Fix rational numbers

$$0 < \varepsilon < S. \quad (8.69)$$

Within an interval of length ε , place the nonextreme potentials in their actual-cut order. These perturbations cannot change a macro inequality whose displayed margin exceeds a fixed multiple of ε .

Witness 1: main ABC.

Place α, ν, ξ in $(S, S + \varepsilon)$ in their required cut order, put $\delta = \xi - \alpha$, and set

$$\beta = K, \quad \eta = K + \delta, \quad \gamma = K - H. \quad (8.70)$$

Choose $H > S + \varepsilon + |\delta|$. For $BY = BAY$, choose $S \ll t \ll H$; for $BY = BCY$, choose $t > H + |\delta|$; and for $BY = Q$, choose t, H larger than all microscopic potentials. Then choose K above H, t, S and all remaining affine lower bounds. Every possibly positive entry of (8.54) is bounded independently of K , while each entry containing $-K$ is negative.

If $AX = P$, take

$$M = K + \frac{t}{2}, \quad t > 2|\delta|;$$

if $AX = ABX$, let Lemma 8.3 supply $M > K + t$. Equations (8.53)–(8.54), (8.64), and (8.65) now give all required strict inequalities. The three choices of BY are separated respectively by $t < \eta - \gamma = H + \delta$, $t > H + \delta$, and the open proxy interval.

Witness 2: via-P, ABC.

Place $\alpha, \nu, \psi(X)$ in $(S, S + \varepsilon)$ in their cut order, put

$$\Delta = \psi(X) - \alpha > 0,$$

and choose

$$M = K, \quad \beta = K - \Delta, \quad \eta = K + R, \quad \gamma = \beta - H. \quad (8.71)$$

First choose R, H, t above the microscopic scale as required by the BY row, and then choose K above R, H, t . Here

$$\eta - \gamma = R + \Delta + H, \quad M - \beta = \Delta, \quad M - 2\alpha - \eta < 0.$$

All entries of (8.54) are bounded independently of K . Choose $t < R + \Delta + H$ for BAY , $t > R + \Delta + H$ for BCY , and put the proxy weight in the nonempty interval (8.64) for $BY = Q$.

Witness 3: via-P, AYC.

Place $\alpha, \nu, \psi(X)$ in $(-S - \varepsilon, -S)$, in their cut order. Let

$$0 < \Delta = \psi(X) - \alpha < \varepsilon$$

and set

$$M = K, \quad \beta = K - \Delta, \quad \eta = K + R, \quad \gamma = \beta + \sigma, \quad |\sigma| < \min\{\varepsilon, \Delta/2\}. \quad (8.72)$$

Then

$$T = -\sigma - \alpha - \nu > 2S - 3\varepsilon,$$

after decreasing ε if necessary, and

$$\xi = \alpha + R + \Delta, \quad \Omega = R + \Delta - \sigma + \alpha - \nu.$$

Choose R larger than $3S + 3\varepsilon$, and then choose K above R . Every positive entry of (8.58) is at most $2S + 2\varepsilon$. For *BAY*, identities (8.66) leave the nonempty interval

$$2S + 2\varepsilon < t < \delta + T;$$

for *BCY*, take $t > \delta + T$. In the proxy case take $t > 2S + 2\varepsilon$ and enlarge R ; both connector totals in (8.62)–(8.63) then exceed every nonextreme potential.

Witness 4: via-P, APXC.

Place $\alpha, \nu, \psi(X)$ in $(S, S + \varepsilon)$, let

$$0 < \Delta = \psi(X) - \alpha < \varepsilon,$$

and set

$$M = K, \quad \beta = K - \Delta, \quad \eta = K + R, \quad \gamma = -K, \quad K \gg R \gg S. \quad (8.73)$$

Choose $h = 2S$ for *BAY*, $h = S/2$ for *BCY*, and $h = S$ in the proxy case, after taking $\varepsilon < S/10$. These choices give the required sign of $2h - (\alpha + \nu)$, while the large K, R terms make all three upper bounds in (8.60) larger than $2h$. Every lower bound in (8.61) is finite; choose t above their maximum. This sector imposes no upper bound on t , so the choice is compatible with all other conditions.

Witness 5: via-Q, ABC.

Place α, ν in $(S, S + \varepsilon)$, choose $L > S + \varepsilon$, and set

$$\beta = K, \quad \xi = L + \rho, \quad \eta = \beta + \xi - \alpha, \quad \gamma = \beta - H, \quad (8.74)$$

where $\rho, H > S + \varepsilon$. Choose

$$t = L - \alpha + T_0, \quad T_0 > S + \varepsilon. \quad (8.75)$$

The proxy margins are

$$K_Q^A - L = T_0, \quad K_Q^C - L = \rho + H. \quad (8.76)$$

Choose K above L, ρ, H, T_0 . With $R_0 = L + \beta - \alpha$, take

$$M = R_0 + \frac{T_0}{2}$$

when $AX = P$, and let Lemma 8.3 supply $M > t + \beta$ when $AX = ABX$. Equations (8.64), (8.67), and (8.76) give the required positive widths. Every vertex on the right plateau of (8.44), not only X , belongs to the full cut and has potential L .

Witness 6: special ABC.

Place α, ν in $(S, S + \varepsilon)$, choose t above the microscopic scale, and put

$$\beta = K, \quad \delta = t + R, \quad \eta = \beta + \delta, \quad \gamma = \beta - H, \quad (8.77)$$

where $R, H > S + \varepsilon$. Choose K above t, R, H ; when $AX = ABX$, let Lemma 8.3 supply $M > \beta + t$. We have

$$\psi(X) = t + \alpha, \quad t < \delta < \eta - \gamma,$$

and every lower bound in (8.54) is strictly below t once the hierarchy is chosen. This gives the special $XBAY$ interface.

The six constructions cover every row of (8.68). Their microscopic placements realize (8.47), and the finite-dominance choices verify (8.51)–(8.64). In particular, the reconstructed weights in (8.52), (8.55), and (8.59) are positive and all prescribed branch choices are strict.

8.8.4. Short edges and non-proxy kernel traces

For the six short edges, the complete endpoint-chordless ledger is

$$\begin{array}{l|l} AB & APXB, AYCB, \\ BC & BAYC, BXC, \\ BX & BAPX, BCX, \\ CX & CBX, CYAPX, \\ AY & APXCY, ABCY, \\ CY & CBAY, CXPAY. \end{array} \quad (8.78)$$

In the same order, competitor weight minus edge weight is

$$\begin{array}{l|l} AB & M + \beta, \quad \alpha + 2g - \nu, \\ BC & t + \alpha + \nu, \quad t + \alpha + \beta + \eta + \nu - 2g, \\ BX & M - \beta, \quad \xi - \nu, \\ CX & 2g - \xi - \nu, \quad 2g + M - \eta, \\ AY & M + \eta, \quad 2g - \alpha - \nu, \\ CY & t + \alpha - \nu, \quad t + 2\alpha + \eta + M - 2g. \end{array} \quad (8.79)$$

Substitution from (8.52), (8.55), or (8.59) makes every expression in (8.79) positive. More explicitly, $M \pm \beta$, $M + \eta$, and $\xi - \nu$ occur in (8.51); the remaining expressions are exactly the positive counterparts of the lower bounds retained in Λ_{ABC} , Λ_{AYC} , or Λ_{PXC} , together with the gap conditions (8.53), (8.57), and (8.60). Thus every short edge is the unique shortest path between its endpoints.

Lemma 8.15 (non-proxy trace exhaustion)

After contracting every maximal segment of P in an endpoint-chordless competitor to its two kernel entrances, its trace is of one of the following four forms:

1. a direct P -segment or its exterior-complement detour;
2. a portal route from a P -vertex to one of B, C, Y ;
3. one of the branch paths AC, AX, BY, XY , or one of the alternative branch routes with the same endpoints;
4. one of the twelve routes in (8.78).

Proof.

The finite kernel has branch vertices A, X, B, C, Y , the contracted A -to- X chain, and the six short edges in (8.3). A simple trace that meets P twice is either the direct segment or

leaves through one end and returns through the other, giving form 1. A trace that meets P once and has its other endpoint among B, C, Y is form 2.

It remains to consider traces whose endpoints are branch vertices outside the interior of P . If their endpoints are one of AC, AX, BY, XY , the induced routes listed in (8.13) give form 3. For adjacent kernel endpoints, deleting their edge leaves exactly the two endpoint-chordless alternatives shown in the corresponding row of (8.78). Any longer simple kernel trace contains one of the six short edges joining two nonconsecutive trace vertices, and hence has a chord; if a branch vertex is repeated, the trace is not simple. These two observations exhaust the finite kernel and give form 4. $\square$

Forms 1–4 are controlled respectively by Lemma 8.3, the portal thresholds, (8.64), and (8.78)–(8.79). Lemma 8.5 therefore handles every non-proxy competitor.

8.8.5. The proxy edge

Consider first a main or via- P interface with $BY = Q$. Construct the strict core without the proxy. A rational perturbation inside an unused open connector gap makes $K_Q^A \neq K_Q^C$ without changing a selected route, and

$$D_0 = \min\{K_Q^A, K_Q^C\} = d(B, Y).$$

For every core vertex v , the triangle inequality gives

$$|\psi(v)| \leq D_0. \quad (8.80)$$

Equality at a nonterminal v would force equality in the relevant triangle inequality. Uniqueness of the three core geodesics would then make the prescribed B -to- v or Y -to- v path contain a full shortest exterior B -to- Y connector. None of the main or via- P routes in (8.17) contains such a subpath. Hence

$$\max_{v \neq B, Y} |\psi(v)| < D_0. \quad (8.81)$$

Choose a rational number

$$\max_{v \neq B, Y} |\psi(v)| < L < D_0 \quad (8.82)$$

and add BY with weight L . This changes no old distance. Indeed, a new path that uses the proxy in the direction B -to- Y has an initial portion

$$d(u, B) + L > d(u, Y)$$

because $\psi(u) < L$; replacing it by the old u -to- Y geodesic strictly shortens the path. The reverse orientation uses $-\psi(u) < L$. Thus the proxy is the unique prescribed B -to- Y path and no other core path changes.

In a via- Q interface the proxy is present from the beginning. Witness 5 gives

$$|\alpha| < L, \quad |\nu| < L, \quad L < \min\{K_Q^A, K_Q^C\}. \quad (8.83)$$

The right plateau in (8.44) legitimately has $\psi = L$: those vertices and B form the same full-cut block. This equality is not an equality between distinct cut blocks and therefore does not conflict with (8.47).

Lemma 8.16 (proxy-trace exhaustion)

Every endpoint-chordless trace containing the proxy belongs to one of the following classes:

1. the edge BY , compared with BAY and BCY ;
2. one of the four proxy triangles

$$A - Y - B, \quad A - B - Y, \quad C - Y - B, \quad C - B - Y,$$

with competitor-minus-edge slacks

$$L + \alpha, \quad L - \alpha, \quad L + \nu, \quad L - \nu; \tag{8.84}$$

3. a right-exit trace $p_iPX - B - Y$, including $X - B - Y$.

Proof.

If both B and Y are internal vertices of an endpoint-chordless trace, leaving Y through A or C creates the chord BA or BC . If B is an endpoint, extending BY to BYA or BYC creates the same chord. If Y is an endpoint, the vertex preceding B is A, C , or X . The first two possibilities are the four triangles in class 2, and the last is a right-exit trace. The case in which the trace has endpoints B, Y is class 1. These alternatives are disjoint and exhaustive. $\square$

In the via- Q sector, a right-exit trace is selected exactly when

$$x_i \geq R_0 = L + \beta - \alpha,$$

the third piece of (8.44); all remaining comparisons are its portal and endpoint margins in (8.64). In a main or via- P proxy case, right-exit traces are unselected and are strictly longer by the P -to- B portal comparison and $L - \psi(p_i) > 0$. Classes 1 and 2 are strict by (8.82)–(8.84). Together with Lemma 8.15 and Lemma 8.5, this proves Proposition 8.14. $\square$

8.9. Restoring Q

Apply Lemma 8.11 to the core metric of Proposition 8.14 with

$$s = B, \quad t = Y, \quad C = Q.$$

Condition (8.47) is precisely the terminal-potential hypothesis (8.26). If $BY \neq Q$, use the non-proxy case. If $BY = Q$, replace the proxy edge of weight L by a chain of the same total weight. In the via- Q sector, the right plateau and B constitute the full-cut block allowed by (8.35).

The long-ear lemma simultaneously realizes:

1. the actual entrance choice from every Q -vertex to every core vertex;
2. every prescribed path between a P -interior and a Q -interior vertex;
3. every direct or core-detour path between two Q -vertices;
4. every prescribed path already present in the restricted core.

These are the original prescribed paths, rather than merely paths with the same endpoint branches: consistency fixes each path from its terminal subpaths and its actual entrance cuts. Hence all eighteen interfaces in (8.17) have positive rational strict realizations.

8.10. Unit-chain boundaries

The branch count 34 is the generic count, but the proof covers all $m, n \geq 1$. The surviving signatures are recorded explicitly below.

If $m = 1$, neighborliness forces $AX = P$. The expansion and direct groups that remain are

class	XY	AC	BY
expansion	XCY	$AYC, APXC$	BAY, BCY, Q
expansion	$XBQY$	$AYC, APXC$	Q
direct	XCY	ABC	BAY, BCY, Q
direct	$XPAY$	$ABC, AYC, APXC$	BAY, BCY, Q
direct	$XBQY$	ABC	Q .

(8.85)

There are 8 expansion and 13 direct signatures. The P -interior order and same- P detour relation are empty.

If $n = 1$, neighborliness forces $BY = Q$. The surviving groups are

class	XY	AC	AX
expansion	XCY	AYC	$ABX, P, AYCX$
expansion	XCY	$APXC$	P
expansion	$XBQY$	AYC	ABX, P
expansion	$XBQY$	$APXC$	P
direct	XCY	ABC	ABX, P
direct	$XPAY$	$ABC, AYC, APXC$	P
direct	$XBQY$	ABC	ABX, P .

(8.86)

There are 7 expansion and 7 direct signatures. The proxy is replaced by the single edge BY , and there is no internal Q -order or same- Q relation.

If $m = n = 1$, both forced choices apply. The four expansion signatures are

XY	AC
XCY	$AYC, APXC$
$XBQY$	$AYC, APXC,$

and the five direct signatures are

XY	AC
XCY	ABC
$XPAY$	$ABC, AYC, APXC$
$XBQY$	ABC .

(8.87)

In (8.85)–(8.87), every expansion signature is handled by the appropriate line of (8.23), and every direct signature is handled by Lemma 8.11 with an empty order where appropriate. Contracted route names such as $XPAY$ and $XBQY$ retain the same prescribed subpaths, so no new branch interface or scalar sector is introduced.

8.11. Proof of Theorem 8.1

Lemmas 8.7 and 8.8 exhaust every generic branch interface and split them as $34 = 16 + 18$. Proposition 8.9 and Corollary 8.10 strictly realize the sixteen expansion interfaces. Proposition 8.14 and Lemma 8.11 strictly realize the eighteen direct interfaces. The explicit boundary tables (8.85)–(8.87) cover every case in which one or both chains have length one. These

alternatives include every neighborly consistent path system on $H_{m,n}^{st}$, and every constructed edge weight is positive and rational. The theorem follows. $\square$

9 Strict neighborly realization of the mother family $U_{m,n}$

Let $m, n \geq 1$, and let

$$P : A = p_0 \cdots p_m = X, \quad Q : B = q_0 \cdots q_n = Y$$

and let the seven short edges

$$AB, AC, BC, BX, CX, AY, CY$$

have weights a, b, c, d, e, f, g , respectively. All inequalities below are strict and all data may be chosen rational.

Theorem 9.1

For every $m, n \geq 1$, every neighborly consistent path system on $U_{m,n}$ has a positive rational strict realization.

The proof occupies Subsections 9.1–9.7. Its four layers are the exact 16-signature branch catalogue, simultaneous portal and cross-order insertion, strict core calibration, and the same-chain and kernel-trace exhaustion.

9.1. Exact branch catalogue

First assume $m, n \geq 2$. Induced-path exhaustion gives

$$P_{AX} \in \{P, ABX, ACX\}, \quad P_{BY} \in \{Q, BAY, BCY\},$$

and

$$P_{XY} \in \{XCY, XBAY, XPAY, XBQY\}.$$

Taking subpaths gives

$$XPAY \Rightarrow P_{AX} = P, \quad XBQY \Rightarrow P_{BY} = Q,$$

and

$$XBAY \Rightarrow (P_{AX}, P_{BY}) = (ABX, BAY).$$

There are therefore exactly

$$9 + 3 + 3 + 1 = 16$$

branch types. No further branch implication is valid: direct inspection of the four displayed route sets shows that each signature satisfying the three implications is closed under taking branch-to-branch subpaths. Hence the catalogue contains all and only the 16 compatible branch signatures.

Two cut inclusions will be used throughout. In the main type $P_{XY} = XCY$, consistency gives

$$I_C \subseteq I_Y, \quad J_C \subseteq J_X. \quad (\text{U1})$$

Indeed, an A -exit to C together with an X -exit to Y would prescribe two different subpaths from the same chain vertex to C ; the Q -side argument is symmetric. The same inclusions hold in the special type $P_{XY} = XBAY$: the offending two exits would instead prescribe two different paths to A , or symmetrically to B . Numerically we may therefore impose $C < Y$

and $V < X$, placing distinct symbols in the same open potential gap when two combinatorial cuts are equal.

If $m = 1$, the unit edge AX is neighborly and hence $P_{AX} = P = AX$. Thus the ABX , ACX , and special $XBAY$ endpoint types disappear, and the P -side same-chain detour relation is empty. The surviving types are the three main types, the three via- P types, and the one via- Q type. For $n = 1$ the symmetric seven types survive. If $m = n = 1$, both unit edges are forced and the only X - Y paths are

$$XCY, \quad XAY, \quad XBY. \quad (\text{U2})$$

where $XPAY = XAY$ and $XBQY = XBY$; these are the surviving main, via- P , and via- Q types. On a unit chain every threshold which compares its two neighborly endpoint exits has the unique nontrivial prefix, and any coincident such thresholds may be represented by distinct rational symbols in the sole open endpoint gap. The P -side Y -cut in a via- P type is full and disappears from the middle catalogue; symmetrically, the Q -side X -cut in a via- Q type is full. Thus a unit side deletes or merges types and cuts, but creates no new one. All constructions below apply to the surviving boundary types; the fixed-margin step on a unit-chain side is simply omitted.

9.2. Main type $P_{XY} = XCY$: portal insertion

Let the three A -side prefix thresholds on P be B, C, Y , and the three B -side prefix thresholds on Q be A, V, X . Consistency gives

$$C < Y, \quad V < X, \quad \delta := Y - B = X - A. \quad (\text{U3})$$

where two cuts which are combinatorially equal are represented by distinct numbers in the same open potential gap. Put

$$u = Y - C > 0, \quad v = X - V > 0, \quad \tau = u + v - \delta. \quad (\text{U4})$$

The six-cut insertion can be chosen so that

$$\tau > 0, \quad (\text{U5})$$

and, when the indicated endpoint connector is prescribed,

$$ABX : 2A + \delta > 0, \quad BAY : 2B + \delta > 0. \quad (\text{U6})$$

Here is a complete order proof. Consistency says that the two pairs (I_B, I_Y) and (J_A, J_X) cannot have opposite strict-inclusion directions. Indeed, if $p \in I_Y \setminus I_B$ and $q \in J_A \setminus J_X$, then the four possible induced cross routes using AB , AY , XB , and XCY contradict, respectively, $p \notin I_B$, $q \in J_A$, $q \notin J_X$, and $p \in I_Y$; hence no route is available. The opposite strict-inclusion pattern is symmetric. If at least one pair is strict, use its direction; if both pairs are equal, either of the following two constructions may be used.

Suppose first that the common direction is positive. Normalize

$$B = A = 0, \quad Y = X = 1, \quad \delta = 1. \quad (\text{U6a})$$

In the central rectangle, compare a P -potential symbol x_i and a Q -potential symbol y_j by

$$x_i \prec y_j \iff P_{p_i q_j} \text{ uses } AY,$$

and put $y_j \prec x_i$ when it uses XB . Taking subpaths shows that the AY -cells form a Ferrers staircase. Equivalently, these comparisons, together with the two internal chain orders, are acyclic and compare every P/Q pair in the central rectangle.

Insert the symbol C into the open gap of the P -order prescribed by I_C , and insert V into its prescribed Q -gap. Each new symbol is constrained only relative to one already ordered chain, so the augmented order remains acyclic. Embed a linear extension rationally. All symbols strictly between the 0- and 1-cuts may be compressed into $(0, 1/2)$. If the C -cut lies weakly before the B -cut, put $C \in (-1/2, 0)$, using their common empty gap in the equality case; otherwise put $C < 1/2$ in its prescribed central gap. Place $V < 1/2$ symmetrically. Thus

$$\tau = 1 - C - V > 0, \quad 2A + \delta = 2B + \delta = 1 > 0. \quad (\text{U6b})$$

Suppose next that the common direction is negative. Normalize

$$B = A = 1, \quad Y = X = 0, \quad \delta = -1. \quad (\text{U6c})$$

The unresolved cells now form the down-set

$$AB \text{ is selected} \iff x_i + y_j < 1.$$

Put $z_j = 1 - y_j$, with the Q -order reversed. The down-set prescribes the acyclic merged order $x_i < z_j$ for its members and $z_j < x_i$ for its nonmembers. Embed that finite order in $(0, 1)$ and recover $y_j = 1 - z_j$. By $C < Y$ and $V < X$, put $C, V < 0$ in their prescribed gaps. Again

$$\tau = 1 - C - V > 0, \quad 2A + \delta = 2B + \delta = 1 > 0. \quad (\text{U6d})$$

If two combinatorial cuts are equal, their distinct threshold symbols may be placed anywhere in the same potential-free open gap. Finally put all unused prefix and suffix potentials outside the displayed central intervals, preserving their chain orders. This constructs all six portals and proves (U5)–(U6) for every cut signature.

Equations (U3)–(U4) also give the required triangle condition:

$$A + Y - C - V = u + v - \delta = \tau > 0. \quad (\text{U7})$$

For completeness, after deleting the common endpoint constants, the four canonical cross-route lengths are

$$\begin{array}{l|l} AB & (x + y)/2 + a \\ AY & (x - y)/2 + f \\ XB & (-x + y)/2 + d \\ XCY & (-x - y)/2 + e + g. \end{array} \quad (\text{U7a})$$

Their adjacent comparisons are exactly the B -, A -, Y -, and X -portal cuts. If $\delta > 0$, the only remaining cells compare AY with XB and

$$AY < XB \iff x - B < y - A;$$

if $\delta < 0$, they compare AB with XCY and

$$AB < XCY \iff (x - Y) + (y - X) < -\delta.$$

Thus the first case is a strict difference-Ferrers insertion and the second a strict additive-Ferrers insertion. If both orientation-defining cut pairs are equal, use either normalization (U6a) or (U6c); no zero parameter or limiting argument is needed.

9.3. Main type: exact core calibration

Solve the threshold equations by

$$\begin{aligned} a &= r + g, & d &= a + B, & f &= a + A, \\ e &= r + A + Y, & b &= e - C, & c &= g - V. \end{aligned} \quad (\text{U8})$$

Then $e + g - d = X$ follows from (U3). Three triangle slacks are already fixed:

$$b + c - a = \tau, \quad c + e - d = v, \quad b + g - f = u. \quad (\text{U9})$$

Define

$$\begin{aligned} L_g &= \max \left\{ 0, V, \frac{A + Y + V - C}{2}, \frac{A + Y + V - B}{2}, \frac{Y - C}{2} \right\}, \\ L_r(g) &= \max \left\{ -g, -g - B, -g - A, -A - Y, C - A - Y, \right. \\ &\quad \left. \frac{C - V - A - Y}{2}, -\frac{A + B + Y + V}{2}, \frac{C - 2A - Y}{2} \right\}. \end{aligned} \quad (\text{U10})$$

The seven weights in (U8) are positive and all three core triangles ABC, BCX, ACY are strict whenever

$$g > L_g, \quad r > L_r(g). \quad (\text{U11})$$

This follows by expanding the six remaining triangle slacks; the five g -bounds and the eight r -bounds in (U10) are exactly positivity and those six inequalities.

The two pairs of exterior connector totals satisfy

$$\begin{aligned} ABX < ACX &\iff g < G_P, \quad G_P = A + \frac{\delta + u}{2}, \\ BAY < BCY &\iff r < G_Q, \quad G_Q = -A - \frac{\delta}{2} + \frac{v}{2}. \end{aligned} \quad (\text{U12})$$

If ABX is prescribed, (U6) and (U4)–(U5) give

$$L_g < G_P.$$

Indeed, in the displayed order in (U10), the five differences are

$$\frac{2A + \delta + u}{2}, \quad \frac{v + \tau}{2}, \quad \frac{v}{2}, \quad \frac{\tau}{2}, \quad \frac{2A + \delta}{2}.$$

If BAY is prescribed, the differences from G_Q to the five g -independent entries of $L_r(g)$, in their displayed order, are

$$\frac{2B + \delta + v}{2}, \quad \frac{u + \tau}{2}, \quad \frac{u}{2}, \quad \frac{2B + \delta}{2}, \quad \frac{\tau}{2}.$$

The three remaining bounds require

$$g > \max \left\{ A + \frac{\delta}{2} - \frac{v}{2}, \frac{\delta}{2} - \frac{v}{2}, A + \frac{\delta}{2} - \frac{v}{2} - B \right\}. \quad (\text{U13})$$

When ABX and BAY are both prescribed, every entry in (U13) is still strictly below G_P ; the three differences are, respectively,

$$\frac{u + v}{2}, \quad A + \frac{u + v}{2}, \quad B + \frac{u + v}{2},$$

and the last two are positive by (U5)–(U6). Thus the choices of g and r are compatible in all nine endpoint branch types. For ACX or BCY , reverse the corresponding inequality in (U12); for a direct chain endpoint there is no connector-order constraint.

Finally

$$w(XBAY) - w(XCY) = 2(r + g) - \delta. \quad (\text{U14})$$

If one of r, g has no upper bound, make it large. If both ABX and BAY are prescribed, take g, r near their upper endpoints. Since

$$G_P + G_Q = \frac{u + v}{2} > \frac{\delta}{2},$$

(U14) is again strict. The other two competitors satisfy

$$w(XPAY) - w(XCY) = M - Y > 0, \quad w(XBQY) - w(XCY) = N - X > 0 \quad (\text{U15})$$

because the portal cuts are internal.

If P is the prescribed endpoint path, choose its centered potentials only after the core calibration and put their endpoints strictly between the six portal symbols and the smaller A – X connector. This interval is nonempty by the same inequalities above. The only case with an upper bound on r is BAY ; as $r \uparrow G_Q$, the ACX total tends to

$$B + u + v.$$

Writing $T_P = B + u + v$, the six gaps are exactly

$$\begin{array}{l|l} T_P - B & u + v \\ T_P + B & 2B + \delta + \tau \\ T_P - C & u + \tau \\ T_P + C & 2B + \delta + v \\ T_P - Y & \tau \\ T_P + Y & 2B + \delta + u + v. \end{array}$$

They are all positive by (U4)–(U6), so T_P exceeds each of $\pm B, \pm C, \pm Y$. Choose $r < G_Q$ sufficiently close to G_Q , then increase g , and put

$$\max\{|B|, |C|, |Y|\} < M < \min\{ABX, ACX\}. \quad (\text{U15a})$$

Thus both connectors can exceed M even though ACX does not grow with g . The symmetric exceptional case is $P_{BY} = Q$ together with $P_{AX} = ABX$: as $g \uparrow G_P$, the BCY total tends to $A + u + v$, which exceeds $\pm A, \pm V, \pm X$ by (U5)–(U6); then increase r and insert N between the portal symbols and both B – Y connectors. If an exterior connector is prescribed, use the fixed-margin same-chain construction with that connector total; (U12) makes the unused connector strictly longer.

9.4. The special type $P_{XY} = XBAY$

Consistency forces $(P_{AX}, P_{BY}) = (ABX, BAY)$. Here

$$B = d - a, \quad A = f - a, \quad Y = d + a, \quad X = f + a,$$

so $Y - B = X - A = 2a > 0$. After a common rational scaling, choose

$$a = d = f = \frac{1}{2}, \quad b = c = 10, \quad e = 10 + C, \quad g = 10 + V, \quad (\text{U16a})$$

with $|C|, |V| < 1/2$. Insert the six cuts with

$$B = A = 0, \quad Y = X = 1$$

as follows. Consistency gives $I_B \subseteq I_Y$ and $J_A \subseteq J_X$, while (U1) gives $I_C \subseteq I_Y$ and $J_C \subseteq J_X$. In the central rectangle the four induced route expressions, after deletion of the common endpoint constants, are

$$\begin{array}{l|l} AB & (x+y)/2 + 1/2 \\ AY & (x-y)/2 + 1/2 \\ XB & (-x+y)/2 + 1/2 \\ XBAY & (-x-y)/2 + 3/2. \end{array} \quad (\text{U16b})$$

The four adjacent comparisons are precisely the 0- and 1-cuts, and the only unresolved comparison is

$$AY < XB \iff x < y.$$

Use the merged-order construction from the positive case of Subsection 9.2, insert C and V into their prescribed single-chain gaps, and compress the central symbols so that $|C|, |V| < 1/2$. Equal cuts use a wide common potential-free gap. This realizes every cross-chain path and all six portals, not only their endpoint cuts.

All three core triangles are strict. Moreover

$$ABX = BAY = 1, \quad XBAY = 3/2,$$

whereas $ACX = 20 + C$, $BCY = 20 + V$, and $XCX = 20 + C + V$ are larger than 19. The fixed-margin construction of Subsection 9.6 chooses $M, N > 1$, so

$$XPAY = M + 1/2 > 3/2, \quad XBQY = N + 1/2 > 3/2.$$

Apply the two fixed-margin extensions with the four comparisons in (U16b) and all six portal cuts included in their auxiliary family. Outward safety keeps this merged order unchanged while the P- and Q-active tails are moved; the pinned construction in Subsection 9.6 leaves the middle cut symbols and core weights numerically fixed. Hence the inequalities $M, N > 1$, the six cuts, and the central table hold simultaneously. Thus every branch competitor is strictly longer than $XBAY$.

9.5. Via- P and via- Q

Suppose $P_{XY} = XPAY$. Then $P_{AX} = P$, every prescribed p_i -to- Y path exits P through A , and the Q -side X portal is

$$X = f + M - d = M + A - B. \quad (\text{U17})$$

Use a core scale K , distinct from the chain total M , and put

$$a = e = K, \quad d = K + B, \quad f = K + A, \quad (\text{U18})$$

and

$$g = K + t, \quad c = K + t - V, \quad b = K - C. \quad (\text{U19})$$

Choose the bounded rational t by

$$\begin{array}{l|l} P_{BY} = BAY & t > (A + V)/2, \\ P_{BY} = BCY & t < (A + V)/2, \end{array} \quad (\text{U20})$$

with no restriction on t when Q is selected. Indeed

$$BCY - BAY = 2t - A - V.$$

After M, A, B, C, V, t have been fixed, choose

$$K > \max\{M + A - t, (M - B)/2, (M + C)/2\} + \varepsilon \quad (\text{U21})$$

and larger than the finitely many bounded positivity and triangle thresholds. The branch margins are then

$$\begin{aligned} ABX - P &= 2K + B - M, \\ ACX - P &= 2K - C - M, \\ XCY - XPAY &= K + t - M - A, \\ XBAY - XPAY &= 2K + B - M, \\ XBQY - XPAY &= N - X. \end{aligned} \quad (\text{U22})$$

The first four are positive by (U21); the last is positive because the X cut in (U17) is strictly internal to Q . Thus one must take $K > M + A - t$, not the equality $K = M + A - t$.

It remains to synchronize the cross table and the endpoint portal in (U17). A selected cross path cannot use XCX or $XBAY$, since either would prescribe an X - Y subpath different from $XPAY$. After deleting the common endpoint constants, the three possible selected routes are exactly

$$\begin{array}{l|l} AB & (x + y)/2 + K \\ AY & (x - y)/2 + K + A \\ XB & (-x + y)/2 + K + B. \end{array} \quad (\text{U22a})$$

Their AB comparisons are the B - and A -cuts, and the remaining rectangle is

$$AY < XB \iff x - B < y - A. \quad (\text{U23})$$

Normalize $A = B = 0$ while constructing the order. Merge the P -symbols $x_i - B$ and Q -symbols $y_j - A$, putting the P -symbol first exactly in the AY -cells. Consistency gives the Ferrers monotonicity and hence an acyclic total comparison system. Include the endpoint P -symbol $x_X = M$. Its comparisons with the Q -symbols are precisely the prescribed Q -side X -cut, so the cut symbol must have coordinate

$$X = M,$$

which is (U17) in this normalization. Insert C into its P -gap and V into its Q -gap; neither creates a cross-order constraint. Undoing the two middle translations gives $X = M + A - B$. Thus the portal formula is supplied by the endpoint symbol itself and is not an additional scalar assumption.

If Q is prescribed, choose

$$\max\{|A|, |V|, |X|\} < N < \min\{BAY, BCY\},$$

which is possible after increasing K . If a Q connector is prescribed, use (U20) and then apply its fixed-margin extension, so that the chosen connector is below N and the unused connector. Increasing N only increases the last margin in (U22).

In that extension the X -cut is included in the fixed middle auxiliary data. Recentering Q translates A, V, X together, while recentering P translates B, C, x_X together; therefore

$$X - A = x_X - B$$

is invariant. The endpoint anchor (U17), every chosen connector comparison, and the full cross table consequently survive the tail placement simultaneously.

The common core scale K cancels from all three comparisons in (U22a).

For the via-Q type $P_{XY} = XBQY$, interchange (P, A, X, M, B, C) with (Q, B, Y, N, A, V) . Concretely, keep $a = K, d = K + B, f = K + A, g = K, c = K - V$, put

$$e = K + s, \quad b = K + s - C,$$

and choose $s > (B + C)/2$ for ABX , $s < (B + C)/2$ for ACX . The P -side Y portal is

$$Y = d + N - f = N + B - A.$$

The five margins symmetric to (U22) are

$$\begin{aligned} BAY - Q &= 2K + A - N, \\ BCY - Q &= 2K - V - N, \\ XCY - XBQY &= K + s - B - N, \\ XBAY - XBQY &= 2K + A - N, \\ XPAY - XBQY &= M - Y. \end{aligned} \tag{U24}$$

Choose $K > N + B - s + \varepsilon$ and then treat the P endpoint by its direct or fixed-margin connector construction. All margins in (U24) are strict.

The endpoint-anchor construction is symmetric. Normalize $A = B = 0$, include the endpoint Q -symbol $y_Y = N$ in the merged difference order, and observe that its comparisons with the P -symbols are exactly the prescribed P -side Y -cut. Hence $Y = N$ in the normalization and

$$Y = N + B - A$$

after undoing the translations. Insert C and V in their respective single-chain gaps. This realizes the full via- Q cross table and all portal cuts simultaneously. During a later P -side fixed-margin extension the same gauge argument preserves $Y - B = y_Y - A$, so the endpoint anchor, chosen connector, and cross table remain synchronized.

9.6. Same-chain and noncanonical paths

We use the following fixed-margin tool, restated here for convenience.

Fixed-margin outward-safe extension. Let

$$F \subseteq \{0, \dots, p\} \times \{p + 1, \dots, m\}$$

be expanding, and take the active prefix to consist exactly of its nonempty rows. Suppose a finite family of already strict affine comparisons is outward safe: every available expression involving an active-prefix label has coefficient $+x_i$, every one involving an active-suffix label has coefficient $-x_j$, and every prescribed middle cut contains the active prefix and avoids the active suffix. For a positive rational connector margin D , after an order-preserving compression of the still-free finite gaps there are centered rational labels

$$-M = x_0 < \dots < x_m = M$$

which preserve all auxiliary comparisons and middle cuts and satisfy

$$(i, j) \in F \iff x_j - x_i > M + D$$

strictly in both directions. The symmetric assertion holds with an active suffix moved right.

This is the centered specialization of Lemma 6.3 and its outward-safety hypothesis. For completeness, keep the right-block labels $x_{p+1}, \dots, x_m$ fixed and let $r_i = \min\{j : (i, j) \in F\}$. Expansion makes the r_i nondecreasing. Choose increasing rational gap labels

$$x_{r_i-1} < H_i < x_{r_i};$$

the left endpoint is interpreted as $-\infty$ when $r_i = p+1$. When several r_i 's agree, choose the corresponding H_i 's as distinct rationals in that one open gap. Before centering put

$$T = 2D + x_m - H_0, \quad x_i = H_i - T \quad (0 \leq i \leq p).$$

Then

$$x_j - x_i > T \iff x_j > H_i \iff j \geq r_i, \quad T - \frac{x_m - x_0}{2} = D. \quad (\text{U25})$$

The preliminary compression makes the prescribed D large relative to the finite inactive order span, so the moved block lies in its outward-safe region. Translate the chain labels and all of its cut symbols together to the centered gauge; differences and every auxiliary comparison are unchanged, and $T = M + D$. This proves the specialized statement using only finite strict order insertions; in particular all labels remain rational. If a global rescaling is used, it is simultaneous:

$$(x, \text{ cut symbols, core weights, } D) \longmapsto \lambda(x, \text{ cut symbols, core weights, } D).$$

Thus $D = w(R)$ is invariant as an identity; no argument rescales the order data while holding a previously chosen connector margin fixed.

For the application below we use a pinned centered form, so that no recalibration follows the same-chain extension. Complete the portal and cross-order insertion first, and then choose the final core weights. If R is a selected exterior connector, put $D = w(R)$. The still-free inactive labels may be compressed so that all fixed middle vertex and cut symbols lie between $-D$ and D ; equality at an endpoint is allowed only in a potential-free gap.

Indeed, the selected endpoint detour implies $(0, m) \in F$. Put $r_i = \min\{j : (i, j) \in F\}$. Expansion gives

$$\{j : (0, j) \in F\} = \{r_0, r_0 + 1, \dots, m\}.$$

Keeping the previously fixed finite order, choose the right-block labels so that, in the final placement,

$$x_j < D \quad (p+1 \leq j < r_0), \quad D < x_j \quad (r_0 \leq j \leq m). \quad (\text{U25a})$$

The first range is empty when $r_0 = p+1$. When $r_0 = m$, first put $H_0 = D$ in the last required gap (with its left end open if $r_0 = p+1$), insert the remaining H_i 's that use this gap, and only then choose $x_m = M$ above all of them. In every other case choose $M = x_m > D$ after the finite middle symbols have been placed. Now choose increasing rational numbers H_i , putting those with $r_i = r_0$ successively above $H_0 = D$, so that

$$x_{r_i-1} < H_i < x_{r_i}, \quad H_0 = D,$$

with the usual open-end convention. Repeated values of r_i are separated inside the same open gap. Put

$$x_m = M, \quad T = M + D, \quad x_i = H_i - T \quad (0 \leq i \leq p).$$

Then $x_0 = -M$, and

$$x_j - x_i > M + D \iff x_j > H_i \iff (i, j) \in F.$$

Taking M sufficiently large moves only the active prefix into its outward-safe region. Thus every fixed portal symbol, auxiliary comparison, and core weight remains numerically unchanged.

The required domination by D is compatible with the scalar calibration. In the main sector an unbounded connector is enlarged by its free parameter. If the P -connector is bounded by the endpoint choice $P_{BY} = BAY$, its limiting value is

$$T_P = B + u + v;$$

if the Q -connector is bounded by $P_{AX} = ABX$, its limiting value is

$$T_Q = A + u + v.$$

The six positive gaps displayed after (U15a), and their symmetric counterparts, show that T_P and T_Q dominate all corresponding portal symbols. The two upper endpoints may be approached simultaneously. In the special sector, the P -side identity is $D = Y = 1$, and the Q -side identity is symmetrically $D = X = 1$. These are not additional calibrations: consistency of the corresponding endpoint-detour subpath says that the connector cut and the row-zero boundary occupy the same gap. Thus $H_0 = D$ is legitimate, and the middle symbols were already compressed into that unit gap. In the via- P and via- Q sectors the common scale K makes the selected connector arbitrarily large. Hence the pinned construction applies in every active sector.

If both chain relations are active, perform the two pinned constructions successively. The P -relation involves only the P -labels and its fixed connector total, and the Q -relation is symmetric; outward safety preserves every cross comparison during the second construction. Thus the first fitted relation is unchanged.

Now let R be a selected exterior A - X connector. Consistency makes its same-chain detour pairs an expanding relation between a prefix P^- and a suffix P^+ . Every portal prefix contains P^- and avoids P^+ : the opposite exit would put a direct $p_i P p_j$ segment inside a prescribed portal or cross path, contradicting the selected detour subpath. Thus all portal cuts lie in the middle block. Moreover the only available routes on P^- exit through A , and those on P^+ exit through X ; their affine expressions have respectively the common coefficients $+x_i$ and $-x_j$. Hence outward safety holds, and the pinned form above with $D = w(R)$ realizes

$$R\text{-detour} \iff x_j - x_i > M + w(R) \quad (\text{U25b})$$

strictly in both directions while preserving every portal and cross-table comparison. The unused A - X connector is strictly longer by the endpoint branch choice, so it cannot create a tie. The Q statement is symmetric. If the direct endpoint chain is selected, consistency selects every one of its subsegments directly; the direct-chain intervals in Subsection 9.3 (and their via-sector analogues) put both connectors above the chain total. On a unit chain this statement is automatic.

We next make the competitor exhaustion explicit.

Kernel-trace lemma. Let W be an induced path in $U_{m,n}$. Contract each maximal subpath of W contained in P or Q . Up to reversal and specialization of an endpoint to a branch vertex, its trace is one of:

1. two vertices of P : the direct P -segment, or its complementary P -segment followed by ABX or ACX ;
2. two vertices of Q : the symmetric three traces through BAY or BCY ;
3. a P -vertex and C : the A - C or X - C portal route; symmetrically, a Q -vertex and C has the B - C or Y - C portal route;

4. an internal P -vertex and an internal Q -vertex: exactly

$$AB, \quad AY, \quad XB, \quad XCY, \quad XBAY; \quad (\text{U26})$$

5. one of the A - X , B - Y , or X - Y branch paths in Subsection 9.1, or a boundary form of one of the preceding traces.

To prove the lemma, an induced same-chain path which leaves its chain must use both chain ends. The only exterior connector traces from A to X are ABX, ACX , and those from B to Y are BAY, BCY ; when the corresponding chain has length greater than one, these are also the induced exterior connectors. A path from a chain to C has the two stated exits. For two internal P/Q endpoints, the four endpoint pairs A/X and B/Y give the three short edges AB, AY, XB ; the X - Y pair has exactly the two induced core paths XCX, XBY . Every longer kernel walk repeats a branch vertex or has one of the seven short core edges as a chord. Specializing an endpoint gives the boundary forms and the four X - Y branch paths. This proves the list for all $m, n \geq 1$.

All traces in the lemma have already been compared. In the main sector the four table routes are AB, AY, XB, XCY ; the fifth trace $XBAY$ contains the nonprescribed X - Y branch and is strictly reduced by (U14). In the special sector the table uses $AB, AY, XB, XBAY$, while XCX is longer by Subsection 9.4. In a via- P or via- Q sector, only AB, AY, XB can be selected; the two core X - Y traces are strictly reduced by (U22) or (U24) to the selected via-chain branch. If that replacement creates a repeated chain segment, deleting the resulting positive closed walk leaves a strictly shorter path. The same-chain traces are controlled by (U25b), the chain-to- C traces by the portal cuts, and all branch traces by Subsections 9.3–9.5.

It remains to verify the edge bases required by the endpoint-chord reduction. The four portal thresholds forced by adjacent branch endpoints are

$$B = d - a, \quad C = e - b, \quad A = f - a, \quad V = g - c.$$

Their placement in open endpoint gaps gives

$$-M < B, C < M, \quad -N < A, V < N, \quad (\text{U27})$$

where M, N are the totals of P, Q . The three core triangles ABC, BCX, ACY satisfy strict triangle inequalities by the core calibration. Every endpoint-chordless detour for a short edge is therefore strictly longer for the reason shown in the following complete ledger:

edge	endpoint-chordless detour	excess quantity
AB	$A - P - X - B$	$M + d - a = M + B > 0$
	$A - C - B$	$b + c - a > 0$
	$A - Y - Q^{-1} - B$	$f + N - a = N + A > 0$
AC	$A - P - X - C$	$M + e - b = M + C > 0$
	$A - B - C$	$a + c - b > 0$
	$A - Y - C$	$f + g - b > 0$
BC	$B - Q - Y - C$	$N + g - c = N + V > 0$
	$B - A - C$	$a + b - c > 0$
	$B - X - C$	$d + e - c > 0$
BX	$B - A - P - X$	$M + a - d = M - B > 0$
	$B - C - X$	$c + e - d > 0$
CX	$C - A - P - X$	$M + b - e = M - C > 0$
	$C - B - X$	$c + d - e > 0$
AY	$A - B - Q - Y$	$N + a - f = N - A > 0$
	$A - C - Y$	$b + g - f > 0$
CY	$C - A - Y$	$b + f - g > 0$
	$C - B - Q - Y$	$N + c - g = N - V > 0.$

(U28)

To see that the table is exhaustive, remove the edge under consideration. An induced detour cannot enter and leave the interior of a suspended chain twice. After suppressing each maximal chain segment, an induced core trace between the endpoints is one of the traces displayed in (U28); every other core trace contains one of the seven short edges as a chord. For a chain edge of P , every endpoint-chordless detour is the complementary chain segment followed by one of the two induced exterior A - X connectors ABX or ACX . The prescribed endpoint path strictly orders both connector totals against the direct chain, and the same-chain Ferrers relation compares every internal segment with the smaller connector. The other connector is strictly longer. The symmetric statement holds on Q , with connectors BAY and BCY .

Lemma 5.1 now proves by induction on the number of edges in a competitor that every graph edge is uniquely shortest: an endpoint-chordless competitor is covered above, while a chorded competitor is shortened by replacing a proper chorded subpath with its chord edge.

Given an arbitrary competitor to a prescribed path, repeatedly replace a proper chorded subpath by its uniquely shorter edge. The result is an induced competitor. Apply the kernel-trace lemma. It is either one of the canonical traces already compared, or it contains a nonprescribed branch trace, which is replaced by its strictly shorter prescribed branch path. Delete any positive closed walk created by an overlap and repeat. Each operation is strict and returns, after cycle deletion, one of finitely many simple paths, so the process terminates at the prescribed canonical trace. Hence every prescribed path is uniquely shortest.

9.7. Completion of the proof

The catalogue, six-cut synchronization, core calibration, special types, same-chain margins, and endpoint-chordless ledger together prove Theorem 9.1. $\square$

10 Proof of Theorem 1.1

We now assemble the structural and realization results.

For a finite graph G , the following statements are equivalent.

1. G is strictly metrizable.
2. G has no minor isomorphic to any graph in

$$\mathcal{F} = \{K_{2,4}, C_3 \square K_2, K_{3,3}, K_4^\Delta, K_4^\star, \Theta_{3,3,3}\},$$

where K_4^Δ is obtained by subdividing once each edge of a triangle of K_4 , and $K_4^\star$ is obtained by subdividing once each edge incident with one vertex of K_4 .

Proof

Suppose first that G is strictly metrizable. Strict metrizability is minor-closed [2, Theorem 1.1], and none of the six graphs in $\mathcal{F}$ is strictly metrizable [2, Figures 13–14 and Appendix A]. Consequently G contains no member of $\mathcal{F}$ as a minor.

Conversely, suppose that a graph with no minor in $\mathcal{F}$ is not strictly metrizable. By Theorem 3.10 there is a minor-minimal such graph carrying a full-support neighborly consistent path system that is not strictly realizable. The same theorem shows that the graph is either K_5 , W_5 , or a subdivision of one of

$$K_{2,3}, \quad K_4, \quad W_4, \quad W'_4 = K_5 - e.$$

Propositions 4.1, 4.3, 4.5, and 4.8 reduce the four subdivided cores, including all unit-parameter boundaries, to the following families:

$$\Theta_{2,b,c}, \quad K_4 \text{ with at most two long core edges}, \quad P_4(p, 2, r), \quad H_{m,n}^{rr}, \quad H_{m,n}^{sr}, \quad U_{m,n}.$$

Theorem 5.3 treats $\Theta_{2,b,c}$; Theorems 6.6 and 6.9 treat the K_4 and split families; and Theorems 7.1, 8.1, and 9.1 treat $H_{m,n}^{rr}$, $H_{m,n}^{sr}$, and $U_{m,n}$, respectively. Each theorem supplies a positive edge weighting that makes every selected path uniquely shortest. Proposition 4.9 does the same for K_5 , and Proposition 4.12 does the same for W_5 . Thus every case produced by the structural reduction is strictly realizable, contradicting the choice of the path system. Hence every $\mathcal{F}$ -minor-free graph is strictly metrizable. $\square$

Theorem 1.1 proves Conjecture 5.1 of Chudnovsky, Cizma, and Linial [2].

Role of finite verification

No finite experiment is used to infer an infinite-family statement. The theta, K_4 , split, H^{rr} , H^{sr} , and U realization theorems are symbolic. The only finite certificate used in the proof is the exact integer certificate for W_5 : consistency leaves 57 words, dihedral symmetry leaves 12 representatives, and the displayed positive integer weights have strictly positive margins against every alternative simple path. The supplementary verifier merely recomputes these integer sums.

Declarations

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Author contributions. Guangfu Wang: conceptualization, methodology, formal analysis, investigation, software, validation, visualization, writing—original draft, and writing—review and editing. The author reviewed and approved the final manuscript.

Data availability statement. No external data sets were generated or analyzed. All mathematical certificate data used in the proof are displayed in the article. The supplementary material contains the standalone exact-integer verifier and its expected output for the W_5 certificate; the proof does not depend on executing that program.

Use of generative artificial intelligence. During preparation of the manuscript, the author used OpenAI Codex for English-language editing, formal organization and clarification of proof arguments, LaTeX preparation, and generation of the supplementary verification script. Every AI-assisted mathematical argument was checked against the stated definitions and reduced to the explicit derivations in the manuscript; the integer certificate was independently recomputed by the supplementary script. The author reviewed the arguments, certificate data, references, and final text and takes full responsibility for the content.

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